UNIT-3

ORDINARY DIFFERENTIAL EQUATION OF 1st ORDER AND 1st DEGREE.

Differential Equation: An equation involving the derivatives of dependent variable with respect to two or more independent variables is known as differential equation.

Ex: $\frac{dy}{dx} + y = 0$

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial \omega}{\partial z} = 5$$

Differential equations are of two types:

1. Ordinary differential equation: An equation involving the derivatives of dependent variable w.r.t one independent variable is known as ordinary differential equation.

Ex: $\frac{d^2y}{dx^2} + 6\frac{dy}{dx} + 5y = e^x$

2. partial differential equation: An equation involving derivates of dependent variable w.r.t one or more independent variables is known as partial differential equation.

Ex: $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 0$.

Order: The order of differential equation is highest derivative in the given differential equation.

Degree: The power of highest order is its degree.

NOTE: Here the equations are free of radicals and fractions.

S.No	Differential equation	order	degree
1.	$\frac{dy}{dx}+e^{x}$	1	1
2.	$\frac{d^2y}{dx^2}+\left(\frac{dy}{dx}\right)^2+2y=0$	2	1
3.	$\left(\frac{d^3y}{dx^3}\right)^2=x^2\frac{dy}{dx}$	3	2
4.	$y=e^{dy/dx}$	1	∞
5	$y=\sin\left(\frac{dy}{dx}\right)$	1	∞
6.	$\frac{\partial^2u}{\partial x^2}+\frac{\partial^2u}{\partial y^2}=1$	2	1
7	$\sqrt{1+\frac{dy}{dx}}=\frac{d^2y}{dx^2}$ (on squaring both sides)	2	2

Formation of D.E:

A D.E is formed by eliminating arbitrary constants from the given equation. Let $f(x,y,c_1,c_2,\dots,c_n)=0 \rightarrow ①$

Differentiate eq① w.r.t x;

we get a differential equation of the form

$$f\left(x,y,\frac{dy}{dx},\frac{d^2y}{dx^2},\dots,\frac{d^{n+1}y}{dx^{n+1}}\right)$$

Solution of D.E:

A solution of integral differential equation is the relation between the variables which satisfies the differential equation.

$\rightarrow$ $x=a\cos(nt+\alpha)$ is the solution of differntial equation

$$\frac{d^2x}{dt^2}+n^2x=0.$$

General / complete solution: General solution / complete solution (G.S / C.S) is a solution of D.E where the no. of orbitrary constants is equal to order of D.E

Example: $x = a\cos(nt+\alpha)$ is general solution of $\frac{d^2x}{dt^2} + n^2x = 0$ where orditrary constants = order of D.E = 2.

particular solution: A particular solution is a solution obtained from general solution by giving particular values to arbitrary ~~so~~ constants.

Example: $x = a\cos(nt+\alpha)$ is general solution of differential eq. $\frac{d^2x}{dt^2} + n^2x = 0$ where $a = 2, \alpha = \frac{\pi}{2}$

$$x = 2\cos\left(nt + \frac{\pi}{2}\right) \text{ is particular solution.}$$

① * Form a D.E by eliminating arbitrary constants A and B from $xy = Ae^x + Be^{-x}$

Sol Given, $xy = Ae^x + Be^{-x} \rightarrow ①$

Differentiating on both sides w.r.t x,

$$x.\frac{dy}{dx} + y.\frac{dx}{dx} = Ae^x - Be^{-x} \quad \left(\because \frac{d}{dx}(uv) = u\frac{dv}{dx} + v.\frac{du}{dx}\right)$$

$$xy' + y = Ae^x - Be^{-x} \rightarrow ②$$

Differentiating on both sides w.r.t 'x',

$$xy'' + y' + y' = Ae^x + Be^{-x}$$

$$\therefore xy'' + 2y' = xy \quad (\because \text{from } ①)$$

$\therefore xy'' + 2y' - xy = 0$ is the required D.E.

② $x = a\sin(\omega t + b)$

Given, $x = a\sin(\omega t + b) \rightarrow ①$

Differentiating on b.s w.r.t 't';

$$\frac{dx}{dt} = a\cos(\omega t + b).\omega$$

Again differentiating w.r.t 't';

$$\frac{d^2x}{dt^2} = a\omega.(-\omega.\sin(\omega t + b)) = -\omega^2 x \quad (\because \text{from } ①)$$

$\therefore \frac{d^2x}{dt^2} + \omega^2 x = 0$ is D.E of $x = a\sin(\omega t + b)$

② Form a D.E by eliminating arbitrary constants from $\log\left(\frac{y}{x}\right) = cx$.

Sol Given, $\log\left(\frac{y}{x}\right) = cx \longrightarrow ①$

Differentiating on both sides w.r.t x;

$$\frac{d}{dx}\left(\log\left(\frac{y}{x}\right)\right) = \frac{d}{dx}(cx)$$

$$\frac{1}{(y/x)} \cdot \frac{x \cdot y' - y \cdot 1}{x^2} = c$$

$$\therefore c = \frac{x}{y} \cdot \left(\frac{xy' - y}{x^2}\right) \longrightarrow ②$$

Substitute ② in ①;

$$\log\left(\frac{y}{x}\right) = \frac{\cancel{x}}{y} \cdot \left(\frac{xy' - y}{x^{\cancel{2}}}\right) \cdot \cancel{x}$$

$$\log \frac{y}{x} = \frac{xy'}{y} - 1$$

$\therefore y \log\left(\frac{y}{x}\right) = xy' - y$ is 1st order D.E.

③ Form a D.E by eliminating arbitrary constants from the family of circles $(x-a)^2 + (y-b)^2 = r^2$.

Sol Given, family of circles equation is $(x-a)^2 + (y-b)^2 = r^2 \longrightarrow ①$.

Differentiating on both sides w.r.t x;

$$2(x-a) + 2(y-b)y' = 0$$

$$(x-a) + y'(y-b) = 0 \longrightarrow ②$$

Again differentiate on both sides w.r.t x;

$$1 + y' \cdot (y' - 0) + (y-b)y'' = 0$$

$$\therefore (y-b)y'' + (y')^2 + 1 = 0$$

$$(y-b)y'' = -(y')^2 - 1$$

$$y - b = \frac{-((y')^2 + 1)}{y''} \longrightarrow ③$$

Substitute ③ in ②;

$$(x-a)-\frac{((y')^2+1)}{y''}y'=0$$

$$x-a=\frac{((y')^2+1)y'}{y''} \longrightarrow (4)$$

Substitute eq's (3) & (4) in eq (1)

$$\frac{((y')^2+1)^2 y'^2}{(y'')^2}+\frac{(1+y'^2)^2}{(y'')^2}=r^2$$

$$(1+y'^2)^2 y'^2+(1+y'^2)^2=(y'')^2 r^2$$

$\therefore (1+y'^2)^2(y'^2+1)=(y'')^2 r^2$ is 2nd order D.E.

(4) Form a D.E by eliminating arbitrary constants from family of ellipse. Here equation of ellipse is not mentioned.

Sol Equation for family of ellipse with 2 diff. arbitrary constants is $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1 \longrightarrow (1)$

Differentiating the above equation on b/s w.r.t 'x';

$$\frac{2x}{a^2}+\frac{2y}{b^2}\cdot y'=0$$

$$\frac{x}{a^2}+\frac{y\cdot y'}{b^2}=0 \longrightarrow (2)$$

Differentiating on b/s w.r.t 'x';

$$\frac{1}{a^2}+\frac{1}{b^2}[y\cdot y''+y'^2]=0$$

$$\frac{1}{a^2}=\frac{-1}{b^2}[y\cdot y''+y'^2] \longrightarrow (3)$$

substitute (3) in (2);

$$\frac{-x}{b^2}[yy''+y'^2]+\frac{yy'}{b^2}=0$$

$$-x\cdot y\cdot y''-xy'^2+yy'=0$$

$\therefore xy\cdot y''+x\cdot y'^2-yy'=0$ is 2nd order D.E.

First order and First degree D.E:

An equation is of the form $f(x,y)=\frac{dy}{dx}$ is known as first order and first degree D.E.

solutions of O.D.E:

1. Linear D.E: A D.E is said to be linear if dependent variable and its derivatives occurs in 1st degree and they are not multiplied

S.NO	D.E	I.F	G.S
1	$\frac{dy}{dx}+P(x)y=Q(x)$ is L.D.E in terms of y.	I. $e^{\int P(x)dx}$	$y(I.F)=\int(I.F)Q(x)dx+C$
2.	$\frac{dx}{dy}+P(y)x=Q(y)$ is L.D.E in terms of x.	$e^{\int P(y)dy}$	$x(I.F)=\int(I.F)Q(y)dy+C$
3	$\frac{dr}{d\theta}+P(\theta).r=Q(\theta)$ is L.D.E in terms of r	$e^{\int P(\theta)d\theta}$	$r(I.F)=\int(I.F)Q(\theta)d\theta+C$
4	$\frac{d\theta}{dr}+P(r)\theta=Q(r)$ is L.D.E interms of r	$e^{\int P(r)dr}$	$\theta(I.F)=\int(I.F).Q(r)dr+C.$

1. Solve $x.\frac{dy}{dx}+y=\log x$

Sol. Given, D.E is $x.\frac{dy}{dx}+y=\log x$

Divide on b/s by x;

$$\frac{dy}{dx}+\frac{y}{x}=\frac{\log x}{x}$$

$\frac{dy}{dx}+\frac{1}{x}(y)=\frac{\log x}{x}$ is linear D.E in y.

$$P(x)=\frac{1}{x};\ Q(x)=\frac{\log x}{x}$$

$$I.F=e^{\int P(x)dx}$$

$$I.F=e^{\int\frac{1}{x}dx}=e^{\log x}=x$$

G.S is $y(I.F)=\int(I.F)Q(x)dx+C$

$$y(x)=\int x.\frac{\log x}{x}dx+C$$

$\therefore\ xy=x\log x-x+C$ is G.S of given D.E.

2. solve $x\cos x \cdot \frac{dy}{dx} + (x\sin x + \cos x)y = 1$

Sol. Given D.E is $x\cos x \cdot \frac{dy}{dx} + (x\sin x + \cos x)y = 1$

Divide throughout by $x\cos x$,

$$\frac{dy}{dx} + \left(\frac{x\sin x + \cos x}{x\cos x}\right)y = \frac{1}{x\cos x}$$

$$\frac{dy}{dx} + \left(\tan x + \frac{1}{x}\right)y = \frac{1}{x\cos x}$$

$$\frac{dy}{dx} + \left(\tan x + \frac{1}{x}\right) = \frac{1}{x}\sec x$$

It is L.D.E in terms of 'y'

$$\frac{dy}{dx} + P(x)y = Q(x)$$

$$P(x) = \tan x + \frac{1}{x} \ ; \ Q(x) = \frac{1}{x}\sec x$$

$$I.F = e^{\int P(x)\,dx}$$

$$I.F = e^{\int(\tan x + \frac{1}{x})\,dx}$$

$$I.F = e^{\log|\sec x| + \log x}$$

$$I.F = e^{\log(x\sec x)} = x\sec x$$

G.S is $y(I.F) = \int (I.F)\,Q(x)\,dx + C$

$$y(x\sec x) = \int \cancel{x}\sec x \cdot \frac{1}{\cancel{x}}\sec x\,dx + C$$

$$xy\sec x = \int \sec^2 x\,dx + C$$

$\therefore xy\sec x = \tan x + C$ is G.S of given D.E.

3. solve $(1+y^2)dx = (\tan^{-1}y - x)dy$

Sol. Given D.E is $(1+y^2)\,dx = (\tan^{-1}y - x)dy$

$$\frac{dx}{dy} = \frac{\tan^{-1}y - x}{1+y^2}$$

$$\frac{dx}{dy} = \frac{\tan^{-1}y}{1+y^2} - \frac{x}{1+y^2}$$

$\frac{dx}{dy} + \left(\frac{1}{1+y^2}\right)x = \frac{\tan^{-1}y}{1+y^2}$ is L.D.E in terms of x.

$$\frac{dx}{dy} + P(y)\cdot x = Q(y)$$

$$P(y) = \frac{1}{1+y^2} ; \quad Q(y) = \frac{\tan^{-1}y}{1+y^2}$$

$$I.F = e^{\int P(y)dy}$$

$$I.F = e^{\int \frac{1}{1+y^2}dy}$$

$$\therefore I.F = e^{\tan^{-1}y}$$

G.S is $x(I.F) = \int (I.F)\, Q(y)\, dy + C$

$$x.e^{\tan^{-1}y} = \int e^{\tan^{-1}y} \cdot \frac{\tan^{-1}y}{1+y^2} dy + C$$

Let $e^{\tan^{-1}y} = t \Rightarrow \tan^{-1}y = \log t$

$$e^{\tan^{-1}y} \cdot \frac{1}{1+y^2} \cdot dy = dt$$

$$x.e^{\tan^{-1}y} = \int e^{\tan^{-1}y} \cdot \frac{1}{1+y^2} \cdot \cancel{\log} \tan^{-1}y \cdot dy + C$$

$$= \int \log t\, dt + C$$

$$= t\log t - t + C$$

$\therefore x.e^{\tan^{-1}y} = e^{\tan^{-1}y}\log e^{\tan^{-1}y} - e^{\tan^{-1}y} + C$ is G.S of given D.E.

$$\therefore x e^{\tan^{-1}y} = e^{\tan^{-1}y}(\log e^{\tan^{-1}y} - 1) + C$$

4. solve $\frac{dy}{dx} + y = e^x$

sol Given D.E is $\frac{dy}{dx} + y = e^x$

It is L.D.E in terms of y i.e., $\frac{dy}{dx} + P(x).y = Q(x)$

$P(x) = 1$ & $Q(x) = e^x$

$$I.F = e^{\int P(x)dx} = e^{\int 1 dx} = e^x$$

G.S is $y(I.F) = \int (I.F)\, Q(x)\, dx + C$

$$y(e^x) = \int e^x \cdot e^x dx + C$$

$$ye^x = \int e^{2x} dx + C$$

$\therefore ye^x = \frac{e^{2x}}{2} + C$ is G.S for given D.E

5. solve $x\log x \cdot \frac{dy}{dx} + y = 2\log x$

Sol, Given D.E is $x\log x \cdot \frac{dy}{dx} + y = 2\log x$

Divide throughout by '$x\log x$';

$$\frac{dy}{dx} + \frac{y}{x\log x} = \frac{2\log x}{x\log x}$$

$$\frac{dy}{dx} + \left(\frac{1}{x\log x}\right) y = \frac{2}{x}$$

It is L.D.E in terms of y.

$$\frac{dy}{dx} + P(x)\cdot y = Q(x)$$

$$P(x) = \frac{1}{x\log x} ; \; Q(x) = \frac{2}{x}$$

$$I.F = e^{\int P(x)dx}$$

$$I.F = e^{\int \frac{1}{x\log x}dx} = e^{\log(\log x)} = \log x \quad \left(\because \int \frac{f'(x)}{f(x)} dx = \log(f(x)) + c\right)$$

G.S is $y\cdot(I.F) = \int (I.F)\cdot Q(x)\cdot dx + C$

$$y(\log x) = \int \log x \cdot \frac{2}{x} dx + C$$

$$y(\log x) = 2\int \log x \cdot \frac{1}{x} dx + C \quad \left(\because \int f^n(x)\cdot f'(x)\, dx = \frac{f^{n+1}(x)}{n+1}\right)$$

$$y(\log x) = 2\frac{(\log x)^2}{2} + C$$

$\therefore$ $y(\log x) = (\log x)^2 + C$ is G.S of given D.E.

6. solve $(x+y+1)\frac{dy}{dx} = 1$

Given D.E is $(x+y+1)\frac{dy}{dx} = 1$

$$\frac{dy}{dx} = \frac{1}{x+y+1}$$

$$\frac{dx}{dy} = x + y + 1$$

$$\frac{dx}{dy} - x = (1+y)$$

It is L.D.E in terms of x i.e., $\frac{dx}{dy} + P(y)\cdot x = Q(y)$

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$$\frac{dx}{dy} + (-1)x = (1+y)$$

where $P(y) = -1$ & $Q(y) = (1+y)$

$I.F = e^{\int P(y)dy} = e^{\int -1 dy} = e^{-y}$

G.S is $x(I.F) = \int (I.F)\, Q(y)\, dy + c$

$x e^{-y} = \int e^{-y}(1+y)\, dy + c$

$x e^{-y} = (1+y)(-e^{-y}) - (1)e^{-y} + c$

$\therefore x e^{-y} = e^{-y}(-(1+y) - 1) + c$

$\therefore x e^{-y} = e^{-y}(-2-y) + c$ is required G.S of given D.E.

7. solve $x\frac{dy}{dx} + y + 4 = 0$

Sol Given D.E is $x\frac{dy}{dx} + y + 4 = 0$

Divide throughout by 'x'

$$\frac{dy}{dx} + \frac{y}{x} + \frac{4}{x} = 0$$

$$\frac{dy}{dx} + \left(\frac{1}{x}\right)y = \frac{-4}{x}$$

It is L.D.E in terms of y i.e., $\frac{dy}{dx} + p(x)y = Q(x)$

where $p(x) = \frac{1}{x}$ and $Q(x) = \frac{-4}{x}$

$I.F = e^{\int p(x)dx}$

$I.F = e^{\int \frac{1}{x}dx} = e^{\log x} = x.$

G.S is $y(I.F) = \int (I.F)\, Q(x)\, dx + c$

$yx = \int x \cdot \frac{-4}{x} dx + c$

$\therefore xy = -4x + c$ is required G.S of given D.E.

8. solve $\frac{dy}{dx} + \frac{y}{x\log x} = \frac{\sin 2x}{\log x}$

Sol Given D.E is $\frac{dy}{dx} + \frac{y}{x\log x} = \frac{\sin 2x}{\log x}$

It is L.D.E interms of y i.e, $\frac{dy}{dx}+P(x)y=Q(x)$

where $P(x)=\frac{1}{x\log x}$ and $Q(x)=\frac{\sin 2x}{\log x}$

$I.F = e^{\int P(x)dx}$

$I.F = e^{\int \frac{1}{x\log x}dx}$

$I.F = e^{\log t} = t = \log x.$

Let $\log x = t$

$\frac{1}{x}dx = dt$

$\int \frac{1}{x\log x}dx = \int \frac{1}{t}dt = \log t$

G.S is $y(I.F) = \int Q(x).I.F\,dx + C$

$y(\log x) = \int \frac{\sin 2x}{\log x}\log x\,dx + C$

$y\log x = \int \sin 2x\,dx + C$

$\therefore y\log x = -\frac{\cos 2x}{2} + C$ is required G.S of given D.E.

9. solve $\frac{dy}{dx}+y=e^{e^x}$

Sol, Given D.E is $\frac{dy}{dx}+(1)(y)=e^{e^x}$

It is L.D.E interms of y i.e, $\frac{dy}{dx}+P(x)y=Q(x)$

where $P(x)=1$ and $Q(x)=e^{e^x}$

$I.F = e^{\int P(x)dx}$

$I.F = e^{\int 1dx} = e^x.$

G.S is $y(I.F) = \int Q(x)\,I.F\,dx + C$

$ye^x = \int e^{e^x}\cdot e^x dx + C$

$\therefore$ Let $e^x = t$

$e^x dx = dt$

$ye^x = \int e^t dt + C$

$ye^x = e^t + C$

$\therefore ye^x = e^{e^x} + C$ is G.S of given D.E

10. solve $(x^2-1)\frac{dy}{dx}+2xy=1$

Sol) Given D.E is $(x^2-1)\frac{dy}{dx} + 2xy = 1$

Divide throughout by 'x^2-1';

$$\frac{dy}{dx} + \frac{2x}{x^2-1} y = \frac{1}{x^2-1}$$

It is L.D.E interms of y i.e, $\frac{dy}{dx} + P(x)y = Q(x)$

where $P(x) = \frac{2x}{x^2-1}$; $Q(x) = \frac{1}{x^2-1}$

$I.F = e^{\int P(x)\,dx}$

$I.F = e^{\int \frac{2x}{x^2-1}dx}$

$I.F = e^{\log|(x^2-1)|} = x^2-1$

$\therefore$ G.S is $y(I.F) = \int \cancel{Q(x)}\, I.F\, Q(x)dx + C$

$y(x^2-1) = \int (x^2-1)\frac{1}{(x^2-1)}\,dx + C$

$y(x^2-1) = \int dx + C$

$\therefore y(x^2-1) = x + C$ is required G.S of given D.E.

11. solve $e^{-y}\sec^2 y\,dy = dx + x\,dy$

Sol, Given D.E is $e^{-y}\sec^2 y\,dy = dx + x\,dy$

$e^{-y}\sec^2 y\,dy - x\,dy = dx$

$dy(e^{-y}\sec^2 y - x) = dx$

$\frac{dx}{dy} = e^{-y}\sec^2 y - x$

$\frac{dx}{dy} + \cancel{e^{-y}\sec^2}\, x = e^{-y}\sec^2 y$

$\frac{dx}{dy} + (1)x = e^{-y}\sec^2 y$

It is L.D.E interms of 'x' i.e, $\frac{dx}{dy} + P(y)x = Q(y)$

where $P(y) = 1$ and $Q(y) = e^{-y}\sec^2 y$

$I.F = e^{\int P(y)dy} = e^{\int 1 dy} = e^{y}$.

G.S is $x(I.F) = \int (I.F)\, Q(y)\, dy + c$

$$x . e^{\tan y} = \int e^{\tan y} . e^{-y} \sec^2 y \, dy + c$$

$$x e^{\tan y} = \int \sec^2 y . e^{2y} dy + c$$

$$x e^{\tan y} = \int \sec^2 y \, dy + c$$

$\therefore x e^{\tan y} = \tan y + c$ is required G.S of given D.E.

12. solve $dr + (2r\cot\theta + \sin 2\theta) d\theta = 0$

Sol, Given $dr + (2r\cot\theta + \sin 2\theta) d\theta = 0$

$$dr = -(2r\cot\theta + \sin 2\theta)\, d\theta$$

$$\frac{dr}{d\theta} = -2r\cot\theta - \sin 2\theta$$

$$\frac{dr}{d\theta} + (2\cot\theta) r = -\sin 2\theta$$

It is L.D.E in terms of r i.e, $\frac{dr}{d\theta} + P(\theta).r = Q(\theta)$

where $P(\theta) = 2\cot\theta$; $Q(\theta) = -\sin 2\theta$

$$I.F = e^{\int P(\theta) d\theta}$$

$$I.F = e^{\int 2\cot\theta d\theta} = e^{2\log(\sin\theta)} = e^{\log(\sin^2\theta)}$$

$\therefore I.F = \sin^2\theta$

$\therefore$ G.S is $r(I.F) = \int (I.F)\, Q(\theta)\, d\theta + c$

$$r . \sin^2\theta = \int \sin^2\theta . (-\sin 2\theta)\, d\theta + c$$

$$r \sin^2\theta = -2\int \sin^3\theta . \cos\theta \, d\theta + c \qquad (\because \sin\theta = t;\ \cos\theta\, d\theta = dt;\ \int t^3 dt = \frac{t^4}{4})$$

$\therefore r\sin^2\theta = \frac{-2\sin^4\theta}{4} + c$ is required G.S of given D.E

13. Solve $\frac{dy}{dx} + y\cot x = 4x\,\mathrm{cosec}\,x$. if $y=0$ and $x = \frac{\pi}{2}$

Sol. Given D.E is $\frac{dy}{dx} + y\cot x = 4x\,\mathrm{cosec}\,x$.

It is L.D.E interms of xy i.e. $\frac{dy}{dx} + P(x)\,y = Q(x)$

where $P(x) = \cot x$; $Q(x) = 4x\,\mathrm{cosec}\,x$

$$I.F = e^{\int P(x)\,dx}$$

$$I.F = e^{\int \cot x\,dx} = e^{\log|\sin x|} = \sin x$$

$\therefore$ G.S is $y(I.F) = \int (I.F)\,Q(x)\,dx + C$

$$y(\sin x) = \int(\sin x)(4x\,\mathrm{cosec}\,x)\,dx + C$$

$$\therefore y\sin x = \frac{4x^2}{2} + C$$

$\therefore$ $y\sin x = 2x^2 + C \rightarrow ①$ is required G.S of given D.E.

put $x = \frac{\pi}{2}$ & $y=0$ in eq ①;

$$0 = 2\left(\frac{\pi^2}{4}\right) + C$$

$$\therefore C = \frac{-\pi^2}{2}$$

$\therefore$ $y\sin x = 2x^2 + \left(\frac{-\pi^2}{2}\right)$ is particular solution.

Bernouli differential equation:

A differential equation which is of the form $\frac{dy}{dx} + P(x)y = Q(x)\cdot y^n$ where $n \neq 0,1$ is known as Bernouli differential equation. we can get G.S of B.D.E by reducing it into L.D.E

Working procedure for reducing B.D.E into L.D.E:

Given equation $\frac{dy}{dx} + P(x)y = Q(x)\,y^n \rightarrow ①$ $(n \neq 0,1)$ is a Bernouli differential equation interms of y.

Divide the eq ① on b/s with y^n,

$$\frac{1}{y^n}\cdot\frac{dy}{dx}+\frac{P(x)\cdot y}{y^n}=\frac{Q(x)\,y^n}{y^n}$$

$$y^{-n}\cdot\frac{dy}{dx}+P(x)\cdot y^{1-n}=Q(x)\longrightarrow ②$$

put $y^{1-n}=z$

Differentiate above eqn w.r.t 'x' on b/s,

$$(1-n)\cdot y^{-n}\cdot\frac{dy}{dx}=\frac{dz}{dx}$$

$$y^{-n}\cdot\frac{dy}{dx}=\frac{1}{1-n}\cdot\frac{dz}{dx}$$

Substitute above equation value in eq ②;

$$\left(\frac{1}{1-n}\right)\frac{dz}{dx}+P(x)\cdot z=Q(x)\longrightarrow ③$$

Multiply eqn with $(1-n)$ on both sides.

$\frac{dz}{dx}+(1-n)\,P(x)\cdot z=Q(x)$ is L.D.E interms of 'z'.

① solve $x\cdot\left(\frac{dy}{dx}\right)+y=x^3y^6$.

sol Given equation is $x\cdot\left(\frac{dy}{dx}\right)+y=x^3y^6\longrightarrow ①$

Eq ① is of the form B.D.E i.e; $\frac{dy}{dx}+P(x)\,y=x\,y^n$

Divide eq ① on B.S with x and y^6

$$\frac{x}{xy^6}\cdot\frac{dy}{dx}+\frac{y}{xy^6}=\frac{x^3y^6}{xy^6}$$

$$y^{-6}\cdot\frac{dy}{dx}+\left(\frac{1}{x}\right)y^{-5}=x^2\longrightarrow ②$$

Let $y^{-5}=z$

Diff eqn on both sides w.r.t x,

$$-5\cdot y^{-6}\frac{dy}{dx}=\frac{dz}{dx}$$

$$y^{-6}\cdot\frac{dy}{dx}=\frac{-1}{5}\cdot\frac{dz}{dx}$$

Sub above eqn value in eq②;

$$\frac{-1}{5}\frac{dz}{dx}+\left(\frac{1}{x}\right)z=x^2$$

Multiply above eqn with (-5) on b.s,

$$\frac{dz}{dx}+\left(\frac{-5}{x}\right)z=-5x^2 \rightarrow ③$$

eqn-③ is of the form $\frac{dz}{dx}+P(x)z=Q(x)$

$$P(x)=\frac{-5}{x} \text{ and } Q(x)=-5x^2$$

$$I.F=e^{\int P(x)\,dx}=e^{\int -5/x\,dx}$$

$$=e^{-5\int\frac{1}{x}dx}$$

$$=e^{-5\log x}$$

$$=e^{\log x^{-5}}$$

$$=x^{-5}$$

$$=\frac{1}{x^5}$$

G.S is $z(I.F)=\int(I.F)\,Q(x)\,dx+c$

$$z\left(\frac{1}{x^5}\right)=\int\frac{1}{x^5}\cdot-5x^2dx+c$$

$$y^{-5}/x^5=-5\int\frac{1}{x^3}\,dx+c$$

$\therefore \frac{1}{x^5y^5}=\frac{5}{2x^2}+c$ is required G.S.

② solve $\frac{dy}{dx}+\frac{y}{x}=y^2x\sin x$

sol. Given equation is

$$\frac{dy}{dx}+\frac{y}{x}=y^2x\sin x \rightarrow ①$$

Divide the above equation by y^2,

$$\frac{1}{y^2}\cdot\frac{dy}{dx}+\frac{1}{xy}=x\sin x \rightarrow ②$$

Let $\frac{1}{y}=z$

$$y^{-1}=z$$

Diff on b.s w.r.t 'x',

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$$\frac{-1}{y^2}\cdot\frac{dy}{dx} = \frac{dz}{dx}$$

$$\frac{1}{y^2}\cdot\frac{dy}{dx} = -\frac{dz}{dx}$$

Substitute above value in eq ②;

$$-\frac{dz}{dx} + \frac{1}{x}(z) = x\sin x$$

Multiply above equation with '-1';

$$\frac{dz}{dx} + \left(\frac{-1}{x}\right)z = -x\sin x \longrightarrow ③$$

eqn-③ is of the form $\frac{dz}{dx} + P(x).z = Q(x)$

$$P(x) = \frac{-1}{x},\ Q(x) = -x\sin x$$

$$I.F = e^{\int P(x)dx} = e^{\int \frac{-1}{x}dx} = e^{-\log x} = e^{\log(1/x)} = \frac{1}{x}$$

G.S is $z(I.F) = \int (I.F)\,Q(x)\,dx + C$

$$\frac{z}{x} = \int \frac{1}{x}(-x\sin x)\,dx + C$$

$$\frac{z}{x} = -\int \sin x\,dx + C$$

$$\frac{1}{xy} = -(-\cos x) + C$$

$\therefore \frac{1}{xy} = \cos x + C$ is G.S of given D.E.

③ solve $\frac{dy}{dx}(x^2y^3 + xy) = 1$.

sol, $\frac{dy}{dx}(x^2y^3 + xy) = 1$

$$\frac{dx}{dy} = x^2y^3 + xy$$

$\frac{dx}{dy} - xy = x^2y^3$ ① is B.D.E in terms of x

Divide the above eqn with x^2 on b.s,

$$\frac{1}{x^2}\cdot\frac{dx}{dy} - \frac{xy}{x^2} = \frac{x^2y^3}{x^2}$$

$$x^{-2}\frac{dx}{dy} - \frac{1}{x}y = y^3$$

$x^{-2}\frac{dx}{dy} - x^{-1}y = y^3 \longrightarrow (2)$

Let $-x^{-1} = z$

Diff on b.s w.r.t y,

$x^{-2}\frac{dx}{dy} = \frac{dz}{dy}$

substitute above value in eqn(2);

$\frac{dz}{dy} + zy = y^3 \longrightarrow (3)$

eq(3) is of form $\frac{dz}{dy} + P(y)\,z = Q(y)$

$P(y) = y$; $Q(y) = y^3$

$I.F = e^{\int P(y)dy} = e^{\int y\,dy} = e^{y^2/2}$

G.S is $z(I.F) = \int Q(y)\, I.F\, dy + c$

$z.e^{+y^2/2} = \int e^{y^2/2} . y^3 . dy + c$

Let $e^{y^2/2} = t \Rightarrow y^2/2 = \log t$

diff. on b.s $\qquad y^2 = 2\log t$

$e^{y^2/2} . \frac{2y}{2} dy = dt$

$e^{y^2/2} . y\,dy = dt$

$z.e^{y^2/2} = \int e^{y^2/2} y . y^2 dy + c$

$= \int 2\log t\, dt + c$

$= 2\int \log t\, dt + c$

$= 2t[\log t - 1] + c$

$(-x^{-1}).e^{y^2/2} = 2e^{y^2/2}.[y^2/2 - 1] + c$

$\therefore \frac{-e^{y^2/2}}{x} = 2e^{y^2/2}\left[\frac{y^2}{2} - 1\right] + c$ is the G.S.

4. Solve $\frac{dy}{dx} + x\sin 2y = x^3\cos^2 y$

Sol, Given D.E is $\frac{dy}{dx} + x\sin 2y = x^3\cos^2 y$

Divide both sides with $\cos^2 y$;

$$\frac{1}{\cos^2 y}\cdot\frac{dy}{dx} + x\cdot\frac{\sin 2y}{\cos^2 y} = \frac{x^3\cos^2 y}{\cos^2 y}$$

$$\sec^2 y\cdot\frac{dy}{dx} + x\cdot\frac{2\sin y\cos y}{\cos^2 y} = x^3$$

$$\sec^2 y\cdot\frac{dy}{dx} + x2\tan y = x^3$$

$$\sec^2 y\cdot\frac{dy}{dx} + 2x\tan y = x^3 \longrightarrow ②$$

Let $\tan y = z$

Diff. on both sides

$$\sec^2 y\cdot\frac{dy}{dx} = \frac{dz}{dx}$$

Substitute eqn in ② eqn

$\frac{dz}{dz} + 2x(z) = x^3$ is LDE interms of z.

$P(x) = 2x$; $Q(x) = x^3$

$$I.F = e^{\int P(x)dx}$$

$$I.F = e^{\int 2x\,dx} = e^{\frac{2x^2}{2}} = e^{x^2}$$

G.S is $z(I.F) = \int (I.F)\,Q(x)\,dx + C$

$$z(e^{x^2}) = \int e^{x^2}\cdot x^3\,dx + C$$

Let $e^{x^2} = t \Rightarrow x^2 = \log t$

diff on b.s

$$e^{x^2}\cdot 2x\,dx = dt$$

$$e^{x^2}\cdot x\,dx = \frac{1}{2}dt$$

$$ze^{x^2} = \int \frac{1}{2}\log t\,dt + C$$

$$ze^{x^2} = \frac{1}{2}\left(t[\log t - 1]\right) + C$$

$$z\cdot e^{x^2} = \frac{1}{2}\cdot e^{x^2}(x^2-1) + C$$

$\tan y\, e^{x^2} = \frac{e^{x^2}}{2}(x^2-1) + C$ is G.S

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NOTE:
Sometimes Bernouli differential equation can be converted into L.D.E by suitable substitution.

5. solve $(1-x^2)\cdot\frac{dy}{dx} + xy = y^3 \sin^{-1}x$

sol Given,

$$(1-x^2)\frac{dy}{dx} + xy = y^3 \sin^{-1}x \rightarrow ① \text{ is B.D.E in terms of } y.$$

Divide b.s by $y^3, (1-x^2)$

$$\frac{(1-x^2)}{y^3(1-x^2)}\cdot\frac{dy}{dx} + \frac{xy}{y^3(1-x^2)} = \frac{y^3\sin^{-1}x}{y^3(1-x^2)}$$

$$y^{-3}\cdot\frac{dy}{dx} + \frac{x}{1-x^2}\cdot y^{-2} = \frac{\sin^{-1}x}{1-x^2} \rightarrow ②$$

put $y^{-2} = z$

$$-2y^{-3}\frac{dy}{dx} = \frac{dz}{dx}$$

$$y^{-3}\frac{dy}{dx} = \frac{-1}{2}\frac{dz}{dx}$$

Substitute the above value in eq②;

$$\frac{-1}{2}\cdot\frac{dz}{dx} + \frac{x}{1-x^2}\cdot z = \frac{\sin^{-1}x}{1-x^2}$$

Multiply the eqn with '-2';

$$\frac{dz}{dx} + \left(\frac{-2x}{1-x^2}\right) = \frac{-2\sin^{-1}x}{1-x^2} \rightarrow ③$$

eq ③ is of the form $\frac{dz}{dx} + P(x)\cdot z = Q(x)$

$$P(x) = \frac{-2x}{1-x^2} \quad \& \quad Q(x) = \frac{-2\sin^{-1}x}{1-x^2}$$

$$I.F = e^{\int P(x)\,dx}$$

$$= e^{\int \frac{-2x}{1-x^2}dx}$$

$$= e^{\log|1-x^2|}$$

$$= 1-x^2$$

G.S is $z(I.F) = \int Q(x)\cdot(I.F)\,dx + C$

$$z(1-x^2) = \int \frac{-2\sin^{-1}x}{(1-x^2)}(1-x^2)\,dx + c$$

$$z(1-x^2) = -2\int \sin^{-1}x(1)\,dx + c$$

$$z(1-x^2) = -2\left(\sin^{-1}x\int 1\,dx - \int \frac{1}{\sqrt{1-x^2}}\cdot x\,dx\right) + c$$

$$= -2x\sin^{-1}x + \left(\frac{2x}{\sqrt{1-x^2}}\,dx\right) + c$$

$$= -2x\sin^{-1}x - 2\sqrt{1-x^2} + c$$

$\therefore \frac{(1-x^2)}{y^2} = -2(x\sin^{-1}x + \sqrt{1-x^2}) + c$ is G.S.

6. Solve $\tan y \cdot \frac{dy}{dx} + \tan x = \cos y \cdot \cos^2 x$.

Sol. Given D.E is $\tan y \cdot \frac{dy}{dx} + \tan x = \cos y \cdot \cos^2 x \rightarrow (1)$

Divide b/s by $\cos y$;

$$\frac{\tan y}{\cos y}\cdot\frac{dy}{dx} + \frac{\tan x}{\cos y} = \frac{\cos y \cdot \cos^2 x}{\cos y}$$

$$\tan y \cdot \sec y \cdot \frac{dy}{dx} + \tan x \cdot \sec y = \cos^2 x \rightarrow (2)$$

put $\sec y = z$

Diff b/s w.r.t x.

$$\tan y \cdot \sec y\,\frac{dy}{dx} = \frac{dz}{dx}$$

Substitute in (2);

$$\frac{dz}{dx} + \tan x \cdot z = \cos^2 x \rightarrow (3)$$

eq (3) is of the form $\frac{dz}{dx} + P(x)z = Q(x)$

where $P(x) = \tan x$; $Q(x) = \cos^2 x$

$$I.F = e^{\int P(x)dx} = e^{\int \tan x\,dx} = e^{\int \log|\sec x|} = \sec x$$

G.S is $(I.F)z = \int (I.F)Q(x)\,dx + C$

$$z \cdot \sec x = \int \sec x \cdot \cos^2 x\,dx + c$$

$$\sec y \cdot \sec x = \int \cos x\,dx + c$$

$\therefore \sec y \cdot \sec x = \sin x + C$ is G.S of given D.E.

7. Solve $\frac{dy}{dx}+x(x+y)=x^3(x+y)^3-1$

Sol, Given D.E is $\frac{dy}{dx}+x(x+y)=x^3(x+y)^3-1 \longrightarrow ①$

Let $(x+y)=z$

Diff on b.s w.r.t 'x';

$$1+\frac{dy}{dx}=\frac{dz}{dx}$$

$$\frac{dy}{dx}=\frac{dz}{dx}-1$$

Substitute in eq①;

$$\frac{dz}{dx}-\cancel{1}+xz=x^3z^3-\cancel{1}$$

$$\frac{dz}{dx}+xz=x^3z^3 \longrightarrow ②$$

It is B.D.E interms of z.

Divide eq② by z^3;

$$\frac{1}{z^3}\cdot\frac{dz}{dx}+\frac{xz}{z^3}=\frac{x^3z^3}{z^3}$$

$$z^{-3}\cdot\frac{dz}{dx}+xz^{-2}=x^3 \longrightarrow ③$$

put $z^{-2}=t$

Diff on b/s w.r.t 'x'

$$-2z^{-3}\frac{dz}{dx}=\frac{dt}{dx}$$

$$\frac{dz}{dx}=\frac{-1}{2}\cdot\frac{dt}{dx}$$

eq③ becomes; $-\frac{1}{2}\cdot\frac{dt}{dx}+xt=x^3$

$$\frac{dt}{dx}-2xt=-2x^3$$

$$\frac{dt}{dx}+(-2x)t=-2x^3 \longrightarrow ④$$

It is L.D.E interms of t.

$P(x)=-2x$; $Q(x)=-2x^3$

$$I.F=e^{\int P(x)\,dx}=e^{\int -2x\,dx}=e^{-x^2}$$

G.S is $t(I.F) = \int (I.F) \cdot Q(x)\, dx + C$

$$t \cdot e^{-x^2} = \int e^{-x^2} (-2x^3)\, dx + c$$

$$t e^{-x^2} = \int e^{-x^2} (2x)\, x^2\, dx + c$$

put $e^{-x^2} = K \Rightarrow -x^2 = \log K$

$-e^{-x^2} \cdot 2x \cdot dx = dK$

$$t e^{-x^2} = \int -\log K\, dK + c$$

$$t e^{-x^2} = -[K \log K - K] + c$$

$$z^{-2} e^{-x^2} = -[e^{-x^2} \log e^{-x^2} - e^{-x^2}] + c$$

$\therefore (x+y)^{-2} \cdot e^{-x^2} = -e^{-x^2}[\log e^{-x^2} - 1] + c$ is required G.S of given D.E.

8. solve $\frac{dy}{dx} = \frac{y}{x+\sqrt{xy}}$

sol, Given $\frac{dy}{dx} = \frac{y}{x+\sqrt{xy}} \rightarrow (1)$

$$\frac{dx}{dy} = \frac{x+\sqrt{xy}}{y}$$

$$\frac{dx}{dy} = \frac{x}{y} + \sqrt{\frac{x}{y}}$$

$$\frac{dx}{dy} - \frac{x}{y} = \sqrt{\frac{x}{y}}$$

Divide throughout $\sqrt{x}$;

$$\frac{1}{\sqrt{x}} \cdot \frac{dx}{dy} - \frac{\sqrt{x}}{y} = \frac{1}{\sqrt{y}} \rightarrow (2)$$

put $\sqrt{x} = z$

Diff on b/s w.r.t y;

$$\frac{1}{2\sqrt{x}} \cdot \frac{dx}{dy} = \frac{dz}{dy}$$

$$\frac{1}{\sqrt{x}} \cdot \frac{dx}{dy} = 2 \cdot \frac{dz}{dy}$$

substitute above value in eq (2);

$$\frac{2\,dz}{dy} - \frac{z}{y} = \frac{1}{\sqrt{y}}$$

Divide throughout by '2';

$$\frac{dz}{dy} - \frac{1}{2}\cdot\frac{z}{y} = \frac{1}{2\sqrt{y}}$$

$$\frac{dz}{dy} + \left(\frac{-1}{2y}\right)z = \frac{1}{2\sqrt{y}} \quad \text{--- (3)}$$

It is L.D.E interms of 'z'.

$$P(y) = \frac{-1}{2y} \; ; \; Q(y) = \frac{1}{2\sqrt{y}}$$

$$I.F = e^{\int P(y)\,dy} = e^{\int \frac{-1}{2y} dy}$$

$$= e^{\frac{-1}{2}\int \frac{1}{y} dy} = e^{\frac{-1}{2}\log y} = e^{\log y^{-1/2}} = y^{-1/2}$$

$$= \frac{1}{\sqrt{y}}$$

$\therefore$ G.S is $z\,(I.F) = \int (I.F)\,Q(y)\,dy + C$

$$z.\frac{1}{\sqrt{y}} = \int \frac{1}{\sqrt{y}}\cdot\frac{1}{2\sqrt{y}}\,dy + C$$

$$z\cdot\frac{1}{\sqrt{y}} = \frac{1}{2}\int \frac{1}{y}\,dy + C$$

$$z.\frac{1}{\sqrt{y}} = \frac{1}{2}\log y + C$$

$$\sqrt{x}.\frac{1}{\sqrt{y}} = \log y^{1/2} + C$$

$\therefore \sqrt{\frac{x}{y}} = \log y^{1/2} + C$ is G.S of given D.E.

9. solve $2y\cos y^2 \frac{dy}{dx} - \cancel{2y}\frac{2}{1+x}\sin y^2 = (1+x)^3$.

Sol. Given D.E is $2y\cos y^2 \cdot \frac{dy}{dx} - \frac{2}{1+x}(\sin y^2) = (1+x)^3$ --- (1)

Let $\sin y^2 = z$

$$\cos y^2\cdot(2y)\frac{dy}{dx} = \frac{dz}{dx}$$

$$2y\cos y^2\cdot\frac{dy}{dx} = \frac{dz}{dx}$$

sub above value in eq(1);

$\frac{dz}{dx} - \frac{2}{1+x}\cdot z = (1+x)^3 \longrightarrow$ (2), it is L.D.E interms of z.

$P(x) = \frac{-2}{1+x}$; $Q(x) = (1+x)^3$

$I.F = e^{\int P(x)dx} = e^{\int \frac{-2}{1+x}dx} = e^{-2\int \frac{1}{1+x}dx} = e^{-2\log(1+x)}$

$I.F = e^{\log(1+x)^{-2}} = (1+x)^{-2}$

G.S is $z(I.F) = \int (I.F) \cdot Q(x)\, dx + C$

$z.(1+x)^{-2} = \int (1+x)^{-2}(1+x)^3 dx + C$

$z(1+x)^{-2} = \int (1+x)\, dx + C$

$z(1+x)^{-2} = x + \frac{x^2}{2} + C$

$\therefore \sin y^2 \cdot (1+x)^{-2} = x + \frac{x^2}{2} + C$ is G.S.

Exact Differential equation:

An equation is of the form $M(x,y)\,dx + N(x,y)\,dy = 0$ is said to be an exact if the left hand side quantity $Mdx + Ndy$ can be expressed exact Differentiation of some function $u(x,y)$ i.e., $M(x,y)\,dx + N(x,y)\,dy = 0$ (or) $d[u(x,y)] = Mdx + Ndy$.

Condition for exactness: The necessary and sufficient condition for D.E is $Mdx + Ndy = 0$ is said to be exact if $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$.

Working procedure:

1. The general form of Exact D.E can be of $Mdx + Ndy = 0$
2. The condition for exactness $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$.
3. General Solution is $\underset{y \text{ constant}}{\int M dx} + \underset{\text{neglect } x \text{ terms}}{\int N dy} = C$

1. Solve $(2x+e^y)dx + xe^y dy = 0$.

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Sol Given D.E is $(2x+e^y)dx + xe^y\,dy = 0 \longrightarrow ①$

eq ① is of form $M\,dx + N\,dy = 0$

$M = 2x+e^y$; $N = xe^y$

$\frac{\partial M}{\partial y} = 0+e^y$ & $\frac{\partial N}{\partial x} = e^y$

$\frac{\partial M}{\partial y} = e^y$

$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

$\therefore$ eq ① is an Exact Differential equation.

G.S is $\underset{y\text{ constant}}{\int M\,dx} + \underset{\text{Neglect }x\text{ terms}}{\int N\,dy} = C$

$\int (2x+e^y)\,dx + \int xe^y\,dy = C$

$\frac{2x^2}{2} + xe^y = C$

$\therefore x^2 + xe^y = C$ is required G.S.

2. Solve $(2x-y+1)dx + (2y-x+1)dy = 0$

Sol Given D.E is $(2x-y+1)dx + (2y-x+1)\,dy = 0 \longrightarrow ①$

Eq ① is of the form $M\,dx + N\,dy = 0$

where $M = 2x-y+1$ & $N = 2y-x+1$

$\frac{\partial M}{\partial y} = 0-1+0 = -1$ & $\frac{\partial N}{\partial x} = 0-1+0 = -1$

$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

$\therefore$ eq ① is an Exact D.E

$\therefore$ G.S is $\underset{y\text{ constant}}{\int M\,dx} + \underset{\text{Neglect }x\text{ terms}}{\int N\,dy} = C$

$\int (2x-y+1)\,dx + \int (2y-x+1)\,dy = C$

$\left(\frac{2\cdot x^2}{2} - xy + x\right) + \left(2\cdot\frac{y^2}{2} + y\right) = C$

$\therefore x^2 - xy + y^2 + y + x = C$ is required G.S

3. $\left[y\left(1+\frac{1}{x}\right)+\cos y\right]dx + \left[x+\log x - x\sin y\right]dy = 0$

Sol. Given D.E is $\left[y\left(1+\frac{1}{x}\right)+\cos y\right]dx + \left[x+\log x - \sin y\right]dy = 0$

eq ① is of the form $Mdx + Ndy = 0$

$M = y\left(1+\frac{1}{x}\right)+\cos y$ | $N = x+\log x - x\sin y$

$= y + \frac{y}{x} + \cos y$

$\frac{\partial M}{\partial y} = 1 + \frac{1}{x} - \sin y$ & $\frac{\partial N}{\partial x} = 1+\frac{1}{x} - \sin y$

$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

$\therefore$ eq ① is ~~of~~ an exact D.E

G.S is $\underset{y\text{ constant}}{\int M dx} + \underset{\text{Neglect } x\text{-terms}}{\int N dy} = c$

$\int \left(y+\frac{y}{x}+\cos y\right)dx + \int (\cancel{x}+\log x - x\sin y)\,dy = c$

$\therefore xy + y\log x + x\cos y = c$ is required G.S.

4. solve $(x.e^{xy}+2y)(dy \cancel{+} dx) + ye^{xy} = 0$

Sol Given $(xe^{xy}+2y)(dy \cancel{+} dx) + ye^{xy} = 0$

$(xe^{xy}+2y)\frac{dy}{dx} + ye^{xy} = 0$

$(xe^{xy}+2y)\,dy + (ye^{xy})\,dx = 0 \longrightarrow ①$

eq ① is of the form $Mdx + Ndy = 0$

$M = ye^{xy}$ and $N = xe^{xy}+2y$

$\frac{\partial M}{\partial y} = y.e^{xy}(x) + e^{xy}$ | $\frac{\partial N}{\partial x} = xe^{xy}(y) + e^{xy} + 0$

$\frac{\partial M}{\partial y} = e^{xy}(xy+1)$ | $\frac{\partial N}{\partial x} = e^{xy}(xy+1)$

$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

$\therefore$ Eq ① is an exact D.E

G.S is $\underset{y\text{ Constant}}{\int M dx} + \underset{\text{neglect } x\text{-terms}}{\int N dy} = c$

$\int (ye^{xy})dx + \int (xe^{xy} + 2y)\, dy = C$

$\frac{2x}{y^3}dx + \left(\frac{y^3 - 3x^2}{y^4}\right)dy = 0$

$y \cdot \frac{e^{xy}}{y} + 2 \cdot \frac{y^2}{2} = C$

$\therefore$ $e^{xy} + y^2 = C$ is required G.S.

5. solve $(x^2 - ay)dx - (ax - y^2)dy = 0$

Sol, Given $(x^2 - ay)\, dx - (ax - y^2)\, dy = 0$

$(x^2 - ay)\, dx + (y^2 - ax)\, dy = 0 \rightarrow (1)$

eq (1) is of the form $Mdx + Ndy = 0$

$M = x^2 - ay \qquad N = y^2 - ax$

$\frac{\partial M}{\partial y} = 0 - a \qquad \frac{\partial N}{\partial x} = 0 - a$

$\frac{\partial M}{\partial y} = -a \qquad \frac{\partial N}{\partial x} = -a$

$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial N}$

$\therefore$ Eq (1) is an exact D.E

G.S is $\underset{y\text{-constant}}{\int M dx} + \underset{\text{Neglect } x\text{-terms}}{\int N dy} = C$

$\int (x^2 - ay)\, dx + \int (y^2 - ax)\, dy = C$

$\frac{x^3}{3} - ayx + \frac{y^3}{3} = C$

$\therefore$ $\frac{x^3}{3} + \frac{y^3}{3} - ayx = c$ is required G.S.

6. Solve $\frac{2x}{y^3}dx + \left(\frac{y^3 - 3x^2}{y^4}\right) dy = 0$

Sol, Given D.E is $\frac{2x}{y^3}dx + \left(\frac{y^3 - 3x^2}{y^4}\right) dy = 0 \rightarrow (1)$

eq (1) is of the form $Mdx + Ndy = 0$

$M = \frac{2x}{y^3} \qquad N = \frac{y^3 - 3x^2}{y^4} = \frac{1}{y} - \frac{3x^2}{y^4}$

$\frac{\partial M}{\partial y} = 2x \cdot \frac{-3}{y^4} = \frac{-6x}{y^4}$ & $\frac{\partial N}{\partial x} = 0 - \frac{3}{y^4} \cdot 2x = \frac{-6x}{y^4}$

$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

$\therefore$ Eq (1) is an exact D.E

G.S is $\int M\,dx + \int N\,dy = 0$
$\quad$ y-constant $\quad$ Neglect x-terms

$$\int \frac{2x}{y^3}\,dx + \int \left(\frac{1}{y} - \frac{3x^2}{y^4}\right) dy = C$$

$$\frac{2}{y^3}\cdot\frac{x^2}{2} + \log y = C$$

$\therefore \frac{x^2}{y^3} + \log y = C$ is required G.S.

Integrating Factor: A differential equation is not an exact differential equation, it can be reducible to exact D.E by multiplying with suitable function. That function is called integrating factor.

D.E reducible to exact D.E.

Methods for Finding Integrating factor:

1. Inspection Method:

Generally an integrating factor can be form by inspection after regrouping the terms of the equation. If necessary the following differentials are after using for finding I.F.

Formulae:

1. $x\,dy + y\,dx = d(x,y)$
2. $x\,dx + y\,dy = d\left[\frac{x^2+y^2}{2}\right]$
3. $\frac{y\,dx - x\,dy}{y^2} = d(x/y)$
4. $\frac{x\,dy - y\,dx}{x^2} = d(y/x)$
5. $\frac{x\,dy + y\,dx}{x^2y^2} = d\left(\frac{-1}{xy}\right)$
6. $\frac{x\,dy + y\,dx}{xy} = d[\log(xy)]$
7. $\frac{x\,dy - y\,dx}{xy} = d[\log(y/x)]$
8. $\frac{y\,dx - x\,dy}{xy} = d[\log(x/y)]$
9. $\frac{x\,dy - y\,dx}{x^2+y^2} = d[\tan^{-1}(y/x)]$
10. $\frac{y\,dx - x\,dy}{x^2+y^2} = d[\tan^{-1}(x/y)]$
11. $\frac{x\,dx + y\,dy}{x^2+y^2} = \frac{1}{2}\log(x^2+y^2)$
12. $\frac{ye^x\,dx - e^x\,dy}{y^2} = d(e^x/y)$

1. Solve $ydx - xdy = a(x^2+y^2)dx$

Sol Given $ydx - xdy = a(x^2+y^2)\,dx \rightarrow ①$

Divide eq① by (x^2+y^2);

$$\frac{ydx - xdy}{x^2+y^2} = a\,dx$$

$$d\left[\mathrm{Tan}^{-1}(x/y)\right] = a\,dx$$

Integrating on B.S

$$\int d\left[\mathrm{Tan}^{-1}(x/y)\right] = \int a\,dx$$

$$\mathrm{Tan}^{-1}(x/y) = a\int dx$$

$\therefore\ \mathrm{Tan}^{-1}(x/y) = ax + c$ is required G.S of given D.E.

2. Solve $y(xy+e^x)\,dx - e^x dy = 0$

Sol. Given D.E $y(xy+e^x)\,dx - e^x dy = 0$

$$xy^2dx + ye^x dx - e^x dy = 0$$

$$ye^x dx - e^x dy = -xy^2 dx$$

Divide b.s by y^2;

$$\frac{ye^x dx - e^x dy}{y^2} = -x\,dx \Rightarrow d(e^x/y) = -x\,dx$$

Integrating on B.S

$$\int d(e^x/y) = -\int x\,dx$$

$$e^x/y = \frac{-x^2}{2} + c$$

$\therefore\ \dfrac{x^2}{2} + \dfrac{e^x}{y} = c$ is required G.S.

3. Solve $ydx - xdy + 3x^2y^2e^{x^3}dx = 0$

Sol. Given D.E is $ydx - xdy + 3x^2y^2e^{x^3}dx = 0$

Divide by 'y^2' on B.S

$$\frac{ydx - ydx}{y^2} + 3x^2e^{x^3}dx = 0$$

$$d\left(\frac{x}{y}\right) + d(e^{x^3}) = 0$$

Integrating on b.s $\Rightarrow \dfrac{x}{y} + e^{x^3} = c$ is required G.S.

Integrating Factors:

S.No	$Mdx + Ndy = 0$	condition	I.F
1	If both M and N are homogeneous functions of same degree	$Mx + Ny \neq 0$	$\frac{1}{Mx+Ny}$
2	If function is of form $yf(x,y)dx + xg(x,y)dy = 0$	$Mx - Ny \neq 0$	$\frac{1}{Mx-Ny}$
3	If both M and N are not homogeneous functions there exists continuous single valued function $f(x)$	$\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = f(x)$	$e^{\int f(x)dx}$
4	If both M and N are not homogeneous functions there exists continuous single valued function $g(y)$	$\frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M} = g(y)$	$e^{\int g(y)dy}$

1. solve $x^2y\,dx - (x^3+y^3)dy = 0$

sol, Given D.E is $x^2y\,dx - (x^3+y^3)\,dy = 0$

$x^2y\,dx + (-x^3-y^3)\,dy = 0 \rightarrow ①$

eq ① is of the form $Mdx + Ndy = 0$

$M = x^2y \;;\; N = -x^3 - y^3$

$\frac{\partial M}{\partial y} = x^2 \;\&\; \frac{\partial N}{\partial x} = -3x^2$

$\therefore \frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$

$\therefore$ Eq ① is not an E.D.E

Eq ① is of homogeneous function of same degree

$Mx + Ny = (x^2y)x + (-x^3-y^3)y$

$= x^3y - x^3y - y^4$

$= -y^4 \neq 0$

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$$I.F = \frac{1}{Mx+Ny} = \frac{-1}{y^4}$$

multiply eq① with I.F;

$$\frac{-1}{y^4}(x^2y)\,dx + \left(\frac{-1}{y^4}\right)(-x^3-y^3)\,dy = 0$$

$$\frac{-x^2y}{y^4}\,dx + \left(\frac{x^3}{y^4}+\frac{y^3}{y^4}\right)dy = 0$$

$$\frac{-x^2}{y^3}\,dx + \left(\frac{x^3}{y^4}+\frac{1}{y}\right)dy = 0 \longrightarrow ②$$

Eq② is of form $M_1dx + N_1dy = 0$

where $M_1 = \frac{-x^2}{y^3}$ & $N_1 = \frac{x^3}{y^4}+\frac{1}{y}$

$$\frac{\partial M_1}{\partial y} = \frac{3x^2}{y^4} \text{ and } \frac{\partial N_1}{\partial x} = \frac{3x^2}{y^4}$$

$$\therefore \frac{\partial M_1}{\partial y} = \frac{\partial N_1}{\partial x}$$

$\therefore$ Eq② is ~~of~~ an exact D.E

G.S is $\underset{y \text{ constant}}{\int M_1 dx} + \underset{\text{Neglect } x\text{-terms}}{\int N_1 dy} =$

$$= \int \frac{-x^2}{y^3}\,dx + \int\left(\frac{x^3}{y^4}+\frac{1}{y}\right)dy$$

$$\Rightarrow \frac{-x^3}{3y^3} + \log y = C$$

$\therefore x^3 - 3y^3\log y = 3Cy^3$ is required G.S.

2. Solve $y^2dx + (x^2-xy-y^2)\,dy = 0$

sol Given D.E is $y^2dx + (x^2-xy-y^2)\,dy = 0 \longrightarrow ①$

Eq① is of the form $Mdx + Ndy = 0$

$$M = y^2 \;;\; N = x^2-xy-y^2$$

$$\frac{\partial M}{\partial y} = 2y \;;\; \frac{\partial N}{\partial x} = 2x-y$$

$$\therefore \frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

$\therefore$ Eq① is not an E.D.E.

Eq① is homogeneous function of same degree.

$$Mx+Ny = y^2x + (x^2-xy-y^2)\,y$$

$$= \cancel{xy^2} + x^2y - \cancel{xy^2} - y^3 = x^2y - y^3 \neq 0$$

$$I.F = \frac{1}{Mx+Ny} = \frac{1}{x^2y-y^3}$$

Multiply eq① with I.F;

$$\frac{y^2}{x^2y-y^3}dx + \frac{x^2-xy-y^2}{x^2y-y^3}dy = 0$$

$$\frac{y}{x^2-y^2}dx + \left[\frac{x^2-y^2}{y(x^2-y^2)} - \frac{xy}{y(x^2-y^2)}\right]dy = 0$$

$$\frac{y}{x^2-y^2}dx + \left[\frac{1}{y} - \frac{x}{x^2-y^2}\right]dy = 0 \rightarrow ②$$

Eq ② is of the form $M_1dx + N_1dy = 0$

$M_1 = \frac{y}{x^2-y^2}$ | $N_1 = \frac{1}{y} - \frac{x}{x^2-y^2}$

$\frac{\partial M_1}{\partial y} = \frac{(x^2-y^2)(1) - y(-2y)}{(x^2-y^2)^2}$ | $\frac{\partial N_1}{\partial x} = 0 - \left[\frac{(x^2-y^2)(1) - x(2x)}{(x^2-y^2)^2}\right]$

$\frac{\partial M_1}{\partial y} = \frac{(x^2+y^2)}{(x^2-y^2)^2}$ | $\frac{\partial N_1}{\partial x} = \frac{(x^2+y^2)}{(x^2-y^2)^2}$

$\therefore \frac{\partial M_1}{\partial y} = \frac{\partial N_1}{\partial x}$; eq② is an E.D.E

$\therefore$ General solution is $\underset{y\text{ constant}}{\int M_1\,dx} + \underset{\text{Neglect }x\text{-terms}}{\int N_1\,dy} = C$

$$\int \frac{y}{x^2-y^2}dx + \int\left[\frac{1}{y} - \frac{x}{x^2-y^2}\right]dy = c$$

$$\Rightarrow y\int \frac{1}{x^2-y^2}dx + \int \frac{1}{y}dy = c$$

$$\Rightarrow y \cdot \frac{1}{2y}\log\left(\frac{x-y}{x+y}\right) + \log y = c \quad \left(\because \int\frac{1}{x^2-a^2}dx = \frac{1}{2a}\log\frac{x-a}{x+a}\right)$$

$$\Rightarrow \frac{1}{2}\log\left(\frac{x-y}{x+y}\right) + \log y = c$$

$\Rightarrow \therefore \log\left[\left(\frac{x-y}{x+y}\right)^{1/2} \cdot y\right] = c$ is required G. S.

3. Solve $y(xy\sin xy + \cos xy)dx + x(xy\sin xy - \cos xy)dy = 0$

Sol. Given $y(xy\sin xy + \cos xy)dx + x(xy\sin xy - \cos xy)dy = 0$ —①

Eq ① is of form $Mdx + Ndy = 0$

$M = xy^2\sin xy + y\cos xy$; $N = x^2y\sin xy - x\cos xy$

$$\frac{\partial M}{\partial y} = xy^2 \cos xy(x) + \sin xy(2xy) + y(-\sin xy)(x) + \cos xy(1)$$

$$\frac{\partial M}{\partial y} = x^2y^2 \cos(xy) + 2xy \sin xy - xy \sin xy + \cos xy$$

& $$\frac{\partial N}{\partial x} = x^2y \cos(xy)(y) + \sin xy(2xy) + x \sin xy(y) - \cos xy(1)$$

$$\frac{\partial N}{\partial x} = x^2y^2 \cos(xy) + 2xy \sin xy + xy \sin xy - \cos xy$$

$$\therefore \frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

Eq ① is not an E.D.E

Eq ① is of form $yf(x,y)\,dx + xg(x,y)\,dy = 0$.

$$Mx - Ny = x^2y^2 \sin xy + xy \cos xy - x^2y^2 \sin xy + xy \cos xy$$

$$Mx - Ny = 2xy \cos xy \neq 0.$$

$$I.F = \frac{1}{Mx - Ny} = \frac{1}{2xy \cos xy}$$

Multiply eq ① with I.F;

$$\frac{1}{2xy \cos xy}\left[(xy^2 \sin xy + y \cos xy)dx + (x^2y \sin xy - x \cos xy)dy\right] = 0$$

$$\Rightarrow \left(\frac{xy^2 \sin xy}{2xy \cos xy} + \frac{y \cos xy}{2xy \cos xy}\right)dx + \left(\frac{x^2y \sin xy}{2xy \cos xy} - \frac{x \cos xy}{2xy \cos xy}\right)dy = 0$$

$$\Rightarrow \left(\frac{y}{2} \tan xy + \frac{1}{2x}\right)dx + \left(\frac{x}{2} \tan xy + \frac{1}{2y}\right)dy = 0 \rightarrow ②$$

Eq ② is of the form $M_1 dx + N_1 dy = 0$

$$M_1 = \frac{y}{2} \tan xy + \frac{1}{2x} \qquad N_1 = \frac{x}{2} \tan xy + \frac{1}{2y}$$

$$\frac{\partial M_1}{\partial y} = \frac{1}{2}\left[y \sec^2 xy(x) + (1)\tan xy + 0\right] \qquad \frac{\partial N_1}{\partial x} = \frac{1}{2}\left[x \sec^2 xy(y) + (1)(\tan xy) + 0\right]$$

$$\frac{\partial M_1}{\partial y} = \frac{1}{2}\left[xy \sec^2 xy + \tan xy\right] \qquad \frac{\partial N_1}{\partial x} = \frac{1}{2}\left[xy \sec^2 xy + \tan xy\right]$$

$$\therefore \frac{\partial M_1}{\partial y} = \frac{\partial N_1}{\partial x}$$

$\therefore$ Eq ② is an exact D.E

G.S is $\int M_1 dx + \int N_1 dy = c$
y-constant   Neglect x-terms

$$\int\left(\frac{y}{2}\tan xy + \frac{1}{2x}\right)dx + \int\left(\frac{x}{2}\tan xy + \frac{1}{2y}\right)dy = c$$

$$\Rightarrow \frac{y}{2}\int \tan xy + \frac{1}{2}\log x + 0 + \frac{1}{2}\log y = c$$

$$\Rightarrow \frac{\cancel{y}}{2}\frac{\log \sec xy \cdot (\cancel{x})}{\cancel{y}} + \frac{1}{2}\log x - \frac{1}{2}\log y = c$$

$$\log\left(\frac{x\sec xy}{y}\right) = 2c$$

$$\therefore \frac{x\sec xy}{y} = e^{2c} \text{ is required G.S}$$

4. Solve $y(xy+2x^2y^2)dx + x(xy - x^2y^2)dy = 0$.

Sol Given $y(xy+2x^2y^2)dx + x(xy-x^2y^2)dy = 0 \rightarrow ①$

Eq ① is of the form $Mdx + Ndy = 0$

$$M = xy^2 + 2x^2y^3; \quad N = x^2y - x^3y^2$$

$$\frac{\partial M}{\partial y} = 2xy + 2x^2(3y^2); \quad \frac{\partial N}{\partial x} = y(2x) - y^2(3x^2)$$

$$\frac{\partial M}{\partial y} = 2xy + 6x^2y^2; \quad \frac{\partial N}{\partial x} = 2xy - 3x^2y^2$$

$$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

Eq① is not an Exact D.E

Eq① is of form $y\cdot f(x,y)dx + x\cdot g(x,y)dy = 0$

$$Mx - Ny = \cancel{x^2y^2} + 2x^3y^3 - \cancel{x^2y^2} + x^3y^3 = 3x^3y^3$$

$$I.F = \frac{1}{Mx-Ny} = \frac{1}{3x^3y^3}$$

Multiply eq① with IF;

$$\frac{y.(xy+2x^2y^2)dx}{3x^3y^3} + \frac{x(xy-x^2y^2)dy}{3x^3y^3} = 0$$

$$\frac{\cancel{x}y^2(1+2xy)dx}{3x^{\cancel{3}2}y^{\cancel{3}1}} + \frac{x^2y(1-xy)dy}{3x^{\cancel{3}1}y^{\cancel{3}2}} = 0$$

$$\left(\frac{1+2xy}{3x^2y}\right)dx + \left(\frac{1-xy}{3xy^2}\right)dy = 0 \longrightarrow (2)$$

Eq (2) is of the form $M_1dx + N_1dy = 0$

$$M_1 = \frac{1+2xy}{3x^2y} \quad\Big|\quad N_1 = \frac{1-xy}{3xy^2}$$

$$M_1 = \frac{1}{3x^2y} + \frac{2}{3x} \quad\Big|\quad N_1 = \frac{1}{3xy^2} - \frac{1}{3y}$$

$$\frac{\partial M_1}{\partial y} = \frac{1}{3x^2}\cdot\frac{-1}{y^2} + 0 \quad\Big|\quad \frac{\partial N_1}{\partial x} = \frac{1}{3y^2}\cdot\frac{-1}{x^2} - 0$$

$$\frac{\partial M_1}{\partial y} = \frac{-1}{3x^2y^2} \quad\Big|\quad \frac{\partial N_1}{\partial x} = \frac{-1}{3x^2y^2}$$

$$\therefore \frac{\partial M_1}{\partial y} = \frac{\partial N_1}{\partial x}$$

$\therefore$ Eq (2) is an exact D.E

G.S is $\underset{y\text{-constant}}{\int M\,dx} + \underset{\text{Neglect } x\text{-terms}}{\int N\,dy} = c$

$$\Rightarrow \int\left(\frac{1}{3x^2y} + \frac{2}{3x}\right)dx + \int\left(\frac{1}{3xy^2} - \frac{1}{3y}\right)dy = c$$

$$\Rightarrow \frac{1}{3y}\cdot\frac{-1}{x} + \frac{2}{3}\cdot\log x - \frac{1}{3}\log y = c$$

$$\Rightarrow 2\log x - \log y - \frac{1}{xy} = 3c$$

$$\Rightarrow \therefore \log\left(\frac{x^2}{y}\right) - \frac{1}{xy} = 3c \text{ is required G.S.}$$

5. Solve $2xydy - (x^2+y^2+1)dx = 0$

Sol. Given $2xydy - (x^2+y^2+1)dx = 0 \longrightarrow (1)$

Eq (1) is of the form $Mdx + Ndy = 0$

$$M = -(x^2+y^2+1) \quad ; \quad N = 2xy$$

$$\frac{\partial M}{\partial y} = -2y \quad ; \quad \frac{\partial N}{\partial x} = 2y$$

$$\therefore \frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

$\therefore$ Eq (1) is not an exact D.E

Eq (1) is of form non-homogeneous

$$\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = f(x)$$

$$\frac{-2y-2y}{2xy} = \frac{-4y}{2xy} = \frac{-2}{x} = f(x)$$

$$I.F = e^{\int f(x)\,dx} = e^{\int \frac{-2}{x}dx} = e^{-2\log x} = e^{\log x^{-2}} = x^{-2} = \frac{1}{x^2}$$

Multiply eq ① with I.F;

$$\frac{1}{x^2}\left[2xy\,dy - (x^2+y^2+1)\,dx\right] = 0$$

$$\Rightarrow \frac{2y}{x}dy - \frac{(x^2+y^2+1)}{x^2}dx = 0$$

$$\Rightarrow \frac{2y}{x}dy - \left(1+\frac{y^2}{x^2}+\frac{1}{x^2}\right)dx = 0$$

$$\Rightarrow \left(-1-\frac{y^2}{x^2}-\frac{1}{x^2}\right)dx + \left(\frac{2y}{x}\right)dy = 0 \longrightarrow ②$$

Eq ② is of the form $M_1dx + N_1dy = 0$

$M_1 = -1-\frac{y^2}{x^2}-\frac{1}{x^2}$ | $N_1 = \frac{2y}{x}$

$\frac{\partial M_1}{\partial y} = \frac{-1}{x^2}.2y = \frac{-2y}{x^2}$ | $\frac{\partial N_1}{\partial x} = 2y.\frac{-1}{x^2} = \frac{-2y}{x^2}$

$$\therefore \frac{\partial M_1}{\partial y} = \frac{\partial N_1}{\partial x}.$$

$\therefore$ Eq ② is an exact D.E.

G.S is $\underset{\text{Y-constant}}{\int M_1\,dx} + \underset{\text{Neglect x-terms}}{\int N_1\,dy} = C$

$$\int\left(-1-\frac{y^2}{x^2}-\frac{1}{x^2}\right)dx + \int \frac{2y}{x}dy = C$$

$$\Rightarrow \left(-x+\frac{y^2}{x}+\frac{1}{x}\right)+0 = C$$

$\Rightarrow \therefore y^2+1-x^2 = Cx$ is required G.S.

6. Solve $(y^4+2y)dx + (xy^3+2y^4-4x)dy = 0$

Sol, Given D.E is $(y^4+2y)dx + (xy^3+2y^4-4x)dy = 0 \rightarrow ①$

Eq① is of form $Mdx + Ndy = 0$

$M = y^4+2y$; $N = xy^3+2y^4-4x$.

$\frac{\partial M}{\partial y} = 4y^3+2$; $\frac{\partial N}{\partial x} = y^3+0-4 = y^3-4$

$$\therefore \frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

$\therefore$ eq ① is not an Exact D.E

eq ① is of form non-homogeneous D.E

$$\frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M} = g(y)$$

$$\Rightarrow \frac{(y^3-4)-(4y^3+2)}{y^4+2y} = g(y)$$

$$\Rightarrow \frac{-3y^3-6}{y^4+2y} = g(y)$$

$$\Rightarrow \frac{-3\cancel{(y^3+2)}}{y\cancel{(y^3+2)}} = g(y)$$

$$g(y) = \frac{-3}{y}$$

$$I.F = e^{\int g(y)\,dy} = e^{\int -\frac{3}{y}dy} = e^{-3\log y} = \frac{1}{y^3}$$

Multiply eq ① with I.F;

$$\frac{y^4+2y}{y^3}\,dx + \frac{xy^3+2y^4-4x}{y^3}\,dy = 0$$

$$\left(y+\frac{2}{y^2}\right)dx + \left(x+2y-\frac{4x}{y^3}\right)dy = 0 \longrightarrow ②$$

eq ② is of the form $M_1\,dx + N_1\,dy = 0$

$M_1 = y + \frac{2}{y^2}$ | $N_1 = x+2y-\frac{4x}{y^3}$

$\frac{\partial M_1}{\partial y} = 1+(2)\cdot\frac{-2}{y^3}$ | $\frac{\partial N_1}{\partial x} = 1+0-\frac{4}{y^3}$

$\frac{\partial M_1}{\partial y} = 1-\frac{4}{y^3}$ | $\frac{\partial N_1}{\partial x} = 1-\frac{4}{y^3}$

$$\therefore \frac{\partial M_1}{\partial y} = \frac{\partial N_1}{\partial x}$$

Eq ② is an exact D.E

G.S is $\underset{y\text{-constant}}{\int M_1\,dx} + \underset{\text{Neglect } x\text{-terms}}{\int N_1\,dy} = C$

$$\Rightarrow \int\left(y+\frac{2}{y^2}\right)dx + \int\left(x+2y-\frac{4x}{y^3}\right)dy = C$$

$$\Rightarrow y(x) + \frac{2}{y^2}(x) + \cancel{2}\cdot\frac{y^2}{\cancel{2}} = C \Rightarrow xy + \frac{2x}{y^2} + y^2 = C \text{ is required G.S.}$$

7. Solve $y(2x^2y+e^x)dx = (e^x+y^3)dy$

Sol, Given D.E is $y(2x^2y+e^x)\,dx = (e^x+y^3)dy$

$$(2x^2y^2+e^xy)\,dx + (-e^x-y^3)dy = 0 \longrightarrow ①$$

Eq ① is of form $Mdx + Ndy = 0$

$M = 2x^2y^2 + ye^x$ | $N = -e^x - y^3$

$\frac{\partial M}{\partial y} = 2x^2(2y) + e^x$ | $\frac{\partial N}{\partial x} = -e^x - 0$

$\frac{\partial M}{\partial y} = 4x^2y + e^x$ | $\frac{\partial N}{\partial x} = -e^x$

$$\therefore \frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

Eq ① is not an Exact D.E

Eq ① is of form non-homogeneous D.E

$$\frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M} = \frac{-e^x - 4x^2y - e^x}{2x^2y^2 + ye^x} = \frac{-2e^x - 4x^2y}{2x^2y^2 + ye^x} = \frac{-2(e^x + 2x^2y)}{y(e^x + 2x^2y)} = \frac{-2}{y} = g(y)$$

$$\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = \frac{4x^2y + e^x + e^x}{-e^x - y^3} = \frac{4x^2y + 2e^x}{-e^x - y^3}$$

$$I.F = e^{\int g(y)\,dy} = e^{\int \frac{-2}{y}dy} = e^{-2\log y} = \frac{1}{y^2}$$

Multiply eq ① with I.F;

$$\frac{2x^2y^2 + e^xy}{y^2}\,dx + \frac{(-e^x - y^3)}{y^2}\,dy = 0$$

$$\left(2x^2 + \frac{e^x}{y}\right)dx + \left(\frac{-e^x}{y^2} - y\right)dy = 0 \longrightarrow ②$$

Eq ② is of the form $M_1dx + N_1dy = 0$

$M_1 = 2x^2 + \frac{e^x}{y}$ | $N_1 = \frac{-e^x}{y^2} - y$

$\frac{\partial M_1}{\partial y} = \cancel{2x^2} + e^x\frac{(-1)}{y^2}$ | $\frac{\partial N_1}{\partial x} = \frac{-e^x}{y^2} \cancel{-y}$

$\frac{\partial M_1}{\partial y} = \cancel{2x^2} - \frac{e^x}{y^2}$ | $\frac{\partial N_1}{\partial x} = \frac{-e^x}{y^2}$

$$\therefore \frac{\partial M_1}{\partial y} = \frac{\partial N_1}{\partial x}$$

$\therefore$ Eq ② is an Exact D E

G.S is $\int M_1\,dx + \int N_1\,dy = c$
y-constant   Neglect x-terms

$\Rightarrow \int \left(2x^2 + \frac{e^x}{y}\right) dx + \int \left(\frac{-e^x}{y} - y\right) dy = c$

$\Rightarrow \therefore \frac{2x^3}{3} + \frac{e^x}{y} - \frac{y^2}{2} = c$ is required G.S.

8. solve $(x^2y - 2xy^2)dx - (x^3 - 3x^2y)dy = 0$.

Sol. Given D.E is $(x^2y - 2xy^2)\,dx + (-x^3 + 3x^2y)dy = 0 \rightarrow$ ①

It is of the form $Mdx + Ndy = 0$

$M = x^2y - 2xy^2 \qquad N = -x^3 + 3x^2y$

$\frac{\partial M}{\partial y} = x^2 - 2x(2y) \qquad \frac{\partial N}{\partial x} = -3x^2 + 3y(2x)$

$\frac{\partial M}{\partial y} = x^2 - 4xy \qquad \frac{\partial N}{\partial x} = -3x^2 + 6xy$

$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$

Eq ① is not an exact D.E

Eq ① is of homogeneous function of same degree.

$Mx + Ny = \cancel{x^3y} - 2x^2y^2 - \cancel{x^3y} + 3x^2y^2 = x^2y^2$

$I.F = \frac{1}{Mx+Ny} = \frac{1}{x^2y^2}$

Multiply eq ① with I.F;

$\left(\frac{x^2y}{x^2y^2} - \frac{2xy^2}{x^2y^2}\right) dx + \left(\frac{-x^3}{x^2y^2} + \frac{3x^2y}{x^2y^2}\right) dy = 0$

$\left(\frac{1}{y} - \frac{2}{x}\right) dx + \left(\frac{-x}{y^2} + \frac{3}{y}\right) dy = 0 \longrightarrow$ ②

Eq ② is of the form $M_1dx + N_1dy = 0$

$M_1 = \frac{1}{y} - \frac{2}{x} \qquad N_1 = \frac{-x}{y^2} + \frac{3}{y}$

$\frac{\partial M_1}{\partial y} = \frac{-1}{y^2} \qquad \frac{\partial N_1}{\partial x} = \frac{-1}{y^2}$

$\therefore \frac{\partial M_1}{\partial y} = \frac{\partial N_1}{\partial x}$

$\therefore$ Eq ② is an Exact D.E

G.S is $\int M_1 dx + \int N_1 dy = c$
y-constant   Neglect x-terms

$\Rightarrow \int \left(\frac{1}{y} - \frac{2}{x}\right) dx + \int \left(\frac{-x}{y^2} + \frac{3}{y}\right) dy = c$

$\Rightarrow \frac{x}{y} - 2\log x + 3\log y = c$

$\Rightarrow \frac{x}{y} - \log x^2 + \log y^3 = c$

$\Rightarrow \therefore \frac{x}{y} + \log\left(\frac{y^3}{x^2}\right) = c$ is required G.S

9. Solve $(2x\log x - xy)dy + 2y\,dx = 0$.

Sol, Given D.E is $(2x\log x - xy)\,dy + 2y\,dx = 0 \longrightarrow ①$

Eq ① is of the form $M dy + N dx = 0$

$N = 2x\log x - xy$ $M = 2y$

$\frac{\partial N}{\partial x} = \cancel{2 + 2\log x - y}$ $\frac{\partial M}{\partial y} = 2$

$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$

$\therefore$ Eq ① is not an exact D.E

Eq ① is a non-homogeneous D.E

$$\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = \frac{+2 - 2 - 2\log x + y}{2x\log x - xy} = \frac{y - 2\log x}{-x(y - 2\log x)} = \frac{-1}{x} = f(x)$$

$I.F = e^{\int f(x)dx} = e^{\int -\frac{1}{x}dx} = e^{-\log x} = \frac{1}{x}$

Multiply equation ① with I.F;

$(2\log x - y)\,dy + \frac{2y}{x}\,dx = 0$

$\frac{2y}{x}\,dx + (2\log x - y)dy = 0 \longrightarrow ②$

Eq ② is of the form $M_1 dx + N_1 dy = 0$

$M_1 = \frac{2y}{x} \Rightarrow \frac{\partial M_1}{\partial y} = \frac{2}{x}$

$$\frac{\partial N_1}{\partial x} = \frac{2}{x}$$

$$\therefore \frac{\partial M_1}{\partial y} = \frac{\partial N_1}{\partial x}$$

$\therefore$ Eq ② is an exact D.E

G.S is $\underset{y\text{-constant}}{\int M_1\,dx} + \underset{\text{Neglect } x\text{-terms}}{\int N_1\,dy} = c$

$\Rightarrow \int \frac{2y}{x}dx + \int (2\log x - y)\,dy = c$

$\Rightarrow 2y\log x - \frac{y^2}{2} = c$

$\therefore 2y\log x - \frac{y^2}{2} = c$ is required G.S.

10. solve $(x^2+y^2+x)dx + xy\,dy = 0$

sol. Given D.E is $(x^2+y^2+x)\,dx + xy\,dy = 0 \rightarrow ①$

Eq ① is of the form ~~x^2+y^2+~~ $M\,dx + N\,dy = 0$

$M = x^2+y^2+x \qquad N = xy$

$\frac{\partial M}{\partial y} = 0+2y+0 \qquad \frac{\partial N}{\partial x} = y$

$\frac{\partial M}{\partial y} = 2y \qquad \frac{\partial N}{\partial x} = y$

$$\therefore \frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

Eq ① is not an exact D.E.

Eq ① is of form Non-homogeneous D.E

$$\frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M} = \frac{y-2y}{x^2+y^2+x}$$

$$\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = \frac{2y-y}{xy} = \frac{y}{xy} = \frac{1}{x} = f(x)$$

$$I.F = e^{\int f(x)dx} = e^{\int \frac{1}{x}dx} = e^{\log x}$$

$I.F = x$

Multiply eq ① with 'x';

$(x^3+xy^2+x^2)\,dx + (x^2y)\,dy = 0 \longrightarrow ②$

Eq ② is of form $M_1\,dx + N_1\,dy = 0$

$M_1 = x^3+xy^2+x^2 \qquad N_1 = x^2y$

$\frac{\partial M_1}{\partial y} = x(2y) = 2xy \qquad \frac{\partial N_1}{\partial x} = (2x)y = 2xy.$

$\therefore \frac{\partial M_1}{\partial y} = \frac{\partial N_1}{\partial x}$

Eq ② is an exact D.E

G.S is $\underset{y\text{-constant}}{\int M_1\,dx} + \underset{\text{Neglect } x\text{-terms}}{\int N_1\,dy} = c$

$\Rightarrow \int (x^3+xy^2+x^2)\,dx + \int x^2y\,dy = c$

$\Rightarrow \cancel{(3x^2+y^2+2x) = c} \quad \Rightarrow \frac{x^4}{4} + \frac{x^2y^2}{2} + \frac{x^3}{3} = c$

$\therefore 3x^4 + 6y^2x^2 + 4x^3 = c$ is required G.S.

11. solve $(xy^3+y)\,dx + 2(x^2y^2+y^4+x)\,dy = 0.$

Sol, Given D.E is $(xy^3+y)\,dx + 2(x^2y^2+y^4+x)\,dy = 0 \longrightarrow ①$

Eq ① is of form $M\,dx + N\,dy = 0$

$M = xy^3+y$	$N = 2x^2y^2+2y^4+2x$
$\frac{\partial M}{\partial y} = x(3y^2)+1$	$\frac{\partial N}{\partial x} = 2y^2(2x)+2$
$\frac{\partial M}{\partial y} = 3xy^2+1$	$\frac{\partial N}{\partial x} = 4xy^2+2$

$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$

Eq ① is not an exact D.E

Eq ① is of form Non-homogeneous D.E

$$\frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M} = \frac{4xy^2+2-3xy^2-1}{xy^3+y} = \frac{1+xy^2}{y(xy^2+1)} = \frac{1}{y} = g(y)$$

$I.F = e^{\int g(y)\,dy} = e^{\int \frac{1}{y}dy} = e^{\log y} = y$

Multiply eq ① with I.F;

$(xy^4+y^2)\,dx + (2x^2y^3+2y^5+2xy)\,dy = 0 \longrightarrow ②$

Eq ② is of form $M_1 dx + N_1 dy = 0$

$M_1 = xy^4 + y^2$ | $N_1 = 2x^2y^3 + 2y^5 + 2xy$

$\frac{\partial M_1}{\partial y} = 4xy^3 + 2y$ | $\frac{\partial N_1}{\partial x} = 4xy^3 + 2y$

$$\therefore \frac{\partial M_1}{\partial y} = \frac{\partial N_1}{\partial x}$$

Eq ② is an exact D.E

G.S is $\underset{y\text{ constant}}{\int M_1\,dx} + \underset{\text{Neglect } x\text{-terms}}{\int N_1\,dy} = c$

$$\Rightarrow \int (xy^4 + y^2)\,dx + \int (2x^2y^3 + 2y^5 + 2xy)\,dy = c$$

$$\Rightarrow y^4 \cdot \frac{x^2}{2} + xy^2 + 2 \cdot \frac{y^6}{6} = c$$

$$\therefore \frac{x^2y^4}{2} + xy^2 + \frac{y^6}{3} = c \text{ is required G.S.}$$

Applications of D.E:

* Newton's law of cooling:

The rate of change of temperature of a body is directly proportional to temperature difference between body and surrounding media (room temperature i.e, air).

Let 'θ' be the temperature of the body and θ_0 be the temperature of the surrounding media.

$$\frac{d\theta}{dt} \propto (\theta - \theta_0)$$

$$\frac{d\theta}{dt} = -K(\theta - \theta_0) \text{ (Negative cooling)}$$

$$\frac{d\theta}{\theta - \theta_0} = -K\,dt$$

on I.B.S

$$\int \frac{d\theta}{\theta - \theta_0} = -K \int dt$$

$$\boxed{\log(\theta - \theta_0) = -Kt + C}$$

(or) $\log(\theta - \theta_0) = -Kt + \log C$

$$\log(\theta - \theta_0) - \log C = -Kt$$

$$\log\left(\frac{\theta - \theta_0}{C}\right) = -Kt$$

$$\frac{\theta - \theta_0}{C} = e^{-Kt}$$

$$\boxed{\theta - \theta_0 = C \cdot e^{-Kt}}$$

Q.1: If the temperature of the body is cooling from 100°C to 70°C in 15 min. Find for which time the body temperature will be

40°C if the temperature of the air is 30°C.

Sol, Given, temperature of the air $= \theta_0 = 30°C$

$\theta = 100°C \quad t = 0 \text{ min} \longrightarrow (1)$

$\theta = 70°C \quad t = 15 \text{ min} \longrightarrow (2)$

$\theta = 40°C \quad t = ? \longrightarrow (3)$

By Newton's law of cooling,

$\log(\theta - \theta_0) = -Kt + C \longrightarrow (4)$

Apply condition (1) in (4);

$\log(100-30) = -K(0) + C$

$\therefore C = \log 70$

Apply condition (2) in (4);

$\log(70-30) = -K(15) + \log 70$

$\log 40 - \log 70 = -15K$

$-15K = \log\left(\frac{40}{70}\right)$

$K = \frac{-1}{15}\log\left(\frac{4}{7}\right)$

Apply condition (3) in (4);

$\log(40-30) = \frac{1}{15}\log\left(\frac{4}{7}\right)t + \log 70$

$\log 10 - \log 70 = \frac{t}{15}\log\left(\frac{4}{7}\right)$

$\log\left(\frac{10}{70}\right) = \frac{t}{15}\log\left(\frac{4}{7}\right)$

$\frac{\log(1/7)}{\log(4/7)} = \frac{t}{15}$

$\frac{t}{15} = \frac{-0.845}{-0.243}$

$\frac{t}{15} = 3.477$

$t = 52.16$

$\therefore t \simeq 52 \text{ min}.$

⑧ The temperature of the body drops from $100°C$ to $75°C$ in 10 min when the surrounding air is at $20°C$. Find for which time the body temperature will be $25°C$ and also find what will be the temperature after half an hour.

sol, Given ; temperature of surrounding air $= \theta_0 = 20°C$

$\theta = 100°C \quad t = 0 \text{ min} \longrightarrow ①$

$\theta = 75°C \quad t = 10 \text{ min} \longrightarrow ②$

$\theta = 25°C \quad t = ? \longrightarrow ③$

$\theta = ? \quad t = 30 \text{ min} \longrightarrow ④$

By Newton's law of cooling,

$\log(\theta - \theta_0) = -Kt + C \longrightarrow ⑤$

Apply condition ① in ⑤;

$\log(100-20) = -K(0) + C$

$\therefore C = \log 80$

Apply condition ② in ⑤;

$\log(75-20) = -K(10) + \log 80$

$\log 55 - \log 80 = -10K$

$\log\left(\frac{55}{80}\right) = -10K$

$-10K = \log\left(\frac{11}{16}\right)$

$K = \frac{-1}{10}\log\left(\frac{11}{16}\right)$

Apply condition ③ in ⑤;

$\log(25-20) = \frac{1}{10}\log\left(\frac{11}{16}\right)t + \log 80$

$\log 5 - \log 80 = \frac{t}{10}\log\left(\frac{11}{16}\right)$

$\log\left(\frac{5}{80}\right) = \frac{t}{10}\log\left(\frac{11}{16}\right)$

$$\frac{t}{10} = \frac{\log\left(\frac{1}{16}\right)}{\log\left(\frac{11}{16}\right)}$$

$$\frac{t}{10} = \frac{-1.204}{-0.163} = 7.386$$

$\therefore t = 73.86$

$\therefore t \simeq 74$ min.

Apply condition ④ in ⑤;

$$\log(\theta-20) = \frac{1}{10}\cdot\log\left(\frac{11}{16}\right)\cdot(30) + \log 80$$

$$\log(\theta-20) = 3\log\left(\frac{11}{16}\right) + \log 80$$

$$\log(\theta-20) = \log\left(\frac{11}{16}\right)^3 + \log 80$$

$$\log(\theta-20) = \log\left(\left(\frac{11}{16}\right)^3\cdot 80\right)$$

$$\log(\theta-20) = \log\left(\frac{1331\times 80}{4096}\right)$$

$$\log(\theta-20) = \log(25.996)$$

$$\theta - 20 = 25.996$$

$$\theta = 45.996$$

$\therefore \theta \simeq 46°C$

③ water is heated to the boiling point of temperature 100°C and it is removed from the heat kept in a room where temperature is 60°C after 3 min, the temperature of water is 90°C. Find what will be the water temperature after 6 min and also find when will be the water temperature is 61°C

Sol Given, $\theta_0 = 60°C$

$\theta = 100°C \quad t = 0$ min $\longrightarrow$ ①

$\theta = 90°C \quad t = 3$ min $\longrightarrow$ ②

$\theta = 61°C \quad t = ?$ $\longrightarrow$ ③

$\theta = ? \quad t = 6$ min $\longrightarrow$ ④

By Newton's law of cooling;

$\log(\theta - \theta_0) = -Kt + C \longrightarrow ⑤$

Apply condition ① in ⑤;

$\log(100-60) = -K(0) + C$

$\therefore C = \log 40$

Apply condition ② in ⑤;

$\log(90-60) = -K(3) + \log 40$

$\log 30 - \log 40 = -3K$

$-3K = \log\left(\frac{30}{40}\right)$

$K = \frac{-1}{3}\log\left(\frac{3}{4}\right)$

Apply condition ③ in ⑤;

$\log(61-60) = \frac{1}{3}\log\left(\frac{3}{4}\right)t + \log 40$

$\log 1 - \log 40 = \frac{t}{3}\cdot\log\left(\frac{3}{4}\right)$

$-\log 40 = \frac{t}{3}\log\left(\frac{3}{4}\right)$

$\frac{t}{3} = \frac{-\log 40}{\log\left(\frac{3}{4}\right)}$

$\frac{t}{3} = \frac{-1.602}{-0.125}$

$\frac{t}{3} = 12.816$

$t = 38.448$

$\therefore t \simeq 39$ min

Apply condition ④ in ⑤;

$\log(\theta-60) = \frac{1}{3}\log\left(\frac{3}{4}\right)\cdot 6 + \log 40$

$\log(\theta-60) = 2\log\left(\frac{3}{4}\right) + \log 40$

$\log(\theta-60) = \log\left(\frac{9}{16}\cdot 40\right) = \log\left(\frac{90}{4}\right)$

$\theta - 60 = 22.5$

$\therefore \theta = 82.5°C$

④ A body is kept at temperature of 25°C and it cools from 140°C to 80°C in 20 min. Find when the body cools from to 35°C

Sol Given, $\theta_0 = 25°C$

$\theta = 140°C, t = 0 \text{ min} \rightarrow ①$

$\theta = 80°C, t = 20 \text{ min} \rightarrow ②$

$\theta = 35°C, t = ? \rightarrow ③$

Apply Newton's law of cooling; $\log(\theta - \theta_0) = -Kt + C \rightarrow ④$

Apply condition ① in ④;

$\log(140-25) = -K(0) + C$

$\therefore C = \log 115$

Apply condition ② in ④;

$\log(80-25) = -K(20) + \log 115$

$\log 55 - \log 115 = -20K$

$K = \frac{-1}{20} \log\left(\frac{55}{115}\right)$

Apply condition ③ in ④;

$\log(35-25) = \frac{1}{20} \log\left(\frac{55}{115}\right) \cdot t + \log 115$

$\log 10 - \log 115 = \frac{t}{20} \log \frac{55}{115}$

$\log\left(\frac{10}{115}\right) = \frac{t}{20} \log\left(\frac{55}{115}\right)$

$\frac{t}{20} = \frac{-1.060}{-0.320} = 3.3125 \Rightarrow t = 66.25$

$\therefore t \simeq 66 \text{ min}.$

* Law of growth and decay:

The rate of change of amount of substance is directly proportional to the amount of substance available at that time.

Let N be the amount of substance at any time t.

$\therefore$ According to growth and decay ; $\frac{dN}{dt} \propto N$

<u>For growth</u>: $\frac{dN}{dt} \propto N \Rightarrow \frac{dN}{dt} = KN$

$\frac{dN}{N} = K dt$; I.O.B.S $\Rightarrow \int \frac{1}{N} dN = K\int dt + C$

$\boxed{\therefore \log N = Kt + C}$

<u>For decay</u>: $\frac{dN}{dt} \propto -N \Rightarrow \frac{dN}{dt} = -KN \Rightarrow \frac{dN}{N} = -K dt$

I.O.B.S $\Rightarrow \int \frac{dN}{N} = -K\int dt + C \Rightarrow \boxed{\therefore \log N = -Kt + C}$

① Bacteria culture growing exponentially increases from 100gr – 400gr in 10 hrs. How much quantity of bacteria is present after 3 hrs.

sol, Given, $N = 100gr \quad t = 0 hrs \rightarrow ①$

$N = 400gr \quad t = 10 hrs \rightarrow ②$

$N = ? \quad t = 3 hrs \rightarrow ③$

By Natural law of growth ; $\log N = Kt + C \rightarrow ④$

Apply ① in ④ ; $\log 100 = K(0) + C \Rightarrow C = \log 100$

Apply ② in ④ ; $\log 400 = K(10) + \log 100 \Rightarrow K = \frac{1}{10} \cdot \log(4)$.

Apply ③ in ④ ; $\log N = \frac{1}{10} \log(4) \cdot 3 + \log 100$

$\log N = \log(4)^{3/10} + \log 100$

$\log N = \log((4)^{3/10} \cdot 100) = \log(151.57)$

$\therefore N = 151.57 \simeq 152 \text{ gr}$ //

② If the population of a country doubles in 50 years. In how many years it will be triple assuming that the rate of increase is proportional to No. of inhabitants

sol, Let 'x' be the population of a country at time 't'.

i.e., $N = x \quad t = 0 \text{ years} \rightarrow ①$

$N = 2x \quad t = 50 \text{ years} \rightarrow ②$

$N = 3x \quad t = ? \rightarrow ③$

By Natural law of growth ; $\log N = Kt + C \rightarrow ④$

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Apply ① in ④ $\Rightarrow \log x = K(0) + C \Rightarrow C = \log x$

Apply ② in ④ $\Rightarrow \log 2x = K(50) + \log x \Rightarrow K = \frac{1}{50}\log 2$

Apply ③ in ④ $\Rightarrow \log 3x = \frac{1}{50}\log 2 (t) + \log x$

$$\log 3 = \frac{t}{50}\log 2$$

$$\frac{t}{50} = \frac{0.477}{0.301} = 1.585$$

$$\therefore t = 79.25 \text{ years} \simeq 79 \text{ years}.$$

③ Radium decompose at a rate proportial to amount present if 5% of original amount disappears in 50 years. How much will remain after 100 years.

Sol, $N = 100 \quad t = 0$ years $\longrightarrow$ ①

$N = 95 \quad t = 50$ years $\longrightarrow$ ②

$N = ? \quad t = 100$ years $\longrightarrow$ ③

By ~~Newton's~~ Natural law of decay; $\log N = -Kt + C \rightarrow$ ④

Apply ① in ④ ; $\log 100 = -K(0) + C \Rightarrow C = \log 100$

Apply ② in ④ ; $\log 95 = -K(50) + \log 100 \Rightarrow K = \frac{-1}{50}\log\left(\frac{95}{100}\right)$

Apply ③ in ④ ; $\log N = \frac{1}{\cancel{50}}\log\left(\frac{95}{100}\right)\cdot \cancel{100}^{2} + \log 100$

$$\log N = \log\left[\left(\frac{95}{100}\right)^2 \cdot 100\right] = \log(90.25)$$

$$\therefore N = 90.25 \simeq 90.$$

$\therefore$ 90% will remain after 100 years.

④ A chemical reaction of given substance is converted into another at a rate proportional to amount of substance unconverted. If $\frac{1}{5}$th of original amount transformed in 4 min, how much time is required to half of the original amount.

Sol, Let $N = x$ at $t = 0$ min $\longrightarrow$ ①

$N = \frac{4x}{5}$ at $t = 4$ min $\longrightarrow$ ②

$N = \frac{x}{2}$ at $t = ?$ $\longrightarrow$ ③

By Natural law of decay; $\log N = -Kt + C \rightarrow (4)$

Apply (1) in (4) $\Rightarrow \log x = -K(0) + C \Rightarrow C = \log x$

Apply (2) in (4) $\Rightarrow \log \frac{4x}{5} = -K(4) + \log x \Rightarrow K = \frac{-1}{4}\log\left(\frac{4}{5}\right)$

Apply (3) in (4) $\Rightarrow \log \frac{x}{2} = \frac{1}{4}\log\frac{4}{5}\cdot(t) + \log x$

$$\log\frac{x}{2} - \log x = \frac{t}{4}\log\frac{4}{5}$$

$$\log\left(\frac{x/2}{x}\right) = \frac{t}{4}\log\frac{4}{5}$$

$$\log\frac{1}{2} = \frac{t}{4}\log\frac{4}{5}$$

$$\frac{t}{4} = \frac{-0.3010}{-0.096} = 3.135$$

$$\therefore t = 12.54 \text{ min} \simeq 12 \text{ min.}$$

(5) In a certain chemical reaction, the rate of change of conversion of a substance at a time t is proportional to the quantity of substance still untransformed at that time. At the end of 1 hour 60 grams remains and at end of 4 hours 21 grams will be there. How many grams of substance will be there initially.

Sol. Given, $N = 60\text{gr. at } t = 1\text{ hour} \rightarrow (1)$

$N = 21\text{gr at } t = 4\text{ hours} \rightarrow (2)$

$N = ? \quad \text{at } t = 0\text{hr} \rightarrow (3)$

By Natural law of decay: $\log N = -Kt + C \rightarrow (4)$

Apply (1) in (4) $\Rightarrow \log 60 = -K(1) + C = -K + C \rightarrow (5)$

Apply (2) in (4) $\Rightarrow \log 21 = -K(4) + C = -4K + C \rightarrow (6)$

From (5) − (6) $\Rightarrow \log\left(\frac{60}{21}\right) = 3K \Rightarrow K = \frac{1}{3}\log\frac{60}{21}$

From (5); $\log 60 = -\frac{1}{3}\log\frac{60}{21} + C \Rightarrow C = \log 60 + \frac{1}{3}\log\frac{60}{21}$

$$C = \log\left[60\times\left(\frac{60}{21}\right)^{1/3}\right]$$

Apply (3) in (4) $\Rightarrow \log N = -K(0) + \log\left[60\times\left(\frac{60}{21}\right)^{1/3}\right]$

$$\therefore N = 1.418 \times 60 = 85.139 \simeq 85\text{gr.}$$

⑥ Uranium disintegrates at a rate proportional to amount present at any instant. If M_1 and M_2 grams of uranium are present at t_1 and t_2 respectively. Show that the half life time of uranium is $\frac{(T_2-T_1)\log 2}{\log(M_1/M_2)}$.

Sol, $N = M$ at $t=0 \longrightarrow ①$

$N = M_1$ at $t=T_1 \longrightarrow ②$

$N = M_2$ at $t=T_2 \longrightarrow ③$

$N = \frac{M}{2}$ at $t=? \longrightarrow ④$

By Natural law of decay $\Rightarrow \log N = -Kt + C \longrightarrow ⑤$

Apply ① in ⑤ $\Rightarrow \log M = -K(0) + C \Rightarrow C = \log M$.

Apply ② in ⑤ $\Rightarrow \log M_1 = -KT_1 + \log M$

$\log M_1 - \log M = -KT_1 \longrightarrow ⑥$

Apply ③ in ⑤ $\Rightarrow \log M_2 = -KT_2 + \log M$

$\log M_2 - \log M = -KT_2 \longrightarrow ⑦$

By solving ⑥ & ⑦;

$\log M_1 - \log M = -KT_1$

$\log M_2 - \log M = -KT_2$

$-\quad + \quad +$

$\log M_1 - \log M_2 = -KT_1 + KT_2$

$K = \frac{\log(M_1/M_2)}{T_2 - T_1}$

Apply ④ in ⑤;

$\log \frac{M}{2} = -\frac{\log(M_1/M_2)}{(T_2-T_1)} . t + \log M$

$\log\left(\frac{M/2}{M}\right) = -\frac{\log(M_1/M_2)}{(T_2-T_1)} . t$

$-\log \frac{1}{2} (T_2 - T_1) = \log(M_1/M_2) \cdot t$

$\log 2 . (T_2 - T_1) = \log(M_1/M_2) . t$

$\therefore t = \frac{(T_2 - T_1)\log 2}{\log(M_1/M_2)}$.

Orthogonal tragectories:

Two family of curves are said to be orthogonal to each other if each member of either family intersect each member of other family at right angle.

Itself orthogonal tragectories:

The given family of curves are said to be self orthogonal tragectories if they intersect itself in right angle.

working procedure for finding the orthogonal tragecto-ries in cartesian form:

→ Let $f(x, y, c) = 0 \longrightarrow ①$ be the given family of curves in cartesian form where 'c' is arbitrary constant.

→ Form a differential equation of ① by eliminating the arbitrary constant c, then eq ① will reduces to

$f(x, y, \frac{dy}{dx}) = 0 \longrightarrow ②$

Eq ② represents differential eqn of eq ①

→ Replace $\frac{dy}{dx}$ as $-\frac{dx}{dy}$ in eq ② then eq ② becomes

$f(x, y, -\frac{dx}{dy}) = 0 \longrightarrow ③$

Eq ③ represents D.E of required family of O.T's.

→ By solving eq ③, we will get the required family of O.T's of eq ①.

working procedure for finding O.T's in polar form:

→ Let $f(r,\theta,c)=0 \longrightarrow (1)$ be the equation of family of curves in polar form where 'c' is arbitrary constant.

→ Form a differential equation of (1) by eliminating the arbitary constant c, the eq(1) reduces to $f\left(r,\theta,\frac{dr}{d\theta}\right)=0$ ↳(2)

Eq (2) represents D.E of (1).

→ Replace $\frac{dr}{d\theta}$ as $\frac{-r^2 d\theta}{dr}$ in eq(2), then eq(2) becomes

$$f\left(r,\theta,-r^2\frac{d\theta}{dr}\right)=0 \longrightarrow (3)$$

Eq (3) represents D.E of required family of O.T's

(1) Find the O.T's of family of parabolas $y^2=4ax$.

Sol, Given family of curves is $y^2=4ax \longrightarrow (1)$

Diff eq(1) on b.s w.r.t 'x';

$$2y.\frac{dy}{dx}=4a$$

$$y.\frac{dy}{dx}=2a$$

$$\frac{y}{2}\cdot\frac{dy}{dx}=a$$

Substitute 'a' in eq(1);

$$y^2=4.\left(\frac{y}{2}\cdot\frac{dy}{dx}\right)\cdot x$$

$$y^2=2xy.\frac{dy}{dx} \longrightarrow (2)$$

Eq (2) represents D.E of eq(1).

Replace $\frac{dy}{dx}$ as $\frac{-dx}{dy}$ in eq(2);

$$y^2=2xy.\left(\frac{-dx}{dy}\right)$$

$$y=-2x.\frac{dx}{dy} \longrightarrow (3)$$

Eq ③ represents the D.E of O.T's ~~of eq~~

on solving eq ③;

$$y\,dy = -2x\,dx$$

I.B.S

$$\int y\,dx = -2\int x\,dx + C$$

$$\frac{y^2}{2} = -2\cdot\frac{x^2}{2} + C$$

$$\frac{y^2}{2} + x^2 = C$$

$\therefore 2x^2 + y^2 = C$ is required equation of O.T's of eq ①.

② Find the O.T's of semicubical parabolas $ay^2 = x^3$.

sol Given family of curves $ay^2 = x^3 \longrightarrow ①$

Diff on b/s w.r.t 'x';

$$a(2y)\frac{dy}{dx} = 3x^2$$

$$a = \frac{3x^2}{2y}\cdot\frac{dx}{dy}$$

substitute 'a' in eq ①;

$$\frac{3x^2}{2y}\cdot\frac{dx}{dy}\cdot y^2 = x^3$$

$$y = \frac{2x}{3}\cdot\frac{dy}{dx} \longrightarrow ②$$

Eq ② represents the D.E of eq ①

Replace $\frac{dy}{dx}$ as $\frac{-dx}{dy}$ in eq ②;

$$y = \frac{2x}{3}\cdot\left(\frac{-dx}{dy}\right) \longrightarrow ③$$

on solving eq ③;

$$3y\,dy = -2x\,dx$$

on I.B.S

$$\int 3y\,dy = -2\int x\,dx + C$$

$\therefore x^2 + \frac{3y^2}{2} = C$ is required family of O.T's of eq ①.

3. Find the OT's of the family of curves $x^{2/3}+y^{2/3}=a^{2/3}$

Sol, Given family of curves $x^{2/3}+y^{2/3}=a^{2/3} \longrightarrow (1)$

Diff. on both sides with respect to 'x'

$$\frac{2}{3}\cdot x^{\frac{2}{3}-1}+\frac{2}{3}y^{\frac{2}{3}-1}\cdot\frac{dy}{dx}=0$$

$$x^{-\frac{1}{3}}+y^{-\frac{1}{3}}\cdot\frac{dy}{dx}=0$$

$$\frac{dy}{dx}=\frac{-x^{-1/3}}{y^{-1/3}} \longrightarrow (2)$$

Eq (2) is D.E of eq (1)

Replace $\frac{dy}{dx}$ as $\frac{-dx}{dy}$

$$\frac{-dx}{dy}=\frac{-x^{-1/3}}{y^{-1/3}}$$

$$\frac{dx}{x^{-1/3}}=\frac{dy}{y^{-1/3}}$$

$x^{1/3}dx=y^{1/3}dy$ is represents the D.E of O.T's of eq(2)

I. O. B. S

$$\int x^{1/3}dx=\int y^{1/3}dy+c$$

$$\frac{x^{4/3}}{\frac{4}{3}}=\frac{y^{4/3}}{\frac{4}{3}}+c$$

$$\frac{3}{4}\cdot x^{4/3}-\frac{3}{4}y^{4/3}=c$$

$\therefore x^{4/3}-y^{4/3}=\frac{4}{3}c$ is required family of O.T's of eq(1).

4. Show that the family of orthoganal tragestories of curves $x^2-y^2=c^2$ and $xy=c^2$ are orthogonal.

Sol, Given curve $x^2-y^2=c^2 \longrightarrow (1)$

Diff. on B.S w.r.t 'x';

$$2x-2y\cdot\frac{dy}{dx}=0$$

$$x=y\cdot\frac{dy}{dx} \Rightarrow \frac{dy}{dx}=\frac{x}{y} \longrightarrow (2)$$

Eq ② is D.E of eq ①

replace $\frac{dy}{dx}$ as $\frac{-dx}{dy}$

$$\frac{-dx}{dy} = \frac{x}{y}$$

$-\frac{1}{x} dx = \frac{1}{y} dy$; which represents D.E of O.T's

I.O.B.S

$$\int \frac{-1}{x} dx = \int \frac{1}{y} dy + \log c$$

$$-\log x = \log y + \log c$$

$$\log x + \log y = \log c$$

$xy = c$ is required family of O.T's of ①

∴ The curves are orthogonal to each other.

5. show that the curve $y^2 = 4ax + 4a^2$ is self orthogonal

Sol Given curve $y^2 = 4ax + 4a^2 \longrightarrow ①$

Diff. on B.S w.r.t 'x';

$$2y.\frac{dy}{dx} = 4a$$

substitute '$4a = \frac{y}{2}.\frac{dy}{dx}$' in eq ①;

$$y^2 = 2y.\frac{dy}{dx}.x + 4\left(\frac{y}{2}.\frac{dy}{dx}\right)^2$$

$$y^2 = 2xy.\frac{dy}{dx} + y^2\left(\frac{dy}{dx}\right)^2 \longrightarrow ②$$

eq ② is D.E of eq ①

replace $\frac{dy}{dx}$ as $\frac{-dx}{dy}$

$$y^2 = 2xy\left(\frac{-dx}{dy}\right) + y^2\left(\frac{-dx}{dy}\right)^2$$

$$y^2 = \frac{dx}{dy}\left(-2xy + y^2.\frac{dx}{dy}\right)$$

$$y^2.\frac{dy}{dx} = -2xy + y^2.\frac{dx}{dy}$$

$$y^2\left(\frac{dy}{dx}\right)^2 = -2xy\left(\frac{dy}{dx}\right) + y^2$$

$$y^2 = 2xy \cdot \frac{dy}{dx} + y^2\left(\frac{dy}{dx}\right)^2 \rightarrow (3)$$

eq (3) represents D.E of O.T's

eq (2) and eq (3) are same.

∴ The given curve is self orthogonal.

6. show that the system of confocal of $\frac{x^2}{a^2+\lambda} + \frac{y^2}{b^2+\lambda} = 1$ where 'λ' is the parameter and is self orthogonal.

sol Given, $\frac{x^2}{a^2+\lambda} + \frac{y^2}{b^2+\lambda} = 1 \rightarrow (1)$

Diff. on b.s w.r.t 'x';

$$\frac{2x}{a^2+\lambda} + \frac{2y}{b^2+\lambda} \cdot \frac{dy}{dx} = 0$$

$$\frac{x}{a^2+\lambda} + \frac{y}{b^2+\lambda} \cdot \frac{dy}{dx} = 0$$

$$\frac{x}{a^2+\lambda} + \frac{yy'}{b^2+\lambda} = 0$$

$$\frac{x(b^2+\lambda) + yy'(a^2+\lambda)}{(a^2+\lambda)(b^2+\lambda)} = 0$$

$$xb^2 + x\lambda + yy'a^2 + yy'\lambda = 0$$

$$\lambda(x+yy') + xb^2 + yy'a^2 = 0$$

$$\lambda(x+yy') = -xb^2 - yy'a^2$$

$$\lambda = \frac{-(xb^2+yy'a^2)}{x+yy'}$$

Now, $a^2+\lambda = a^2 - \frac{(xb^2+yy'a^2)}{x+yy'}$

$$= \frac{a^2x + \cancel{yy'a^2} - xb^2 - \cancel{yy'a^2}}{x+yy'}$$

$$= \frac{x(a^2-b^2)}{x+yy'}$$

$b^2+\lambda = b^2 - \left(\frac{xb^2+yy'a^2}{x+yy'}\right)$

$$= \frac{b^2x+b^2yy'-xb^2-yy'a^2}{x+yy'}$$

$$= \frac{yy'(b^2-a^2)}{x+yy'}$$

substitute '$a^2+\lambda$' and '$b^2+\lambda$' in eq ①;

$$\frac{x^2}{\frac{x(a^2-b^2)}{x+yy'}} + \frac{y^2}{\frac{yy'(b^2-a^2)}{x+yy'}} = 1$$

$$-\frac{x(x+yy')}{b^2-a^2} + \frac{y(x+yy')}{y'(b^2-a^2)} = 1$$

$$\frac{-xy'(x+yy')+xy+y^2y'}{y'(b^2-a^2)} = 1$$

$-x^2y'-x$

$$\frac{(x+yy')[-xy'+y]}{y'(b^2-a^2)} = 1$$

$$(x+yy')[-xy'+y] = y'(b^2-a^2) \longrightarrow ②$$

Eq ② represents D.E of eq ①;

replace y' as $\frac{-1}{y'}$;

$$\left[x+y\left(\frac{-1}{y'}\right)\right]\left[-x\cdot\left(\frac{-1}{y'}\right)+y\right] = \frac{-1}{y'}(b^2-a^2)$$

$$\frac{(xy'-y)\cdot(x+yy')}{y'^2} = \frac{-1}{y'}\cdot(b^2-a^2)$$

$$(x+yy')(-xy'+y) = y'(b^2-a^2) \longrightarrow ③$$

Eq ③ represents D.E of O.T's

Eq ② and eq ③ are same

∴ Given system is self-orthogonal.

7. Find the O.T's of family of card $r=a(1-\cos\theta)$

sol. where 'a' is arbitrary constant.

Given, $r=a(1-\cos\theta) \longrightarrow ①$

Apply 'log' on both sides;

$\log r = \log[a(1-\cos\theta)]$

$\log r = \log a + \log(1-\cos\theta)$

Diff. on both sides w.r.t 'θ'

$\frac{1}{r}\cdot\frac{dr}{d\theta} = 0 + \frac{1}{1-\cos\theta}(\sin\theta)$

$\frac{1}{r}\cdot\frac{dr}{d\theta} = \frac{2\sin\theta/2\cdot\cos\theta/2}{2\sin^2\theta/2}$

$\frac{dr}{d\theta} = \frac{r\cdot\cos\theta/2}{\sin\theta/2}$

$\frac{dr}{d\theta} = r\cot\frac{\theta}{2} \rightarrow (2)$; which represents D.E of eq (1).

replace $\frac{dr}{d\theta}$ as $-r^2\frac{d\theta}{dr}$

$-r^2\frac{d\theta}{dr} = r\cdot\cot\frac{\theta}{2}$

$-r\cdot\frac{d\theta}{dr} = \cot\frac{\theta}{2} \rightarrow (3)$ which represents D.E of O.T's

$\frac{d\theta}{\cot\frac{\theta}{2}} = \frac{-dr}{r}$

$\tan\frac{\theta}{2}\cdot d\theta = \frac{-dr}{r}$

I.O.B.S

$\int\tan\frac{\theta}{2}\cdot d\theta + \int\frac{dr}{r} = c$

$\frac{\log\left|\sec\frac{\theta}{2}\right|}{\frac{1}{2}} + \log r = \log c$

$2\log(\sec\theta/2) + \log r = \log c$

$\log(\sec^2\theta/2\cdot r) = \log c$

$\therefore c = r\sec^2\frac{\theta}{2}$ is required family of O.T's.

8. Find the O.T's of the family of curves $r^n\sin n\theta = a^n$

where 'a' is the parameter. $r^n \sin n\theta = a^n$ $r^n = a^n \cos n\theta$

Sol. Given family of curves, $r^n \sin n\theta = a^n \longrightarrow (1)$

Diff. on B.S w.r.t 'θ'.

$$r^n.(\cos n\theta)n + \sin n\theta (r^{n-1})n.\frac{dr}{d\theta} = 0$$

$$n.r^n\left[\cos n\theta + \frac{1}{r}\sin n\theta.\frac{dr}{d\theta}\right] = 0$$

$$\cos n\theta + \frac{1}{r}.\sin n\theta.\frac{dr}{d\theta} = 0 \longrightarrow (2) \quad \frac{dr}{d\theta} = -r.\frac{\cos n\theta}{\sin n\theta}$$

Eq (2) represents D.E of eq (1)

replace $\frac{dr}{d\theta}$ as $-r^2\frac{d\theta}{dr}$

$$\cos n\theta + \frac{1}{r}.\sin n\theta . -r^2\frac{d\theta}{dr} = 0$$

$\cos n\theta - r.\sin n\theta.\frac{d\theta}{dr} = 0 \rightarrow (3)$; represents D.E of O.T's

$$\cos n\theta = r\sin\theta.\frac{d\theta}{dr}$$

$$\frac{d\theta}{dr} = \frac{1}{r}.\frac{\cos n\theta}{\sin n\theta}$$

$$\frac{d\theta}{dr} = \frac{1}{r}.\cot n\theta$$

$$\frac{d\theta}{\cot n\theta} = \frac{dr}{r}$$

$$\tan n\theta \, d\theta = \frac{dr}{r}$$

I.O.B.S

$$\int \tan n\theta \, d\theta = \int\frac{dr}{r} + c$$

$$\frac{\log|\sec n\theta|}{n} = \log r + \log c$$

$$\frac{1}{n}\log|\sec n\theta| = \log(cr)$$

$$\log(\sec n\theta)^{1/n} = \log(cr)$$

$\therefore \quad c = \frac{\log(\sec n\theta)^{1/n}}{r}$; represents family of O.T's.

Find the O.T's of family of curves $r = \frac{2a}{1+\cos\theta}$

Sol. Given family of curves is $r = \frac{2a}{1+\cos\theta} \longrightarrow (1)$

Diff. on B.S w.r.t 'θ'

$$\frac{dr}{d\theta} = 2a\, \cancel{\log}\left(\frac{-1}{(1+\cos\theta)^2}\right)(-\sin\theta)$$

$$\frac{dr}{d\theta} = \frac{2a}{1+\cos\theta} \cdot \frac{\sin\theta}{1+\cos\theta}$$

$$\frac{dr}{d\theta} = r \cdot \frac{\sin\theta}{1+\cos\theta} \quad (\because \text{from } (1))$$

$$\frac{dr}{d\theta} = r \cdot \frac{2\sin\theta/2 \cdot \cos\theta/2}{2\cos^2\theta/2}$$

$$\frac{dr}{d\theta} = r \cdot \tan\frac{\theta}{2} \longrightarrow (2)$$; represents D.E of eq (1)

replace $\frac{dr}{d\theta}$ as $-r^2 \frac{d\theta}{dr}$

$$-r^2 \frac{d\theta}{dr} = r\tan\frac{\theta}{2}$$

$$-r \cdot \frac{d\theta}{dr} = \tan\frac{\theta}{2} \longrightarrow (3)$$; represents D.E of O.T's

$$\frac{d\theta}{\tan\theta/2} = \frac{-dr}{r}$$

$$\cot\frac{\theta}{2} \cdot d\theta = -\frac{1}{r}\, dr.$$

I.O.B.S

$$\int \cot\frac{\theta}{2}\, d\theta + \int \frac{1}{r}\, dr = c$$

$$\frac{\log\left|\sin\frac{\theta}{2}\right|}{\frac{1}{2}} + \log r = \log c$$

$$2\log\left|\sin\frac{\theta}{2}\right| + \log r = \log c$$

$$\log(\sin^2\theta/2) + \log r = \log c$$

$$\log(r\sin^2\theta/2) = \log c$$

$\therefore r\sin^2\theta/2 = c \Rightarrow c = r\sin^2\theta/2$ is required family of O.T's

Find O.T's of family of curves $r = a(\sec\theta + \tan\theta)$ where 'a' is the parameter.

Sol: Given family of curves $r = a(\sec\theta + \tan\theta) \longrightarrow ①$

Diff. on B.S w.r.t 'θ';

$$\frac{dr}{d\theta} = a[\sec\theta\tan\theta + \sec^2\theta]$$

$$\frac{dr}{d\theta} = a\sec\theta[\sec\theta + \tan\theta]$$

$$\frac{dr}{d\theta} = \sec\theta[a(\sec\theta + \tan\theta)]$$

$$\frac{dr}{d\theta} = r\sec\theta \quad (\because \text{From } ①)$$

$\frac{dr}{d\theta} = r\sec\theta \longrightarrow ②$; represents D.E of eq①

replace $\frac{dr}{d\theta}$ as $-r^2\frac{d\theta}{dr}$

$$\frac{-r^2 d\theta}{dr} = r\sec\theta$$

$-r\frac{d\theta}{dr} = \sec\theta \longrightarrow ③$; represents D.E of O.T's

$$\frac{d\theta}{\sec\theta} = \frac{-dr}{r}$$

I.O.B.S

$$\int \frac{d\theta}{\sec\theta} + \int \frac{dr}{r} = c$$

$$\int \cos\theta \, d\theta + \int \frac{dr}{r} = c$$

$$\sin\theta + \log r = c$$

$$\sin\theta + \log r = \log c$$

$\sin\theta = \log\left(\frac{c}{r}\right)$; is required family of O.T's.

Find the O.T's of family of curves $r^n \sin n\theta = a^n$.

Given family of curves, $r^n \sin n\theta = a^n \longrightarrow ①$

Taking "log" on B.S

$\log(r^n \sin n\theta) = \log a^n$

$\log r^n + \log \sin n\theta = \log a^n$

$n \log r + \log(\sin n\theta) = \log a^n$

Diff on B.S w.r.t 'θ'

$n \cdot \frac{1}{r} \cdot \frac{dr}{d\theta} + \frac{1(n)}{\sin n\theta}(\cos n\theta) = \frac{1}{a^n} \cdot (0)$

$\frac{n}{r} \cdot \frac{dr}{d\theta} + n\cot n\theta = 0$

$\frac{dr}{d\theta} = \frac{-r}{\cancel{n}} \cancel{n} \cot n\theta \longrightarrow ②$

Eq ② represents D.E of eq ①

replace $\frac{dr}{d\theta}$ as $-r^2 \cdot \frac{d\theta}{dr}$

$-r^2 \cdot \frac{d\theta}{dr} = -r \cot n\theta$

$+r \cdot \frac{d\theta}{dr} = \cot n\theta \longrightarrow ③$

Eq ③ represents D.E of O.Ts

$\frac{1}{\cot n\theta} \cdot d\theta = \frac{dr}{r}$

$\tan n\theta \cdot d\theta = \frac{1}{r} dr$

I. O. B. S

$\int \tan n\theta \, d\theta = \int \frac{1}{r} dr.$

$\frac{\log|\sec n\theta|}{n} = \log r + c$

UNIT-4 : Linear Differential equation of higher order

The general form of n^{th} order D.E with constants co-efficient is

$$\frac{d^n y}{dx^n} + K_1 \frac{d^{n-1} y}{dx^{n-1}} + K_2 \frac{d^{n-2} y}{dx^{n-2}} + \dots + K_n y = X(x) \quad \text{--- (1)}$$

where X is a function of x (or) constant

where $K_1, K_2 \dots K_n$ are constants.

let us put $D = \frac{d}{dx}$, $D^2 = \frac{d^2}{dx^2}$, $D^n = \frac{d^n}{dx^n}$

$$Dy = \frac{dy}{dx} \quad D^2 y = \frac{d^2 y}{dx^2} \quad D^n y = \frac{d^n y}{dx^n}$$

$$D^n y + K_1 D^{n-1} y + K_2 D^{n-2} y + \dots + K_n Y = X(x)$$

$$(D^n + K_1 D^{n-1} + K_2 D^{n-2} + \dots + K_n) Y = X(x)$$

$$f(D) Y = X(x) \quad \text{--- (2)}$$

where $f(D) = D^n + K \cdot D^{n-1} + \dots + K_n$ is a polynomial of D of order n.

The G.S (or) complete soln. of eqn (2) contains 2 parts.

$$y = y_c + y_p$$

$$C.S = C.F + P.I$$

where C.F = complementary function and

P.I = particular integral.

1. complementary function:

The C.F is the complete solⁿ of $f(D)y = 0$.

and it has n orbitary constants.

2. particular integral:

particular integral is the solⁿ of $f(D)y = X(x)$

and has no orbitary constants.

i.e, $y_p = \frac{1}{f(D)} X(x)$

Auxilliary eqⁿ:

It is used to find C.F, the auxilliary

eqⁿ of $f(D)y = 0$ is $f(m) = 0$.

i.e, $m^n + k_1 m^{n-1} + k_2 m^{n-2} + \cdots + k_n = 0$.

which is a polynomial eqⁿ of order n and

has 'n' roots.

The C.F depends upon nature of the roots of

auxilliary eqⁿ.

let us assume $m_1, m_2 \ldots m_n$

S.NO	Nature of roots i.e, A.E $f(m)=0$	C.F
1.	If all the roots $m_1, m_2, m_3 \ldots m_n$ are real & distinct.	$y_c = c_1 e^{m_1 x} + c_2 e^{m_2 x} + c_3 e^{m_3 x} + \ldots + c_n e^{m_n x}$.
2.	If two roots are real & equal & remaining are real and distinct	$y_c = (c_1 + c_2 x) e^{mx} + c_3 e^{m_3 x} + c_4 e^{m_4 x} + \ldots + c_n e^{m_n x}$.
3	If three roots are real & equal & remaining are real and distinct	$y_c = (c_1 + c_2 x + c_3 x^2) e^{mx} + c_4 e^{m_4 x} + c_5 e^{m_5 x} + \ldots + c_n e^{m_n x}$.
4.	If $\alpha \pm i\beta$ are 2 complex conjugate roots and remaining are real & distinct	$y_c = e^{\alpha x}[c_1 \cos\beta x + c_2 \sin\beta x] + c_3 e^{m_3 x} + c_4 e^{m_4 x} + \ldots + c_n e^{m_n x}$.
5.	If two pairs of complex conjugate roots are repeated $(\alpha \pm i\beta, \alpha \pm i\beta)$ i.e,	

	$m_1 = m_2 = \alpha + i\beta$, $m_3 = m_4 = \alpha - i\beta$ & remaining are real and distinct.	$y_c = e^{\alpha x}[(c_1 + c_2 x)\cos\beta x + (c_3 + c_4 x)\sin\beta x] + c_5 e^{m_5 x} + \dots + c_n e^{m_n x}$.
6.	If $\alpha + \sqrt{\beta}$, $\alpha - \sqrt{\beta}$ are two irrational roots & remaining are real & distinct	$y_c = e^{\alpha x}[c_1 \cosh\sqrt{\beta}\,x + c_2 \sinh\sqrt{\beta}\,x] + c_3 e^{m_3 x} + c_4 e^{m_4 x} + \dots + c_n e^{m_n x}$.

1, Solve $\frac{d^2y}{dx^2} + y = 0$

Sol: Given eqn in operator form be

$D^2y + y = 0$

$(D^2+1)y = 0$

Given eqn is of form $f(D)y = 0$

It's A.E be $f(m) = 0$

i.e, $m^2+1 = 0$

$m^2 = -1$

$m = \pm i$

$y_c = e^{0x}[c_1 \cos x + c_2 \sin x]$.

2, Solve $(D^3 - 6D^2 + 11D - 6)y = 0$

Sol: Given eqn is of form $f(D)y = 0$

It's A.E be $f(m) = 0$

i.e, $m^3 - 6m^2 + 11m - 6 = 0$

$$\begin{array}{c|cccc} 1 & 1 & -6 & 11 & -6 \\ & 0 & 1 & -5 & 6 \\ \hline & 1 & -5 & 6 & \boxed{0} \end{array}$$

$(m-1)(m^2-5m+6) = 0$

$m-1 = 0$ & $m^2 - 5m + 6 = 0$

$m^2 - 3m - 2m + 6 = 0$

$m = 1$ & $m(m-3) - 2(m-3) = 0$

$(m-2)(m-3) = 0$

$m = 1, 2, 3$

$y_c = c_1 e^{x} + c_2 e^{2x} + c_3 e^{3x}$

3, $(D^4 + 4) y = 0$

sol: Given eqn is of form $f(D) y = 0$.

It's A.E be $f(m) = 0$

$m^4 + 4 = 0$

$(m^2)^2 + (2)^2 + (2m)^2 - (2m)^2 = 0$

$(m^2)^2 + (2m)^2 + (2)^2 - (2m)^2 = 0$

$(m^2 + 2)^2 - (2m)^2 = 0$

$(m^2 + 2 + 2m)(m^2 + 2 - 2m) = 0$

$(m^2 + 2m + 2) = 0,\ (m^2 - 2m + 2) = 0$

$m = \frac{-2 \pm \sqrt{4-8}}{2}$ $\quad$ $m = \frac{2 \pm \sqrt{4-8}}{2}$

$m = \frac{-2 \pm \sqrt{-4}}{2}$ $\quad$ $m = \frac{2 \pm \sqrt{-4}}{2}$

$m = \frac{-2 \pm 2i}{2}$ $\quad$ $m = \frac{2 \pm 2i}{2}$

$m = -1 \pm i$ $\quad$ $m = 1 \pm i$

$m = -1+i, -1-i$ $\quad$ $m = 1+i, 1-i$

$y_c = e^{-x}[c_1 \cos x + c_2 \sin x] + e^{x}[c_3 \cos x + c_4 \sin x]$

4, $(D^6 + 1) y = 0$

sol: Given eqn is of form $f(D) \cdot y = 0$.

It's A.E be $f(m) = 0$

$$m^6 + 1 = 0$$

$$(m^2)^3 + (1)^3 = 6$$

$$(m^2+1)\left((m^2)^2 - m^2(1) + 1\right) = 0$$

$$(m^2+1)(m^4 - m^2 + 1) = 0$$

$$(m^2+1)\left((m^2)^2 + 1 - m^2\right) = 0$$

$$(m^2+1)\left((m^2+1)^2 - 3m^2\right) = 0$$

$$m^2+1 = 0 \quad \& \quad \left[\underbrace{(m^2+1)^2}_{a} - (\sqrt{3}m)^2\right]$$

$$m^2 = -1 \quad \& \quad (m^2+1-\sqrt{3}m)(m^2+\sqrt{3}m+1) = 0$$

$$m = \pm i \quad \& \quad m^2 - \sqrt{3}m + 1 = 0 \quad m^2 + \sqrt{3}m + 1 = 0$$

$$m = \pm i \quad m = \frac{\sqrt{3} \pm \sqrt{3-4}}{2} \quad m = \frac{-\sqrt{3} \pm \sqrt{3-4}}{2}$$

$$m = \frac{\sqrt{3} \pm i}{2} \quad m = \frac{-\sqrt{3} \pm i}{2}$$

$$m = \frac{\sqrt{3}}{2} + \frac{i}{2}, \quad m = -\frac{\sqrt{3}}{2} + \frac{i}{2},$$

$$\frac{\sqrt{3}}{2} - \frac{i}{2} \quad -\frac{\sqrt{3}}{2} - \frac{i}{2}$$

$$m = +i, -i, \frac{\sqrt{3}}{2} + \frac{i}{2}, \frac{\sqrt{3}}{2} - \frac{i}{2}, -\frac{\sqrt{3}}{2} + \frac{i}{2}, -\frac{\sqrt{3}}{2} - \frac{i}{2}$$

$$y_c = e^{0x}\left[c_1 \cos x + c_2 \sin x\right] + e^{\sqrt{3}/2x}\left[c_3 \cos \frac{x}{2} + c_4 \sin \frac{x}{2}\right]$$

$$+ e^{-\sqrt{3}/2x}\left[c_5 \cos \frac{x}{2} + c_6 \cos \frac{x}{2}\right].$$

5, Solve $\frac{d^2y}{dx^2} - 8\frac{dy}{dx} + 15y = 0.$

Sol: Given eqn in operator form be

$Dy^2 - 8Dy + 15y = 0$

$(D^2 - 8D + 15)y = 0$

Given eq^n is of form $f(D)y = 0$

It's auxiliary eq^n be $f(m) = 0$.

$m^2 - 8m + 15$.

$m^2 - 5m - 3m + 15 = 0$

$m(m-5) - 3(m-5) = 0$

$(m-3)(m-5) = 0$

$m = 3, 5$.

$y_c = c_1 e^{3x} + c_2 e^{5x}$.

6. solve $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + y = 0$.

sol:- The given eq^n in operator form be

$D^2y - 3Dy + y = 0$

$(D^2 - 3D + 1)y = 0$

Given eq^n is of form $f(D)y = 0$

It's auxiliary eq^n be $f(m) = 0$.

$m^2 - 3m + 1 = 0$

$m = \frac{3 \pm \sqrt{9-4}}{2}$

$= \frac{3 \pm \sqrt{5}}{2}$

$= \frac{3+\sqrt{5}}{2}, \frac{3-\sqrt{5}}{2}$.

$= \frac{3}{2} + \frac{\sqrt{5}}{2}, \frac{3}{2} - \frac{\sqrt{5}}{2}$

$y_c = e^{3/2x}\left[c_1 \cosh \frac{\sqrt{5}}{2}x + c_2 \sinh \frac{\sqrt{5}}{2}x\right]$.

particular integral:-

Type 1:- To find P.I of $f(D)y = X(x)$ where

$X(x) = e^{ax}$

$$y_p = \frac{1}{f(D)} X(x)$$

$$= \frac{1}{f(D)} e^{ax} \quad \begin{bmatrix} \text{put } D=a \\ f(D) = f(a) \neq 0 \end{bmatrix}$$

$$= \frac{1}{f(a)} e^{ax} \text{ if } f(a) \neq 0.$$

If $f(a) = 0$ then $y_p = \frac{x}{f'(D)} e^{ax} = \frac{x}{f'(a)} e^{ax}$

if $f'(a) \neq 0$.

If $f'(a) = 0$ then $y_p = \frac{x^2}{f''(D)} e^{ax} = \frac{x^2}{f''(a)}$ if $f''(a) \neq 0$

similarly - - -

1, solve $(D^2 - 6D + 3)y = e^{3x}$

sol: To find C.F:-

Given eq^n is of form $f(D)y = 0$

It's A.E i.e, $f(m) = 0$

$$m^2 - 6m + 3 = 0$$

$$m = \frac{6 \pm \sqrt{36-12}}{2} = \frac{6 \pm \sqrt{24}}{2} = \frac{6 \pm 2\sqrt{6}}{2} = 3 \pm \sqrt{6}.$$

$$y_c = e^{3x}[c_1 \cosh\sqrt{6}x + c_2 \sinh\sqrt{6}x]$$

To find P.I:

$$y_p = \frac{1}{f(D)} X(x)$$

$$= \frac{1}{D^2 - 6D + 3} e^{3x} \quad \begin{bmatrix} a = 3 \\ \text{put } D = a = 3 \\ f(D) = f(a) = -6 \neq 0 \end{bmatrix}$$

$= \frac{-1}{6} e^{3x}$

$y = y_c + y_p$

$= e^{3x}\{c_1 \cosh\sqrt{6}x + c_2 \sinh\sqrt{6}x] - \frac{1}{6} e^{3x}$

2, solve $(D^2 + D + 1) y = (1 + e^x)^2$

sol:- To find C.F:

Given eq^n is of form $f(D) y = 0$

It's A.E i.e, $f(m) = 0$

$m^2 + m + 1 = 0$

$m = \frac{-1 \pm \sqrt{1-4}}{2} = \frac{-1 \pm \sqrt{-3}}{2} = \frac{-1}{2} \pm \frac{i\sqrt{3}}{2}$

$y_c = e^{-1/2 x} [c_1 \cos\frac{\sqrt{3}}{2}x + c_2 \sin\frac{\sqrt{3}}{2}x]$

$= e^{-x/2} [c_1 \cos\frac{\sqrt{3}}{2}x + c_2 \sin\frac{\sqrt{3}}{2}x]$

To find P.I:

$y_p = \frac{1}{f(D)} X(x)$

$= \frac{1}{D^2 + D + 1} (1 + e^x)^2$

$= \frac{1}{D^2 + D + 1} (1 + e^{2x} + 2e^x)$

$= \underbrace{\frac{1}{D^2+D+1} 1 e^{0x}}_{PI_1} + \underbrace{\frac{1}{D^2+D+1} e^{2x}}_{PI_2} + \underbrace{\frac{1}{D^2+D+1} 2e^x}_{PI_3}$ —— ①

consider PI_1

$y_{P_1} = \frac{1}{D^2 + D + 1} e^{0x}$ $\left[\begin{array}{l} a = 0 \\ \text{put } D = a = 0 \\ f(D) = f(a) = 1 \neq 0 \end{array}\right]$

$= \frac{1}{1} e^{0x}$

$= 1$

consider PI_2

$$y_{P_2} = \frac{1}{D^2+D+1} e^{2x} \qquad \left[\begin{array}{l} a=2 \\ \text{put } D=a=2 \\ f(D)=f(a)=7 \neq 0 \end{array}\right]$$

$$= \frac{1}{7} e^{2x}$$

consider PI_3

$$y_{P_3} = \frac{1}{D^2+D+1} 2e^{x} \qquad \left[\begin{array}{l} a=1 \\ \text{put } D=a=1 \\ f(D)=f(a)=3 \neq 0 \end{array}\right]$$

$$= \frac{1}{3} 2e^{x} = \frac{2}{3}e^{x}.$$

sub PI_1, PI_2, PI_3 in ①

$$y_p = 1 + \frac{1}{7}e^{2x} + \frac{2}{3}e^{x}$$

$$\therefore y = y_c + y_p$$

$$= e^{-x/2}\left[c_1 \cos\frac{\sqrt{3}}{2}x + c_2 \sin\frac{\sqrt{3}}{2}x\right] + 1 + \frac{1}{7}e^{2x} + \frac{2}{3}e^{x}.$$

3, solve $(D^3 - 6D^2 + 11D - 6)y = (e^{-2x} + e^{-3x})$

sol: To find C.F:

Given eq^n is of form $f(D)y = 0$

It's A.E i.e, $f(m) = 0$

$$m^3 - 6m^2 + 11m - 6 = 0$$

$$\begin{array}{r|rrrr} 1 & 1 & -6 & 11 & -6 \\ & 0 & 1 & -5 & 6 \\ \hline & 1 & -5 & 6 & \boxed{0} \end{array}$$

$$(m-1)(m^2 - 5m + 6) = 0$$

$$(m-1) = 0 \qquad m^2 - 5m + 6 = 0$$

$$m^2 - 2m - 3m + 6 = 0$$

$$m(m-2) - 3(m-2) = 0$$

$$(m-2)(m-3) = 0$$

$$m = 2, 3.$$

$m = 1, 2, 3$

$y_c = c_1 e^{x} + c_2 e^{2x} + c_3 e^{3x}$

To find P.I:-

$$y_p = \frac{1}{f(D)} X(x)$$

$$= \frac{1}{f(D)} \left(e^{-2x} + e^{-3x}\right)$$

$$= \frac{1}{D^3 - 6D^2 + 11D - 6} \left(e^{-2x} + e^{-3x}\right)$$

$$= \underbrace{\frac{1}{D^3 - 6D^2 + 11D - 6} e^{-2x}}_{PI_1} + \underbrace{\frac{1}{D^3 - 6D^2 + 11D - 6} e^{-3x}}_{PI_2} \quad \text{--- (1)}$$

consider PI_1

$$y_{p_1} = \frac{1}{D^3 - 6D^2 + 11D - 6} e^{-2x} \quad \left[\begin{array}{l} a = -2 \\ \text{put } D = a = -2 \\ f(D) = f(a) = -60 \neq 0 \end{array}\right]$$

$$= \frac{1}{-60} e^{-2x}$$

$$= -1/60 \, e^{-2x}$$

consider PI_2

$$y_{p_2} = \frac{1}{D^3 - 6D^2 + 11D - 6} e^{-3x} \quad \left[\begin{array}{l} a = -3 \\ \text{put } D = a = -3 \\ f(D) = f(a) = -120 \neq 0 \end{array}\right]$$

$$= \frac{1}{-120} e^{-3x}$$

$$= -1/120 \, e^{-3x}$$

Sub PI_1, PI_2 in (1)

$$y_p = \frac{-1}{60} e^{-2x} - \frac{1}{120} e^{-3x}$$

$$y = y_c + y_p$$

$$= c_1 e^{x} + c_2 e^{2x} + c_3 e^{3x} - \frac{1}{60} e^{-2x} - \frac{1}{120} e^{-3x}$$

4. solve $(D^2-D-6)y = e^x \cosh 2x$.

sol: <u>To find C.F:</u>

Given eqn is of form $f(D)y=0$

It's A.E i.e, $f(m)=0$

$$m^2-m-6=0$$

$$m^2-3m+2m-6=0$$

$$m(m-3)+2(m-3)=0$$

$$(m-3)(m+2)=0$$

$$m=-2,3.$$

$$y_c = c_1e^{-2x}+c_2e^{3x}.$$

<u>To find P.I:</u>

$$y_p = \frac{1}{f(D)} X(x)$$

$e^x = \cosh x + \sinh x$

$e^{2x} = \cosh 2x + \sinh 2x$

$e^{-2x} = \cosh 2x - \sinh 2x$

$\frac{e^{2x}+e^{-2x}}{2} = \cosh 2x.$

$$= \frac{1}{D^2-D-6} e^x \cosh 2x$$

$$= \frac{1}{D^2-D-6} e^x \left(\frac{e^{2x}+e^{-2x}}{2}\right)$$

$$= \frac{1}{2}\left[\underbrace{\frac{1}{D^2-D-6}e^{3x}}_{PI_1} + \underbrace{\frac{1}{D^2-D-6}e^{-x}}_{PI_2}\right] \quad \text{--- (1)}$$

consider PI_1

$$y_{p_1} = \frac{1}{D^2-D-6} e^{3x}$$

$\left[\begin{array}{l} a=3 \\ \text{put } D=a=3 \\ f(D)=f(a)=0 \end{array}\right]$

$$= \frac{x}{2D-1} e^{3x}$$

[put $D=a=3$]

$$= x/5\, e^{3x}$$

consider PI_2

$$y_{p_2} = \frac{1}{D^2-D-6} e^{-x}$$

$= \frac{-1}{4} e^{-x}$

$$\begin{bmatrix} a=-1 \\ \text{put } D=a=-1 \\ f(D)=f(a)=-4\neq 0 \end{bmatrix}$$

sub PI_1, PI_2 in ①

$$y_p = \frac{1}{2}\left[\frac{x}{5}e^{3x} - \frac{1}{4}e^{-x}\right]$$

$$y = y_c + y_p$$

$$= c_1 e^{-2x} + c_2 e^{3x} + \frac{x}{10}e^{3x} - \frac{1}{8}e^{-x}.$$

5, Solve $(D^2-6D+9)y = 6e^{3x} + 7e^{-3x} - \log 2$.

Sol:- To find C.F:

Given eqn is of form $f(D)y=0$

It's A.E i.e, $f(m)=0$.

$$m^2-6m+9=0.$$

$$m^2-3m-3m+9=0$$

$$m(m-3)-3(m-3)=0$$

$$(m-3)(m-3)=0$$

$$m=3,3$$

$$y_c = (c_1+c_2x)e^{3x}.$$

To find P.I:

$$y_p = \frac{1}{f(D)} X(x)$$

$$= \frac{1}{D^2-6D+9}(6e^{3x} + 7e^{-3x} - \log 2)$$

$$= \underbrace{\frac{1}{D^2-6D+9}6e^{3x}}_{PI_1} + \underbrace{\frac{1}{D^2-6D+9}7e^{-3x}}_{PI_2} - \underbrace{\frac{1}{D^2-6D+9}e^{0x}\log 2}_{PI_3} \quad ①$$

consider PI_1

$$y_{P_1} = \frac{1}{D^2-6D+9} 6e^{3x} \quad \begin{bmatrix} a=3 \\ \text{put } D=a=3 \\ f(D)=f(a)=0 \end{bmatrix}$$

$$= \frac{x}{2D-6} 6e^{3x}$$

$$= \frac{x^2}{2} 6e^{3x} = 3x^2 e^{3x}.$$

consider PI_2

$$y_{P_2} = \frac{1}{D^2-6D+9} 7e^{-3x} \quad \begin{bmatrix} a=-3 \\ \text{put } D=a=-3 \\ f(D)=f(a)\neq 0 \end{bmatrix}$$

$$= \frac{1}{36} 7e^{-3x}$$

consider PI_3

$$y_{P_3} = \frac{1}{D^2-6D+9} \log 2\, e^{0x} \quad \begin{bmatrix} a=0 \\ \text{put } D=a=0 \\ f(D)=f(a)\neq 0 \end{bmatrix}$$

$$= \frac{1}{9} \log 2\, e^{0x}$$

$$= \frac{1}{9} \log 2$$

sub PI_1, PI_2, PI_3 in ①

$$y_p = 3x^2 e^{3x} + \frac{7}{36} e^{-3x} - \frac{1}{9} \log 2$$

$$y = y_c + y_p$$

$$= [c_1 + c_2 x] e^{3x} + 3x^2 e^{3x} + \frac{7}{36} e^{-3x} - \frac{1}{9} \log 2.$$

6. solve $(D^2+2D+1)y = 2^x$

sol: <u>To find C.F:</u>

Given eq^n is of form $f(D)y=0$.

It's A.E i.e, $f(m)=0$

$$m^2+2m+1=0$$

$$m^2+m+m+1=0$$

$$m(m+1)+1(m+1)=0.$$

$$(m+1)(m+1)=0$$
$$m=-1,-1$$
$$y_c=[c_1+c_2x]e^{-x}$$

To find P.I:

$$y_p=\frac{1}{f(D)}X(x)$$
$$=\frac{1}{D^2+2D+1}2^x$$
$$=\frac{1}{D^2+2D+1}e^{\log 2^x}$$
$$=\frac{1}{D^2+2D+1}e^{x\log 2}$$
$$=\frac{1}{D^2+2D+1}e^{(\log 2)x} \qquad \begin{bmatrix} a=\log 2 \\ \text{put } D=a=\log 2 \end{bmatrix}$$
$$=\frac{1}{(\log 2)^2+2\log 2+1}e^{(\log 2)x}=\frac{1}{(1+\log 2)^2}e^{(\log 2)x}.$$

$$y=y_c+y_p$$
$$=[c_1+c_2x]e^{-x}+\frac{1}{(1+\log 2)^2}e^{(\log 2)x}.$$

Type 2:-

$f(D)y=X(x)$ where $X(x)=\sin ax$ (or) $\cos x$

$$y_p=\frac{1}{f(D)}X(x)$$
$$=\frac{1}{f(D)}\sin ax \text{ (or) } \cos ax$$
$$=\frac{1}{f(D^2)}\sin ax \text{ (or) } \cos ax=\frac{1}{f(-a^2)}\sin ax \text{ (or) } \cos ax$$
if $f(-a^2)\neq$

if $f(-a^2)=0$ then $\frac{x}{f'(D^2)}\sin ax$ (or) $\cos ax = \frac{x}{f'(-a^2)}\sin ax$ (or) $\cos ax$ if $f'(-a^2)\neq 0$

if $f'(-a^2)=0$ then $\frac{x^2}{f''(D^2)}\sin ax$ (or) $\cos ax =$

$\frac{x^2}{f''(-a^2)}\sin ax$ (or) $\cos ax$ if $f''(-a^2)\neq 0$.

similarly - - -

1) Find the P.I's of

i) $\frac{1}{D^2+1}\sin x$

Sol:- $y_p = \frac{1}{f(D)}X(x)$

$= \frac{1}{D^2+1}\sin x \quad \left[\begin{array}{l} a=1 \\ \text{put } D^2=-a^2=-1 \\ f(D^2)=f(-a^2)=0 \end{array}\right]$

$= \frac{x}{2D}\sin x$

$= \frac{x}{2}\cdot\frac{1}{D}\sin x$

$= \frac{x}{2}\int \sin x\,dx$

$= \frac{-x}{2}\cos x.$

ii) $\frac{1}{D^3+D+1}\sin x.$

Sol:- $y_p = \frac{1}{D^3+D+1}\sin x$

$= \frac{1}{\cancel{-D}+\cancel{D}+1}\sin x \quad \left[\begin{array}{l} a=1 \\ \text{put } D^2=a^2=-1 \end{array}\right]$

$= \sin x.$

iii) $\frac{1}{D^4+D+1}\sin x$

Sol:- $y_p = \frac{1}{D^4+D+1}\sin x$

$= \frac{1}{D^2 \cdot D^2 + D+1}\sin x$ $\left[\begin{array}{l} a=1 \\ D^2 = a^2 = -1. \end{array}\right]$

$= \frac{1}{(-1)(-1)+D+1}\sin x$

$= \frac{1}{1+D+1}\sin x$

$= \frac{1}{D+2}\sin x.$

$= \frac{1}{D+2} \times \frac{D-2}{D-2}\sin x$

$= \frac{D-2}{D^2-4}\sin x.$

$= \frac{(D-2)\sin x}{D^2-4}$ $\left[\begin{array}{l} a=1 \\ D^2 = a^2 = -1 \end{array}\right]$

$= \frac{(D-2)\sin x}{-1-4}$

$= \frac{(D-2)\sin x}{-5}$

$= \frac{D\sin x - 2\sin x}{-5}$

$= \frac{\cos x - 2\sin x}{-5}$

$= \frac{2\sin x - \cos x}{5}.$

2, Solve ~~(B~~ $(D^3+1)y = \cos(2x+1)$

sol:- <u>To find C.F:-</u>

Given eqn is of form $f(D)y=0$

It's A.E i.e, $f(m)=0$.

$$m^3+1=0$$

$$m^3+1^3=0$$

$$(m+1)(m^2-m+1)=0$$

$$m+1=0 \qquad m^2-m+1=0$$

$$m=-1 \qquad m=\frac{1\pm\sqrt{1-4}}{2}$$

$$=\frac{1\pm\sqrt{-3}}{2} = \frac{1}{2}\pm i\frac{\sqrt{3}}{2}$$

$$y_c = c_1e^{-x}+e^{x/2}\left[c_2\cos\frac{\sqrt{3}}{2}x + c_3\sin\frac{\sqrt{3}}{2}x\right]$$

<u>To find P.I:-</u>

$$y_p = \frac{1}{f(D)}X(x)$$

$$=\frac{1}{D^3+1}\cos(2x-1) \quad \left[\begin{array}{c}a=2\\ \text{put } D^2=-a^2=-4\end{array}\right]$$

$$=\frac{1}{-4D+1}\cos(2x-1)$$

$$=\frac{1}{1-4D}\cos(2x-1)$$

$$=\frac{1}{1-4D}\times\frac{1+4D}{1+4D}\cos(2x-1)$$

$$=\frac{(1+4D)\cos(2x-1)}{1-16D^2} \quad \left[\begin{array}{c}a=2\\ \text{put } D^2=-a^2=-4\end{array}\right]$$

$$\frac{(1+4D)\cos(2x-1)}{1-16(-4)}$$

$$= \frac{1}{65}(1+4D)\cdot\cos(2x-1)$$

$$= \frac{\cos(2x-1)+4D\cos(2x-1)}{65}$$

$$= \frac{\cos(2x-1)-8\sin(2x-1)}{65}$$

$$\therefore\ y = y_c + y_p$$

$$= c_1 e^{-x} + e^{x/2}\left[c_2\cos\frac{\sqrt{3}}{2}x + c_3\sin\frac{\sqrt{3}}{2}x\right] + \frac{\cos(2x-1)-8\sin(2x-1)}{65}$$

3. Solve $(D^3+4D)y = \sin 2x$.

Sol: <u>To find C.F:</u>

The given eqn is of form $f(D)y=0$

It's A.E i.e, $f(m)=0$

$$m^3+4m=0$$

$$m(m^2+4)=0$$

$$m=0 \qquad m^2+4=0$$

$$m^2=-4$$

$$m=\pm 2i$$

$$y_c = c_1 e^{0x} + e^{0x}[c_2\cos 2x + c_3\sin 2x]$$

<u>To find P.I:</u>

$$y_p = \frac{1}{f(D)} X(x)$$

$$= \frac{1}{D^3+4D} \sin 2x$$

$$= \frac{1}{D^2 \cdot D + 4D} \sin 2x \quad \left[\begin{array}{l} a=2 \\ \text{put } D^2 = -a^2 = -4 \end{array}\right]$$

$$= \frac{1}{-4D+4D} \sin 2x$$

$$= \frac{x}{3D^2+4} \sin 2x \quad \left[\begin{array}{l} a=2 \\ \text{put } D^2 = -a^2 = -4 \end{array}\right]$$

$$= \frac{x}{3(-4)+4} \sin 2x = \frac{x}{-8} \sin 2x$$

$$= \frac{-x}{8} \sin 2x$$

$$\therefore y = y_c + y_p$$

$$= c_1 e^{0x} + e^{0x}[c_2 \cos 2x + c_3 \sin 2x] - \frac{x}{8} \sin 2x$$

4, solve $(D^2+1)y = \cos x \cos 2x$.

sol: To find C.F:

Given eqn is of form $f(D)y = 0$

It's A.E i.e, $f(m) = 0$

$$m^2+1=0$$

$$m^2 = -1$$

$$m = \pm\sqrt{-1}$$

$$m = \pm i$$

$$y_c = e^{0x}[c_1 \cos x + c_2 \sin x]$$

To find P.I.:

$$y_p = \frac{1}{f(D)} X(x)$$

$$= \frac{1}{D^2+1} \cos x \cos 2x$$

$2 \cos A \cos B = \cos(A+B) + \cos(A-B)$

$2\cos x \cos 2x = \cos(x+2x) + \cos(x-2x)$

$= \cos 3x + \cos x$

$$= \frac{1}{D^2+1}\left[\frac{\cos 3x + \cos x}{2}\right]$$

$$= \frac{1}{2}\left[\underbrace{\frac{1}{D^2+1}\cos 3x}_{PI_1} + \underbrace{\frac{1}{D^2+1}\cos x}_{PI_2}\right] \quad \text{①}$$

$2\cos x \cos 2x = \frac{\cos 3x + \cos x}{2}$

consider PI_1

$$y_{p_1} = \frac{1}{D^2+1}\cos 3x \quad \left[\begin{array}{l} a=3 \\ \text{put } D^2 = -a^2 = -9 \end{array}\right]$$

$$= \frac{1}{-9+1}\cos 3x$$

$$= \frac{-1}{8}\cos 3x.$$

consider PI_2

$$y_{p_2} = \frac{1}{D^2+1}\cos x \quad \left[\begin{array}{l} a=1 \\ \text{put } D^2 = -a^2 = -1 \end{array}\right]$$

$$= \frac{x}{2D}\cos x$$

$$= \frac{x}{2}\int \cos x \, dx$$

$$= \frac{x}{2}\sin x.$$

sub PI_1, PI_2 in ①

$$y_p = \frac{1}{2}\left[\frac{-1}{8}\cos 3x + \frac{x}{2}\sin x\right]$$

$\therefore y = y_c + y_p$

$= e^{0x}[c_1\cos x + c_2\sin x] + \frac{1}{2}\left[\frac{-1}{8}\cos 3x + \frac{x}{2}\sin x\right].$

5. solve $(D^2 - 4D + 4)y = 8e^{2x} + 8\sin 2x$.

Sol: To find C.F:-

The given eq^n is of form $f(D)y = 0$

It's A.E i.e, $f(m) = 0$.

$$m^2 - 4m + 4 = 0$$

$$m^2 - 2m - 2m + 4 = 0$$

$$m(m-2) - 2(m-2) = 0$$

$$(m-2)(m-2) = 0$$

$$m = 2, 2.$$

$y_c = [c_1 + c_2 x]e^{2x}$.

To find P.I:-

$$y_p = \frac{1}{f(D)} X(x)$$

$$= \frac{1}{D^2 - 4D + 4}(8e^{2x} + 8\sin 2x)$$

$$= \underbrace{\frac{1}{D^2 - 4D + 2} 8e^{2x}}_{PI_1} + \underbrace{\frac{1}{D^2 - 4D + 4} 8\sin 2x}_{PI_2} \quad \text{--- (1)}$$

consider PI_1

$$y_{P_1} = \frac{1}{D^2 - 4D + 4} 8e^{2x} \quad \left[\begin{array}{l} a = 2 \\ \text{put } D = a = 2 \\ f(D) = f(a) = 0 \end{array}\right]$$

$$= \frac{x}{2D - 4} 8e^{2x} \quad [\text{put } D = a = 2]$$

$= \frac{x^2}{2} \cdot 8e^{2x}$

$= 4x^2 e^{2x}$

consider PI_2

$$y_{P_2} = \frac{1}{D^2-4D+4} 8\sin 2x \quad \left[\begin{array}{l} a=2 \\ \text{put } D^2 = -a^2 = -4 \end{array}\right]$$

$$= \frac{1}{-4D} 8\sin 2x$$

$$= \frac{-2}{D}\sin 2x$$

$$= -2\int \sin 2x \, dx$$

$$= -\cancel{2}\left(\frac{-\cos 2x}{\cancel{2}}\right)$$

$$= \cos 2x$$

sub PI_1, PI_2 in ①

$$y_p = 4x^2 e^{2x} + \cos 2x$$

$\therefore y = y_c + y_p$

$$= (c_1 + c_2 x)e^{2x} + 4x^2 e^{2x} + \cos 2x.$$

6, Solve $(D^2+9)y = \sin^2 x$.

sol:- To find C.F:-

Given eqn is of form $f(D) \cdot y = 0$.

It's A.E i.e, $f(m) = 0$.

$$m^2 + 9 = 0$$

$$m^2 = -9$$

$$m = \sqrt{-9}$$

$$= \pm 3i$$

$y_C = e^{0x}[c_1\cos 3x + c_2 \sin 3x]$

To find P.I:

$y_P = \frac{1}{f(D)} x(x)$

$= \frac{1}{D^2+9} \sin^2 x.$

$= \frac{1}{D^2+9}\left[\frac{1-\cos 2x}{2}\right].$

$= \frac{1}{2}\left[\underbrace{\frac{1}{D^2+9} 1 e^{0x}}_{PI_1} - \underbrace{\frac{1}{D^2+9}\cos 2x}_{PI_2}\right] \rightarrow (1)$

consider PI_1:

$y_{P_1} = \frac{1}{D^2+9} e^{0x}$ $\left[\begin{array}{l} a=0 \\ \text{put } D=a=0 \end{array}\right]$

$= \frac{1}{9} e^{0x} = \frac{1}{9}.$

consider PI_2

$y_{P_2} = \frac{1}{D^2+9} \cos 2x$ $\left[\begin{array}{l} a=2 \\ \text{put } D^2=-a^2=-4 \end{array}\right]$

$= \frac{1}{-4+9} \cos 2x,$

$= \frac{1}{5}\cos 2x$

sub PI_1, PI_2 in (1)

$y_P = \frac{1}{2}\left[\frac{1}{9} - \frac{1}{5}\cos 2x\right]$

$= \frac{1}{18} - \frac{1}{10}\cos 2x$

$\therefore y = y_c + y_p$

$= e^{0x}[c_1\cos 3x + c_2\sin 3x] + \frac{1}{18} - \frac{1}{10}\cos 2x.$

Note:-

1. $2\cos A\cos B = \cos(A+B) + \cos(A-B)$
2. $2\sin A\sin B = \cos(A-B) - \cos(A+B)$
3. $2\cos A\sin B = \sin(A+B) - \sin(A-B)$
4. $2\sin A\cos B = \sin(A+B) + \sin(A-B)$.

Type 3:

$f(D)y = X(x)$ where $X(x) = x^m$

$$y_p = \frac{1}{f(D)} X(x)$$

$$= \frac{1}{f(D)} x^m$$

$$= \frac{1}{1 \pm \phi(D)} x^m$$

$$= [1 \pm \phi(D)]^{-1} x^m.$$

1. Expand the terms by biomomial expansion.

working procedure:-

→ Express $f(D)$ in the form of $1+\phi(D)$ or $1-\phi(D)$ by taking the lowest degree term common from $f(D)$ to make the 1st term unity.

→ Bring this factor into numerator i.e, $[1+\phi(D)]^{-1}$ (or) $[1-\phi(D)]^{-1}$.

→ Expand the terms $[1+\phi(D)]^{-1}$ (or) $[1-\phi(D)]^{-1}$

by using binomial theorem upto the terms containing degree 'm'.

Note:

1. $(1+x)^{-1} = 1-x+x^2-x^3+x^4+\cdots$

2. $(1-x)^{-1} = 1+x+x^2+x^3+\cdots$

3. $(1+x)^{-2} = 1-2x+3x^2-4x^3+\cdots$

4. $(1-x)^{-2} = 1+2x+3x^2+4x^3+\cdots$

5. $(1+x)^{-3} = 1-3x+6x^2-10x^3+\cdots$

6. $(1-x)^{-3} = 1+3x+6x^2+10x^3+\cdots$

1. Solve $(D^2+1)y = x^4$

Sol:- To find C.F:

Given eq^n is of form $f(D)y=0$

It's A.E i.e, $f(m)=0$

$$m^2+1=0$$

$$m^2=-1$$

$$m = \pm\sqrt{-1} = \pm i.$$

$$y_c = e^{0x}[c_1\cos x + c_2\sin x]$$

To find P.I:

$$y_p = \frac{1}{f(D)} X(x)$$

$$= \frac{1}{D^2+1} x^4$$

$$= \frac{1}{1+D^2} x^4$$

$$= [1+D^2]^{-1} x^4 \quad \left[(1+x)^{-1} = 1-x+x^2-x^3+x^4+\cdots \text{ put } x=D^2 \right]$$

$$= [1 - D^2 + (D^2)^2 - (D^2)^3 + (D^2)^4 + \cdots] x^4$$

$$= [1 - D^2 + D^4] x^4 \quad [\text{neglect powers} > 4]$$

$$= x^4 - D^2(x^4) + D^4(x^4) \qquad \left[\begin{array}{l} D(x^4) = 4x^3 \\ D^2(x^4) = 12x^2 \\ D^3(x^4) = 24x \\ D^4(x^4) = 24 \end{array}\right]$$

$$= x^4 - 12x^2 + 24$$

$$\therefore y = y_c + y_p$$

$$= e^{0x}[c_1 \cos x + c_2 \sin x] + x^4 - 12x^2 + 24.$$

2, Solve $(D^3 + 3D^2 + 3D + 1) y = e^{-x} + x^2$.

Sol:- To find C.F:

The given eq^n is of form $f(D) y = 0$

It's A.E i.e, $f(m) = 0$

$$m^3 + 3m^2 + 3m + 1 = 0$$

$$(m+1)^3 = 0$$

$$\begin{array}{c|cccc} m & 1 & 3 & 3 & 1 \\ -1 & 0 & -1 & -2 & -1 \\ \hline & 1 & 2 & 1 & \boxed{0} \end{array}$$

$$(m+1)(m^2 + 2m + 1) = 0$$

$$(m+1)(m^2 + m + m + 1) = 0$$

$$m(m+1) + 1(m+1) = 0$$

$$(m+1)(m+1) = 0$$

$m = -1$ $\qquad$ $m = -1, -1$

$m = -1, -1, -1$

$$y_c = [c_1 + c_2 x + c_3 x^2] e^{-x}$$

To find P.I:-

$$y_p = \frac{1}{D^3+3D^2+3D+1}(e^{-x}+x^2)$$

$$= \underbrace{\frac{1}{D^3+3D^2+3D+1}e^{-x}}_{PI_1} + \underbrace{\frac{1}{D^3+3D^2+3D+1}x^2}_{PI_2} \quad \text{--- (1)}$$

consider PI_1

$$y_{p_1} = \frac{1}{D^3+3D^2+3D+1}e^{-x} \quad \left\{\begin{array}{l} a=-1 \\ \text{put } D=a=-1 \\ f(D)=0 \end{array}\right\}$$

$$= \frac{x}{3D^2+6D+3}e^{-x}$$

$$= \frac{x^2}{6D+6}e^{-x} \quad \left[\begin{array}{l} a=-1 \\ \text{put } D=a=-1 \end{array}\right]$$

$$= \frac{x^3}{6}e^{-x}$$

consider PI_2

$$y_{p_2} = \frac{1}{D^3+3D^2+3D+1}x^2$$

$$= \frac{1}{[1+(D^3+3D^2+3D)]}x^2$$

$$= [1+(D^3+3D^2+3D)]^{-1}x^2 \quad [(1+x)^{-1} = 1-x+x^2-x^3+\cdots]$$

$$= [1-(D^3+3D^2+3D)+(D^3+3D^2+3D)^2+\cdots]x^2$$

$$= [1-3D^2-3D+9D^2]x^2 \quad [\text{neglect powers} > 2]$$

$$= [1-3D+6D^2]x^2 \quad \left[\begin{array}{l} D(x^2)=2x \\ D^2(x^2)=2 \end{array}\right]$$

$$= x^2-3D(x^2)+6D^2(x^2)$$

$= x^2 - 3(2x) + 6(2)$

$= x^2 - 6x + 12.$

Sub PI_1, PI_2 in ①.

$$y_p = \frac{x^3}{6} e^{-x} + x^2 - 6x + 12.$$

$\therefore y = y_c + y_p.$

$$= [c_1 + c_2 x + c_3 x^2] e^{-x} + \frac{x^3}{6} e^{-x} + x^2 - 6x + 12$$

3. Solve $(D^2 + 2D + 3) y = 5x^2 + e^{3x}$

Sol: To find C.F:

The given eq^n is of form $f(D) y = 0$

It's A.E i.e, $f(m) = 0$.

$$m^2 + 2m + 3 = 0$$

$$m = \frac{-2 \pm \sqrt{4 - 12}}{2}$$

$$= \frac{-2 \pm \sqrt{-8}}{2} = \frac{-2 \pm i\sqrt{4 \times 2}}{2}$$

$$= \frac{-2 \pm i2\sqrt{2}}{2} = -1 \pm i\sqrt{2}$$

$$y_c = e^{-x} [c_1 \cos\sqrt{2}x + c_2 \cancel{\cos} \sin\sqrt{2}x]$$

To find P.I:

$$y_p = \frac{1}{f(D)} X(x)$$

$$= \frac{1}{D^2 + 2D + 3} [5x^2 + e^{3x}]$$

$$= \underbrace{\frac{1}{D^2+2D+3}(5x^2)}_{PI_1} + \underbrace{\frac{1}{D^2+2D+3} e^{3x}}_{PI_2} \quad \text{—— (1)}$$

consider PI_1:

$$y_{P_1} = \frac{1}{D^2+2D+3} 5x^2$$

$$= \frac{1}{3\left[1+\frac{D^2+2D}{3}\right]} 5x^2$$

$$= \frac{5}{3}\left[1+\left(\frac{D^2+2D}{3}\right)\right]^{-1} x^2$$

$$= \frac{5}{3}\left[1-\left(\frac{D^2+2D}{3}\right)+\left(\frac{D^2+2D}{3}\right)^2 - \cdots\right] x^2$$

$$= \frac{5}{3}\left[1-\frac{D^2}{3}-\frac{2D}{3}+\frac{4D^2}{9}\right] x^2 \quad [\text{neglect powers} > 2]$$

$$= \frac{5}{3}\left[x^2 - \frac{2}{3}D(x^2) + \frac{1}{9}D^2(x^2)\right]$$

$$= \frac{5}{3}\left[x^2 - \frac{4x}{3} + \frac{2}{9}\right]$$

consider PI_2

$$y_{P_2} = \frac{1}{D^2+2D+3} e^{3x} \quad \left[\begin{array}{l} a=3 \\ \text{put } D=a=3 \end{array}\right]$$

$$= \frac{1}{18} e^{3x}$$

sub PI_1, PI_2 in (1)

$$y_p = \frac{5}{3}\left[x^2 - \frac{4x}{3} + \frac{2}{9}\right] + \frac{1}{18} e^{3x}.$$

$\therefore y = y_c + y_p.$

$$= e^{-x}\left[c_1 \cos\sqrt{2}x + c_2 \sin\sqrt{2}x\right] + \frac{5}{3}\left[x^2 - \frac{4x}{3} + \frac{2}{9}\right] + \frac{1}{18} e^{3x}.$$

Q. Solve $(D^2-6D+9)y = 2x^2-x+3$.

Sol: To find C.F:

The given eqⁿ is of form $f(D)y=0$

It's A.E i.e, $f(m)=0$

$$m^2-6m+9=0$$

$$m^2-3m-3m+9=0$$

$$m(m-3)-3(m-3)=0$$

$$(m-3)(m-3)=0$$

$$m=3,3.$$

$$y_c = [c_1+c_2x]e^{3x}$$

To find P.I:

$$y_p = \frac{1}{D^2-6D+9} 2x^2-x+3.$$

$$= \underbrace{\frac{1}{D^2-6D+9}2x^2}_{PI_1} - \underbrace{\frac{1}{D^2-6D+9}x}_{PI_2} + \underbrace{\frac{1}{D^2-6D+9}3e^{0x}}_{PI_3} \quad \text{①}$$

consider PI_1

$$y_{p_1} = \frac{1}{D^2-6D+9} 2x^2$$

$$= \frac{1}{9\left[1+\frac{D^2-6D}{9}\right]} 2x^2$$

$$= \frac{2}{9}\left[1+\frac{D^2-6D}{9}\right]^{-1} x^2$$

$[(1+x)^{-1} = 1-x+x^2+x^3+\ldots]$

$$= \frac{2}{9}\left[1-\left(\frac{D^2-6D}{9}\right)+\left(\frac{D^2-6D}{9}\right)^2 \ldots\ldots\right]x^2$$

$$= \frac{2}{9}\left[1 - \frac{D^2}{9} + \frac{2D}{3} + \frac{4D^2}{9}\right]x^2 \quad [\text{neglect terms} > 2]$$

$$= \frac{2}{9}\left[1 + \frac{2D}{3} + \frac{D^2}{3}\right]x^2$$

$$= \frac{2}{9}\left[x^2 + \frac{2}{3}D(x^2) + \frac{1}{3}D^2(x^2)\right]$$

$$= \frac{2}{9}\left[x^2 + \frac{2}{3}(2x) + \frac{1}{3}2\right]$$

$$= \frac{2}{9}\left[x^2 + \frac{4x}{3} + \frac{2}{3}\right]$$

consider PI_2

$$y_{p_2} = \frac{1}{D^2 - 6D + 9}x$$

$$= \frac{1}{9\left[1 + \frac{D^2 - 6D}{9}\right]}x$$

$$= \frac{1}{9}\left[1 + \left(\frac{D^2 - 6D}{9}\right)\right]^{-1}x$$

$$= \frac{1}{9}\left[1 - \left(\frac{D^2 - 6D}{9}\right) + \left(\frac{D^2 - 6D}{9}\right)^2 - \dots\right]x.$$

$$= \frac{1}{9}\left[1 + \frac{6D}{9}\right]x$$

$$= \frac{1}{9}\left[x + \frac{2}{3}D(x)\right] = \frac{1}{9}\left[x + \frac{2}{3}\right].$$

consider PI_3

$$y_{p_3} = \frac{1}{D^2 - 6D + 9}3e^{0x} \quad \left[\begin{array}{l} a = 0 \\ \text{put } D = a = 0 \end{array}\right]$$

$$= \frac{1}{9}3e^{0x} = \frac{1}{3}$$

Sub PI_1, PI_2, PI_3 in ①:

$$y_p = \frac{2}{9}\left[x^2 + \frac{4x}{3} + \frac{2}{3}\right] - \frac{1}{9}\left[x + \frac{2}{3}\right] + \frac{1}{3}.$$

$$\therefore y = y_c + y_p.$$

$$= (c_1 + c_2 x)\, e^{3x} + \frac{2}{9}\left[x^2 + \frac{4x}{3} + \frac{2}{3}\right] - \frac{1}{9}\left(x + \frac{2}{3}\right) + \frac{1}{3}.$$

Type 4:

$f(D)y = X(x)$ where $X(x) = e^{ax}\, V(x)$

where $V(x) = \sin ax$ (or) $\cos ax$ (or) x^m.

$$y_p = \frac{1}{f(D)}\, X(x)$$

$$= \frac{1}{f(D)}\, e^{ax}\, V(x)$$

$$= e^{ax}\, \frac{1}{f(D+a)}\, V(x).$$

procedure:

→ To find P.I of $e^{ax} V(x)$ takeout e^{ax} to leftside of $\frac{1}{f(D)}$ and D is replaced by $D+a$. so that $f(D)$ becomes $f(D+a)$. No

→ Now operate $\frac{1}{f(D+a)} V(x)$ by using the previous cases.

1, Solve $(D^2 - 2D + 4)y = e^x \cos x$.

sol: To find C.F:

The given eq^n is of form $f(D)y = 0$

It's A.E i.e, $f(m)=0$

$$m^2-2m+4=0$$

$$m=\frac{2\pm\sqrt{4-16}}{2}$$

$$=\frac{2\pm\sqrt{-12}}{2}=\frac{2\pm i2\sqrt{3}}{2}$$

$$=1\pm i\sqrt{3}$$

$$y_c=e^x[c_1\cos\sqrt{3}x+c_2\sin\sqrt{3}x]$$

To find P.I:

$$y_p=\frac{1}{f(D)}X(x)$$

$$=\frac{1}{D^2-2D+4}e^x\cos x.$$

$$=e^x\frac{1}{(D+1)^2-2(D+1)+4}\cos x.$$

$$=e^x\frac{1}{D^2+1+\cancel{2D}-\cancel{2D}-2+4}\cos x$$

$$=e^x\frac{1}{D^2+3}\cos x.$$

$$=e^x\frac{1}{-1+3}\cos x$$

$\left[\begin{array}{l}a=1\\ \text{put } D^2=-a^2=-1\end{array}\right]$

$$=\frac{e^x}{2}\cos x.$$

$$\therefore y=y_c+y_p$$

$$=e^x[c_1\cos\sqrt{3}x+c_2\sin\sqrt{3}x]+\frac{e^x}{2}\cos x.$$

3) $(D^3-7D^2+14D-8)y=e^x\cos 2x$

Sol: To find C.F:

The given eq^n is of form $f(D)y=0$

It's A.E i.e, $f(m)=0$

$$m^3-7m^2+14m-8=0$$

$$\begin{array}{c|cccc} 1 & 1 & -7 & 14 & -8 \\ & 0 & 1 & -6 & 8 \\ \hline & 1 & -6 & 8 & \boxed{0} \end{array}$$

$$(m-1)(m^2-6m+8)=0$$

$$m=1, \quad m^2-6m+8=0$$

$$m^2-2m-4m+8=0$$

$$m(m-2)-4(m-2)=0$$

$$(m-2)(m-4)=0$$

$$m=1,2,4.$$

$$y_c=c_1e^x+c_2e^{2x}+c_3e^{4x}$$

To find P.I:

$$y_p=\frac{1}{f(D)}X(x)$$

$$=\frac{1}{D^3-7D^2+14D-8}e^x\cos 2x.$$

$$=e^x\frac{1}{(D+1)^3-7(D+1)^2+14(D+1)-8}\cos 2x.$$

$$=e^x\frac{1}{D^3+1+3D^2+3D-7D^2-7-\cancel{14D}+\cancel{14D}+14-8}\cos$$

$$e^x \frac{1}{D^3-4D^2+3D} \cos 2x$$

$$= e^x \frac{1}{(-4)D-4(-4)+3D} \cos 2x \quad \left[\begin{matrix} a=2 \\ \text{put } D^2=-a^2=-4 \end{matrix}\right]$$

$$= e^x \frac{1}{16-D} \cos 2x.$$

$$= e^x \frac{1}{16-D} \times \frac{16+D}{16+D} \cos 2x$$

$$= e^x \frac{16+D}{256-D^2} \cos 2x \quad \left[\begin{matrix}\text{put} \\ D^2=-a^2=-4\end{matrix}\right]$$

$$= e^x \frac{16+D}{256-(-4)} \cos 2x.$$

$$= e^x \left(\frac{16\cos 2x + D\cos 2x}{260}\right)$$

$$= e^x \left[\frac{16\cos 2x - 2\sin 2x}{260}\right]$$

$\therefore\ y = y_c + y_p$

$$\therefore\ y = c_1 e^x + c_2 e^{2x} + c_3 e^{4x} + e^x \left[\frac{16\cos 2x - 2\sin 2x}{260}\right]$$

3, $(D^2+2)y = e^{3x} x^2 + e^x \cos 2x.$

Sol: To find C.F:

Given eq^n is of form $f(D)y=0$

It's A.E i.e, $f(m)=0$

$m^2+2=0$

$m^2=-2$

$m=\pm\sqrt{2}i$

$y_c = e^{0x}[c_1 \cos\sqrt{2}x + c_2 \sin\sqrt{2}x]$

To find P.I:

$$y_p = \frac{1}{f(D)} X(x)$$

$$= \frac{1}{D^2+2}\left(e^{3x}x^2 + e^{x}\cos 2x\right)$$

$$= \underbrace{\frac{1}{D^2+2} e^{3x}x^2}_{PI_1} + \underbrace{\frac{1}{D^2+2} e^{x}\cos 2x}_{PI_2} \quad \text{— (1)}$$

consider PI_1:

$$y_{p_1} = \frac{1}{D^2+2} e^{3x}x^2$$

$$= e^{3x}\frac{1}{(D+3)^2+2} x^2$$

$$= e^{3x}\frac{1}{D^2+9+6D+2} x^2$$

$$= e^{3x}\frac{1}{D^2+6D+11} x^2$$

$$= e^{3x}\frac{1}{11\left[1+\frac{D^2+6D}{11}\right]} x^2$$

$$= \frac{e^{3x}}{11}\left[1+\left(\frac{D^2+6D}{11}\right)\right]^{-1} x^2$$

$$= \frac{e^{3x}}{11}\left[1-\left(\frac{D^2+6D}{11}\right)+\left(\frac{D^2+6D}{11}\right)^2 - \cdots\right] x^2$$

$$= \frac{e^{3x}}{11}\left[1-\frac{D^2}{11}-\frac{6D}{11}+\frac{36D^2}{121}\right] x^2 \quad \text{[neglect powers} > 2]$$

$$= \frac{e^{3x}}{11}\left[1-\frac{6D}{11}+\frac{25D^2}{121}\right] x^2$$

$$= \frac{e^{3x}}{11}\left[x^2 - \frac{6}{11}D(x^2) + \frac{25}{121}D^2(x^2)\right]$$

$$= \frac{e^{3x}}{11}\left[x^2 - \frac{6}{11}(2x) + \frac{25}{121}(2)\right]$$

$$= \frac{e^{3x}}{11}\left[x^2 - \frac{12x}{11} + \frac{50}{121}\right]$$

consider PI_2

$$y_{P_2} = \frac{1}{D^2+2} e^x \cos 2x$$

$$= e^x \frac{1}{(D+1)^2+2} \cos 2x$$

$$= e^x \frac{1}{D^2+3+2D} \cos 2x \qquad \left[\begin{array}{l} a=2 \\ \text{put } D^2=-a^2=-4 \end{array}\right]$$

$$= e^x \frac{1}{-4+3+2D} \cos 2x.$$

$$= \frac{e^x}{2D-1} \cos 2x.$$

$$= \frac{e^x}{2D-1} \times \frac{2D+1}{2D+1} \cos 2x$$

$$= \frac{e^x(2D+1)}{4D^2-1} \cos 2x$$

$$= \frac{e^x(2D+1)}{4(-4)-1} \cos 2x.$$

$$= \frac{e^x}{-17}(2D\cos 2x + \cos 2x)$$

$$= \frac{e^x}{-17}\left[-4\sin 2x + \cos 2x\right]$$

$$= \frac{e^x}{17}\left[4\sin 2x - \cos 2x\right]$$

y_p sub PI_1, PI_2 in ①

$$y_p = \frac{e^{3x}}{11}\left[x^2 - \frac{12x}{11} + \frac{50}{121}\right] + \frac{e^x}{17}\left[4\sin 2x - \cos 2x\right]$$

$\therefore y = y_c + y_p$

$$= e^{0x}[c_1 \cos\sqrt{2}x + c_2 \sin\sqrt{2}x] + \frac{e^{3x}}{11}\left[x^2 - \frac{12x}{11} + \frac{50}{121}\right]$$

$$+ \frac{e^x}{17}[4\sin 2x - \cos 2x]$$

4) solve $(D^2+4)y = e^x \sin^2 x$.

Sol: To find C.F:

The given eqn is of form $f(D)y = 0$

It's A.E i.e, $f(m) = 0$

$$m^2 + 4 = 0$$

$$m^2 = -4$$

$$m = \pm 2i$$

$$y_c = e^{0x}[c_1 \cos 2x + c_2 \sin 2x]$$

To find P.I:

$$y_p = \frac{1}{f(D)} X(x)$$

$$= \frac{1}{D^2+4} e^x \sin^2 x$$

$$= e^x \frac{1}{(D+1)^2+4} \sin^2 x$$

$$= e^x \frac{1}{D^2+5+2D} \sin^2 x.$$

$$= e^x \frac{1}{D^2+2D+5}\left[\frac{1-\cos 2x}{2}\right]$$

$$= \frac{e^x}{2}\Big[\underbrace{\frac{1}{D^2+2D+5} 1 e^{0x}}_{PI_1} - \underbrace{\frac{1}{D^2+2D+5} \cos 2x}_{PI_2}\Big] \quad \text{--- (1)}$$

consider PI_1

$$y_{P_1} = \frac{1}{D^2+2D+5} e^{0x} \quad \begin{bmatrix} a=0 \\ \text{put } D=a=0 \end{bmatrix}$$

$$= 1/5\, e^{0x} = 1/5$$

consider PI_2

$$y_{P_2} = \frac{1}{D^2+2D+5} \cos 2x \quad \begin{bmatrix} a=2 \\ \text{put } D^2=-a^2=-4 \end{bmatrix}$$

$$= \frac{1}{-4+2D+5} \cos 2x = \frac{1}{2D+1} \cos 2x$$

$$= \frac{1}{2D+1} \times \frac{2D-1}{2D-1} \cos 2x$$

$$= \frac{2D-1}{4D^2-1} \cos 2x$$

$$= \frac{(2D-1)}{4(-4)-1} \cos 2x \quad [\text{put } D^2=-a^2=-4]$$

$$= \frac{(2D-1)\cos 2x}{-17}$$

$$= \frac{2D\cos 2x - \cos 2x}{-17}$$

$$= \frac{-4\sin 2x - \cos 2x}{-17} = \frac{4\sin 2x + \cos 2x}{17}$$

sub PI_1, PI_2 in ①

$$y_P = \frac{e^x}{2}\left[\frac{1}{5} \mp \frac{4\sin 2x + \cos 2x}{17}\right]$$

$$\therefore y = y_c + y_p$$

$$= e^{0x}[c_1\cos 2x + c_2 \sin 2x] + \frac{e^x}{2}\left[\frac{1}{5} - \frac{4\sin 2x + \cos 2x}{17}\right]$$

5. Solve $(D^2+1)y = \sin x \sinh x$.

Sol: To find C.F.:

The given eqn is of form $f(D)y=0$.

It's A.E i.e $f(m)=0$

$$m^2+1=0$$
$$m^2=-1$$
$$m=\pm i$$

$$y_c = e^{0x}[c_1\cos x + c_2 \sin x]$$

To find P.I:

$$y_p = \frac{1}{f(D)} X(x)$$

$$= \frac{1}{D^2+1}(\sin x \sinh x)$$

$$= \frac{1}{D^2+1}\left[\sin x\left(\frac{e^x-e^{-x}}{2}\right)\right]$$

$$= \frac{1}{2}\frac{1}{D^2+1}\left[\sin x\, e^x - \sin x\, e^{-x}\right]$$

$$= \frac{1}{2}\left[\underbrace{\frac{1}{D^2+1}\sin x\, e^x}_{PI_1} - \underbrace{\frac{1}{D^2+1}\sin x\, e^{-x}}_{PI_2}\right] \quad -① $$

consider PI_1

$$y_{P_1} = \frac{1}{D^2+1} e^x \sin x$$

$$= e^x \frac{1}{(D+1)^2+1}\sin x$$

$$= e^x \frac{1}{D^2+2+2D}\sin x.$$

$$= e^{x} \frac{1}{D^2+2D+2} \sin x \qquad \left[\begin{array}{l} a=1 \\ \text{put } D^2=-a^2=-1 \end{array}\right]$$

$$= e^{x} \frac{1}{-1+2D+2} \sin x$$

$$= e^{x} \frac{1}{2D+1} \sin x$$

$$= e^{x} \frac{1}{2D+1} \times \frac{2D-1}{2D-1} \sin x$$

$$= e^{x} \frac{2D-1}{4D^2-1} \sin x \qquad [\text{put } D^2=-1]$$

$$= e^{x} \frac{2D-1}{-5} \sin x$$

$$= \frac{-e^{x}}{5} [2D \sin x - \sin x]$$

$$= \frac{-e^{x}}{5} [2 \cos x - \sin x]$$

consider PI_2

$$y_{p_2} = \frac{1}{D^2+1} e^{-x} \sin x.$$

$$= e^{-x} \frac{1}{(D-1)^2+1} \sin x$$

$$= e^{-x} \frac{1}{D^2+1-2D} \sin x$$

$$= e^{-x} \frac{1}{D^2-2D+2} \sin x \qquad \left[\begin{array}{l} a=1 \\ \text{put } D^2=-a^2=-1 \end{array}\right]$$

$$= e^{-x} \frac{1}{-1-2D+2} \sin x$$

$$= e^{-x} \frac{1}{\cancel{2D+1}\ 1-2D} \sin x.$$

$$= e^{-x} \frac{1}{\cancel{2D+1}\ 1-2D} \times \frac{2D+1}{\cancel{2D-1}\ 1+2D} \sin x$$

$$= e^{-x} \frac{2D+1}{\cancel{4D^2-1}\ 1-4D^2} \sin x \qquad [\text{put } D^2=-1]$$

$$= e^{-x}\frac{2D+1}{+5}\sin x$$

$$= e^{-x}\frac{(1+2D)\sin x}{5}$$

$$= \frac{e^{-x}.(\sin x + 2D\sin x)}{5}$$

$$= \frac{e^{-x}(\sin x + 2\cos x)}{5}$$

Sub PI_1, PI_2 in ①

$$y_p = \frac{1}{2}\left[\frac{-e^{x}}{5}(2\cos x - \sin x) - \frac{e^{-x}}{5}(\sin x + 2\cos x)\right]$$

$$\therefore y = y_c + y_p$$

$$y = e^{0x}[c_1\cos x + c_2\sin x] + \frac{1}{2}\left[\frac{-e^{x}}{5}(2\cos x - \sin x) - \frac{e^{-x}}{5}(\sin x - 2\cos x)\right].$$

** ⑥ Find P.I of $(D^2-6D+13)y = 8e^{3x}\sin 4x + 2^x + x^2$

Sol: $$y_p = \frac{1}{D^2-6D+13}[8e^{3x}\sin 4x + 2^x + x^2]$$

$$= \underbrace{\frac{1}{D^2-6D+13}8e^{3x}\sin 4x}_{PI_1} + \underbrace{\frac{1}{D^2-6D+13}2^x}_{PI_2} + \underbrace{\frac{1}{D^2-6D+13}x^2}_{PI_3}.$$

consider PI_1

$$y_{P_1} = \frac{1}{D^2-6D+13}8e^{3x}\sin 4x.$$

$$= 8e^{3x}\frac{1}{(D+3)^2-6(D+3)+13}\sin 4x$$

$$= 8e^{3x}\frac{1}{D^2+9+\cancel{6D}-\cancel{6D}-18+13}\sin 4x$$

$$= 8e^{3x}\frac{1}{D^2+4}\sin 4x \quad \left[\begin{array}{l} a=4 \\ \text{put } D^2=-a^2=-16 \end{array}\right]$$

$$= 8e^{3x}\frac{1}{-16+4}\sin 4x$$

$$= \frac{-8e^{3x}}{12}\sin 4x = \frac{-2}{3}e^{3x}\sin 4x$$

consider PI_2

$$y_{P_2} = \frac{1}{D^2-6D+13}2^x$$

$$= \frac{1}{D^2-6D+13}e^{\log 2^x}$$

$$= \frac{1}{D^2-6D+13}e^{x\log 2} \quad \left[\begin{array}{l} a=\log 2 \\ \text{put } D=a \\ \quad =\log 2 \end{array}\right]$$

$$= \frac{1}{(\log 2)^2-6\log 2+13}e^{x\log 2}$$

$$= \frac{1}{13-4\log 2}e^{x\log 2}$$

consider PI_3

$$y_{P_3} = \frac{1}{D^2-6D+13}x^2$$

$$= \frac{1}{13\left[1+\left(\frac{D^2-6D}{13}\right)\right]}x^2$$

$$= \frac{1}{13}\left[1+\left(\frac{D^2-6D}{13}\right)\right]^{-1} x^2$$

$$= \frac{1}{13}\left[1-\left(\frac{D^2-6D}{13}\right)+\left(\frac{D^2-6D}{13}\right)^2+\cdots\right]$$

$$= \frac{1}{13}\left[1-\frac{D^2}{13}+\frac{6D}{13}+\frac{36D^2}{169}\right]x^2 \quad [\text{neglect powers} > 2]$$

$$= \frac{1}{13}\left[1+\frac{6D}{13}+\frac{23D^2}{169}\right]x^2$$

$$= \frac{1}{13}\left[x^2+\frac{6}{13}D(x^2)+\frac{23}{169}D^2(x^2)\right]$$

$$= \frac{1}{13}\left[x^2+\frac{12x}{13}+\frac{46}{169}\right]$$

sub PI_1, PI_2, PI_3 in ①

$$y_p = -\frac{2}{3}e^{3x}\sin 4x + \frac{1}{13-4\log 2}e^{x\log 2} + \frac{1}{13}\left[x^2+\frac{12x}{13}+\frac{46}{169}\right]$$

__Type 5:__

$f(D)y = X(x)$ where $X(x) = x\,V(x)$

where $V(x) = \sin ax$ (or) $\cos ax$ (or) e^{ax}.

$$y_p = \frac{1}{f(D)}X(x)$$

$$= \frac{1}{f(D)}x\,V(x)$$

$$= \left[\frac{x}{f(D)} - \frac{f'(D)}{[f(D)]^2}\right]V(x)$$

1. Solve $(D^2+2D+1)y = x\cos x$

Sol: __To find C.F:__

The given eqn is of form $f(D)y=0$

It's A.E i.e, $f(m)=0$

$$m^2+2m+1=0$$

$$m^2+m+m+1=0$$

$$m(m+1)+1(m+1)=0$$

$$(m+1)(m+1)=0$$

$$m=-1,-1$$

$$y_c=[c_1+c_2x]e^{-x}.$$

To find P.I:

$$y_p=\frac{1}{f(D)}X(x)$$

$$=\frac{1}{D^2+2D+1}x\cos x.$$

$$=\left[\frac{x}{f(D)}-\frac{f'(D)}{[f(D)]^2}\right]V(x).$$

$$=\left[\frac{x}{D^2+2D+1}-\frac{2D+2}{(D^2+2D+1)^2}\right]\cos x.$$

$$=\underbrace{\frac{x}{D^2+2D+1}\cos x}_{PI_1}-\underbrace{\frac{2(D+1)}{(D^2+2D+1)^2}\cos x}_{PI_2} \quad \text{--- (1)}$$

consider PI_1

$$y_{P_1}=\frac{x}{D^2+2D+1}\cos x.$$

$$=\frac{x}{-1+2D+1}\cos x \quad \left[\begin{array}{l}a=1\\ \text{put } D^2=-a^2=-1\end{array}\right]$$

$= \frac{x}{2D}\cos x$

$= \frac{x}{2}\int \cos x \, dx = \frac{x}{2}\sin x.$

consider PI_2

$y_{P_2} = \frac{2D+2}{(D^2+2D+1)^2}\cos x$ $\left[\begin{array}{l} a=1 \\ \text{put } D^2=-a^2=-1 \end{array}\right]$

$= \frac{2(D+1)}{(-1+2D+1)^2}\cos x$

$= \frac{2(D+1)}{4D^2}\cos x.$

$= \frac{D+1}{2D^2}\cos x$ [put $D^2=-1$]

$= \frac{D+1}{-2}\cos x.$

$= \frac{D\cos x+\cos x}{-2} = \frac{-\sin x+\cos x}{-2}$

$= \frac{\sin x-\cos x}{2}$

sub PI_1, PI_2 in ①

$y_P = \frac{x}{2}\sin x - \frac{\sin x-\cos x}{2}$

$\therefore y = y_c + y_P$

$= [c_1+c_2x]e^{-x} + \frac{x}{2}\sin x - \frac{\sin x-\cos x}{2}$

2, $(D^2+9)y = x\sin 2x$

sol: To find C.F:

The given eqⁿ is of form $f(D)y=0$

it's A.F i.e, $f(m)=0$.

$$m^2+9=0$$

$$m^2=-9$$

$$m=\pm 3i$$

$$y_c = e^{0x}[c_1\cos 3x + c_2\sin 3x]$$

To find P.I:

$$y_p = \frac{1}{f(D)} x(x)$$

$$= \frac{1}{D^2+9} x\sin 2x.$$

$$= \left[\frac{x}{f(D)} - \frac{f'(D)}{[f(D)]^2}\right] V(x)$$

$$= \left[\frac{x}{D^2+9} - \frac{2D}{(D^2+9)^2}\right]\sin 2x.$$

$$= \left[\underbrace{\frac{x}{D^2+9}\sin 2x}_{PI_1} - \underbrace{\frac{2D}{(D^2+9)^2}\sin 2x}_{PI_2}\right] \text{——} (1)$$

consider PI_1

$$y_{P_1} = \frac{x}{D^2+9}\sin 2x$$

$$\left[\begin{array}{l} a=2 \\ \text{put } D^2=-a^2=-4 \end{array}\right]$$

$$= \frac{x}{-4+9}\sin 2x$$

$$= \frac{x}{5}\sin 2x.$$

consider PI_2

$$y_{P_2} = \frac{2D}{(D^2+9)^2}\sin 2x \qquad \left[\begin{array}{l} a=2 \\ \text{put } D^2=-a^2=-4 \end{array}\right]$$

$$= \frac{2D}{(-4+9)^2}\sin 2x$$

$$= \frac{2D}{25} \sin 2x$$

$$= \frac{2D\sin 2x}{25}$$

$$= \frac{2\cos 2x \times 2}{25} = \frac{4\cos 2x}{25}$$

Sub $P.I_1, P.I_2$ in ①

$$y_p = \frac{x}{5}\sin 2x \mp \frac{4\cos 2x}{25}$$

$$\therefore y = y_c + y_p$$

$$= e^{0x}\left[c_1\cos 3x + c_2\sin 3x\right] + \frac{x\sin 2x}{5} \mp \frac{4\cos 2x}{25}$$

3, Solve $(D^2-2D+1)y = xe^x\sin x$.

Sol: To find C.F:

The given eqn is of form $f(D)y = 0$

It's A.E i.e, $f(m) = 0$.

$$m^2-2m+1=0.$$

$$(m-1)^2 = 0$$

$$(m-1)(m-1) = 0$$

$$m = 1, 1$$

$$y_c = [c_1+c_2x]e^x.$$

To find P.I:

$$y_p = \frac{1}{f(D)} X(x)$$

$$= \frac{1}{D^2-2D+1} xe^x\sin x$$

$$= \frac{1}{D^2-2D+1} e^x \underbrace{x\sin x}_{V(x)}.$$

$$= e^x \frac{1}{(D+1)^2 - 2(D+1) + 1} x \sin x$$

$$= e^x \frac{1}{D^2 + 1 + 2D - 2D - 2 + 1} x \sin x$$

$$= e^x \frac{1}{D^2} x \sin x$$

$$\therefore \left[\frac{x}{f(D)} - \frac{f'(D)}{[f(D)]^2}\right] V(x)$$

$$= e^x \left[\frac{x}{D^2} - \frac{2D}{(D^2)^2}\right] \sin x$$

$$= e^x \left[\frac{x}{D^2} \sin x - \frac{2D}{D^4} \sin x\right]$$

$$\left[\begin{array}{l} a = 1 \\ \text{put } D^2 = -a^2 = -1 \end{array}\right]$$

$$= e^x \left[\frac{x}{-1} \sin x - \frac{2D}{(-1)(-1)} \sin x\right]$$

$$= e^x [-x \sin x - 2\cos x]$$

$$\therefore y = y_c + y_p$$

$$= [c_1 + c_2 x] e^x + e^x [-x \sin x - 2\cos x].$$

Type 6:-

$f(D) y = X(x)$ where $X(x) = x^n \sin ax$ (or) $x^n \cos ax$.

when $X(x) = x^n \sin ax$ (or) $x^n \cos ax$ (where $n > 1$)

consider $\frac{1}{f(D)} x^n (\cos ax + i \sin ax)$

$$= \frac{1}{f(D)} x^n e^{iax}$$

$$\boxed{= \frac{1}{f(D)} x^n \cos ax + i \frac{1}{f(D)} x^n \sin ax}$$

$$= e^{iax} \frac{1}{f(D + ia)} x^n$$

equating real and imaginary parts on b/s.

$$\frac{1}{f(D)} x^n \cos ax = \text{R.P of } e^{iax} \frac{1}{f(D+ia)} x^n.$$

$$\frac{1}{f(D)} x^n \sin ax = \text{Imaginary part of } e^{iax} \frac{1}{f(D+ia)} x^n$$

1, solve $(D^4+2D^2+1)y = x^2\cos x$.

sol: <u>To find C.F</u>:

The eqn is of form $f(D)y=0$

It's A.E i.e, $f(m)=0$

$$m^4+2m^2+1=0$$

$$(m^2)^2+2m^2+1=0$$

$$(m^2+1)^2=0$$

$$(m^2+1)(m^2+1)=0$$

$$m^2=-1 \qquad m^2=-1$$

$$m=\pm i \qquad m=\pm i$$

$$y_c = e^{0x}\left[(c_1+c_2x)\cos x + (c_3+c_4x)\sin x\right]$$

<u>To find P.I</u>:

$$y_p = \frac{1}{f(D)} X(x)$$

$$= \frac{1}{D^4+2D^2+1} x^2\cos x.$$

$$= \frac{1}{D^4+2D^2+1} x^2\left[\text{Real part of } e^{ix}\right]$$

$$= \text{Real part of } e^{ix} \frac{1}{(D+i)^4+2(D+i)^2+1} x^2,$$

$$= R.P \text{ of } e^{ix} \frac{1}{D^4 + 4D^3 i + 6D^2 (i)^2 + 4Di^3 + (i)^4 + 2D^2 + 2i^2 + 4Di + 1} x^2$$

$$= R.P \text{ of } e^{ix} \frac{1}{D^4 + \cancel{4B}4D^3 i - 6D^2 - \cancel{4Di} + 1 + 2D^2 - 2 + \cancel{4Di} + 1} x^2$$

$$= R.P \text{ of } e^{ix} \frac{1}{D^4 + 4D^3 i - 4D^2} x^2$$

$$= R.P \text{ of } e^{ix} \frac{1}{-4D^2\left[1 - \left(\frac{D^4 + 4D^3 i}{4D^2}\right)\right]} x^2$$

$$= R.P \text{ of } e^{ix} \cdot \frac{1}{-4D^2} \left[1 - \left(\frac{D^2}{4} + Di\right)\right]^{-1} x^2$$

$$= R.P \text{ of } e^{ix} \frac{1}{-4D^2} \left[1 + \left(\frac{D^2}{4} + Di\right) + \left(\frac{D^2}{4} + Di\right)^2 + \cdots\right] x^2$$

$$= R.P \text{ of } e^{ix} \frac{1}{-4D^2} \left[1 + \frac{D^2}{4} + Di + D^2 i^2\right] x^2 \quad [\text{neglect powers} > 2]$$

$$= R.P \text{ of } e^{ix} \frac{1}{-4D^2} \left[1 - \frac{3D^2}{4} + Di\right] x^2$$

$$= R.P \text{ of } \frac{e^{ix}}{-4D^2} \left[x^2 - \frac{3}{4} D^2 (x^2) + i D (x^2)\right]$$

$$= R.P \text{ of } \frac{e^{ix}}{-4D^2} \left[x^2 - \frac{6}{4} + 2ix\right]$$

$$= R.P \text{ of } \frac{e^{ix}}{-4D} \int x^2 - \frac{6}{4} + 2ix \, dx.$$

$$= R.P \text{ of } \frac{e^{ix}}{-4D} \left[x^3/3 - \frac{3}{2} x + \cancel{2} i \frac{x^2}{\cancel{2}}\right]$$

$$= R.P \text{ of } \frac{e^{ix}}{-4} \int \frac{x^3}{3} - \frac{3}{2} x + ix^2 \, dx.$$

$$= R.P \text{ of } e^{ix} \frac{1}{D^4 + 4D^3 i + 6D^2 (i)^2 + 4Di^3 + (i)^4 + 2D^2 + 2i^2 + 4Di + 1} x^2$$

$$= R.P \text{ of } e^{ix} \frac{1}{D^4 + \cancel{4B} 4D^3 i - 6D^2 - \cancel{4Di} + 1 + 2D^2 - 2 + \cancel{4Di} + 1} x^2$$

$$= R.P \text{ of } e^{ix} \frac{1}{D^4 + 4D^3 i - 4D^2} x^2$$

$$= R.P \text{ of } e^{ix} \frac{1}{-4D^2 \left[1 - \left(\frac{D^4 + 4D^3 i}{4D^2}\right)\right]} x^2$$

$$= R.P \text{ of } e^{ix} \frac{1}{-4D^2} \left\{1 - \left(\frac{D^2}{4} + Di\right)\right]^{-1} x^2$$

$$= R.P \text{ of } e^{ix} \frac{1}{-4D^2} \left[1 + \left(\frac{D^2}{4} + Di\right) + \left(\frac{D^2}{4} + Di\right)^2 + \cdots\right] x^2$$

$$= R.P \text{ of } e^{ix} \frac{1}{-4D^2} \left[1 + \frac{D^2}{4} + Di + D^2 i^2\right] x^2 \quad \left[\text{neglect powers } > 2\right]$$

$$= R.P \text{ of } e^{ix} \frac{1}{-4D^2} \left[1 - \frac{3D^2}{4} + Di\right] x^2$$

$$x^2 = R.P \text{ of } \frac{e^{ix}}{-4D^2} \left[x^2 - \frac{3}{4} D^2 (x^2) + i D (x^2)\right]$$

$$= R.P \text{ of } \frac{e^{ix}}{-4D^2} \left[x^2 - \frac{6}{4} + 2ix\right]$$

$$= R.P \text{ of } \frac{e^{ix}}{-4D} \int x^2 - \frac{6}{4} + 2ix \, dx.$$

$$= R.P \text{ of } \frac{e^{ix}}{-4D} \left[x^3/3 - \frac{3}{2} x + \cancel{2} i \frac{x^2}{\cancel{2}}\right]$$

$$= R.P \text{ of } \frac{e^{ix}}{-4} \int \frac{x^3}{3} - \frac{3}{2} x + ix^2 \, dx.$$

$= R.P \text{ of } \frac{e^{ix}}{-4}\left[\frac{x^4}{12} - \frac{3}{4}x^2 + \frac{ix^3}{3}\right]$

$= \frac{-1}{4}\left\{R.P \text{ of } (\cos x + i\sin x)\left[\frac{x^4}{12} - \frac{3}{4}x^2 + \frac{ix^3}{3}\right]\right]$

$= \frac{-1}{4}\left[\frac{x^4\cos x}{12} - \frac{3}{4}x^2\cos x - \frac{x^3}{3}\sin x\right]$

$\therefore y = y_c + y_p$

$= e^{0x}\left[(c_1 + c_2x)\cos x + (c_3 + c_4x)\sin x\right] - \frac{1}{4}\left[\frac{x^4\cos x}{12} - \frac{3}{4}x^2\cos x - \frac{x^3}{3}\sin x\right]$

***
2, solve $(D^2 - 4D + 4)y = 8x^2e^{2x}\sin 2x$.
**

sol: <u>To find C.F</u>:-

the given eqn is of form $f(D)y = 0$

It's A.E i.e, $f(m) = 0$

$m^2 - 4m + 4 = 0$

$(m-2)^2 = 0 \Rightarrow (m-2)(m-2) = 0$

$m = 2, 2$

$y_c = (c_1 + c_2x)e^{2x}$

<u>To find P.I</u>:-

$y_p = \frac{1}{f(D)} X(x)$

$= \frac{1}{D^2 - 4D + 4} 8x^2e^{2x}\sin 2x.$

$$= \frac{1}{D^2-4D+4}\, 8e^{2x}\, \frac{x^2 \sin 2x}{V(x)}$$

$$= 8e^{2x} \frac{1}{(D+2)^2-4(D+2)+4}\, x^2 \sin 2x.$$

$$= 8e^{2x} \frac{1}{D^2+4+\cancel{4D}-\cancel{4D}-\cancel{8}+\cancel{4}}\, x^2 \sin 2x.$$

$$= 8e^{2x} \frac{1}{D^2}\, x^2 \sin 2x.$$

$$= 8e^{2x} \frac{1}{D^2}\, x^2 \text{ Imaginary part of } e^{i2x}$$

$$= 8e^{2x} \text{ IP of } e^{i2x} \frac{1}{(D+2i)^2}\, x^2$$

$$= 8e^{2x} \text{ IP of } e^{i2x} \frac{1}{D^2+4Di-4}\, x^2.$$

$$= 8e^{2x} \text{ IP of } e^{i2x} \frac{1}{-4\left[1-\left(\frac{D^2+4Di}{4}\right)\right]}\, x^2$$

$$= \frac{8e^{2x}}{-4} \text{ IP of } e^{i2x}\left[1-\left(\frac{D^2}{4}+Di\right)\right]^{-1} x^2$$

$$= -2e^{2x} \text{ IP of } e^{i2x}\left[1+\left(\frac{D^2}{4}+Di\right)+\left(\frac{D^2}{4}+Di\right)^2+\cdots\right] x^2$$

$$= -2e^{2x} \text{ IP of } e^{i2x}\left[1+\frac{D^2}{4}+Di-D^2\right] x^2 \quad [\text{neglect powers} > 2]$$

$$= -2e^{2x} \text{ IP of } e^{i2x}\left[1-\frac{3D^2}{4}+Di\right] x^2$$

$$= -2e^{2x} \text{ IP of } e^{i2x}\left[x^2-\frac{3}{4}D^2(x^2)+iD(x^2)\right].$$

$$= -2e^{2x} \text{ IP of } e^{i2x}\left[x^2-\frac{3}{2}+i2x\right]$$

$$= -2e^{2x}\ \text{I.P of } (\cos 2x + i\sin 2x)\left[x^2 - \frac{3}{2} + i2x\right]$$

$$= -2e^{2x}\left[2x\cos 2x + x^2\sin 2x - \frac{3}{2}\sin 2x\right]$$

$$\therefore y = y_c + y_p$$

$$= (c_1 + c_2)e^{2x} - 2e^{2x}\left[2x\cos 2x + x^2\sin 2x - \frac{3}{2}\sin 2x\right].$$

General Method for finding the P.I:-

$$\frac{1}{D-m} X(x) = e^{mx}\int e^{-mx} X(x)\,dx.$$

This method is applicable for all kinds of functions. We know that $P.I = \frac{1}{f(D)} X(x)$. Assume that $f(D)$ is a polynomial in D and having n roots. i.e, $m_1, m_2 - - - m_n$.

$$\therefore f(D) = (D-m_1)(D-m_2) - - - (D-m_n)$$

$$\therefore y_p = \frac{1}{(D-m_1)(D-m_2)\cdots(D-m_n)} X(x)$$

$$= \left[\frac{A_1}{D-m_1} + \frac{A_2}{D-m_2} + \frac{A_3}{D-m_3} - - - + \frac{A_n}{D-m_n}\right] X(x)$$

$$= \frac{A_1}{D-m_1} X(x) + \frac{A_2}{D-m_2} X(x) + - - - + \frac{A_n}{D-m_n} X(x)$$

$$= A_1 e^{m_1(x)}\int e^{-m_1 x} X(x)\,dx + A_2 e^{m_2 x}\int e^{-m_2 x} X(x)\,dx + - - - + A_n e^{m_n x}\int e^{-m_n x} X(x)\,dx.$$

similarly, $\frac{1}{D+m} X(x) = e^{-mx}\int e^{mx} X(x)\,dx.$

1, solve $\cancel{d^2}$ $\cancel{D^2+A^2}$ $(D^2+a^2)y = \sec ax.$

Sol: To find C.F:

Given eqn is of form $f(D)y=0$

It's A.E i.e, $f(m)=0$

$m^2+a^2=0$

$m^2=-a^2$, $m=\pm ia.$

$y_c = e^{0x}[c_1\cos ax + c_2\sin ax]$

To find P.I:

$y_P = \frac{1}{f(D)} X(x)$

$= \frac{1}{D^2+a^2}\sec ax.$

$= \frac{1}{(D+ia)(D-ia)}\sec ax.$

consider $\frac{1}{(D+ia)(D-ia)}$ ← ①

By resolving into partial fractions:

$\frac{1}{(D+ia)(D-ia)} = \frac{A}{D+ia} + \frac{B}{D-ia}$ — ②

$\frac{1}{(D+ia)(D-ia)} = \frac{A(D-ia)+B(D+ia)}{(D+ia)(D-ia)}$

$1 = A(D-ia)+B(D+ia)$

put $D=ia$ put $D=-ia.$

$1 = B\,2ia$ $\qquad$ $1 = -2Aia$

$B = 1/2ia$ $\qquad$ $A = -1/2ia$.

sub (A) and (B) in (2)

$$\frac{1}{(D+ia)(D-ia)} = \frac{-1/2ia}{D+ia} + \frac{1/2ia}{D-ia}$$

$$= \frac{1}{2ia}\left[\frac{1}{D-ia} - \frac{1}{D+ia}\right]$$

$$y_p = \frac{1}{(D+ia)(D-ia)} \sec ax.$$

$$= \frac{1}{2ia}\left[\frac{1}{D-ia} - \frac{1}{D+ia}\right] \sec ax.$$

$$= \frac{1}{2ia}\left[\underset{PI_1}{\frac{1}{D-ia} \sec ax} - \underset{PI_2}{\frac{1}{D+ia} \sec ax}\right] \quad \text{--- (3)}$$

consider PI_1:

$$y_{P_1} = \frac{1}{D-ia} \sec ax \quad \left\{ \frac{1}{D-m} X(x) = e^{mx}\int e^{-mx} X(x)\, d\right.$$

$$= e^{iax}\int e^{-iax} \sec ax \, dx.$$

$$= e^{iax}\int (\cos ax - i\sin ax) \sec ax \, dx.$$

$$= e^{iax}\int (\cos ax \sec ax - i \sin ax \sec ax)\, dx.$$

$$= e^{iax}\int 1 - i\tan ax \, dx.$$

$$= e^{iax}\left[x + i\log \frac{\cos ax}{a}\right]$$

consider PI_2:

$$y_{P_2} = \frac{1}{D+ia} \sec ax.$$

$$= e^{-iax}\int e^{iax} \sec ax \, dx.$$

Replace 'i' as (-i) in PI_1

$$y_{P_2} = e^{-iax}\left\{x - i\log \frac{\cos ax}{a}\right\}.$$

sub PI_1, PI_2 in eqn ③

$$y_P = \frac{1}{2ia}\left[e^{iax}\left(x+i\log\frac{\cos ax}{a}\right) - e^{-iax}\left(x - i\log\frac{\cos ax}{a}\right)\right]$$

$$= \frac{1}{2ia^2}\left[axe^{iax} + ie^{iax}\log\cos ax - axe^{-iax} + ie^{-iax}\log\cos ax\right]$$

$$= \frac{1}{2ia^2}\left[ax\left(e^{iax} - e^{-iax}\right) + i\log\cos ax\left(e^{iax} + e^{-iax}\right)\right]$$

$$= \frac{1}{2ia^2}\left[ax\,2i\sin ax + i\log\cos ax\,(2\cos ax)\right]$$

$$= \frac{\cancel{2i}}{\cancel{2i}a^2}\left[\cancel{ax\,2i\sin ax + i\log\cos ax}\ ax\sin ax + \cos ax \log \cos ax\right]$$

$$= \frac{1}{a^2}\left[ax\sin ax + \cos ax\log\cos ax\right]$$

$$\therefore\ y = y_c + y_p$$

$$= e^{0x}\left[c_1\cos ax + c_2\sin ax\right] + \frac{1}{a^2}\left[ax\sin ax + \cos ax \log\cos ax\right]$$

3, solve $(D^2 + a^2)y = \operatorname{cosec} ax$

Sol: <u>To find C.F</u>:

It's A

The given eq^n is of form $f(D)y=0$

It's A.E i.e, $f(m)=0$

$$m^2+a^2=0$$

$$m^2=-a^2$$

$$m=\pm ia.$$

$$y_c=e^{0x}\left[c_1\cos ax+c_2\sin ax\right]$$

<u>To find P.I:</u>

$$y_p=\frac{1}{f(D)}X(x)$$

$$=\frac{1}{D^2+a^2}\operatorname{cosec}ax$$

$$=\frac{1}{(D+ia)(D-ia)}\operatorname{cosec}ax$$

consider $\dfrac{1}{(D+ia)(D-ia)}$ — ①

By resolving into partial functions

$$\frac{1}{(D+ia)(D-ia)}=\frac{A}{(D+ia)}+\frac{B}{D-ia} \quad — ②$$

$$\frac{1}{(D+ia)(D-ia)}=\frac{A(D-ia)+B(D+ia)}{(D+ia)(D-ia)}$$

$$1=A(D-ia)+B(D+ia)$$

put $\cancel{D-ia=0}$ $D=ia$

$1=B2ia$

put $D=-ia$

$1=-2Aia$

$B = 1/2ia$ $\quad\quad\quad\quad\quad\quad$ $A = -1/2ia$.

sub (A) and (B) in (2)

$$\frac{1}{(D+ia)(D-ia)} = \frac{-1/2ia}{D+ia} + \frac{1/2ia}{D-ia}$$

$$= \frac{1}{2ia}\left[\frac{1}{D-ia} - \frac{1}{D+ia}\right]$$

$$y_p = \frac{1}{(D+ia)(D-ia)} \operatorname{cosec} ax.$$

$$= \frac{1}{2ia}\left[\frac{1}{D-ia} - \frac{1}{D+ia}\right] \operatorname{cosec} ax.$$

$$= \frac{1}{2ia}\left[\underbrace{\frac{1}{D-ia}\operatorname{cosec} ax}_{PI_1} - \underbrace{\frac{1}{D+ia}\operatorname{cosec} ax}_{PI_2}\right] \text{ --- (3)}$$

Consider PI_1:

$$y_{P_1} = \frac{1}{D-ia} \operatorname{cosec} ax \left[\frac{1}{D-m} X(x) = e^{mx}\int e^{-mx} X(x)\, dx\right]$$

$$= e^{iax}\int e^{-iax} \operatorname{cosec} ax\, dx.$$

$$= e^{iax}\int (\cos ax - i\sin ax) \operatorname{cosec} ax\, dx.$$

$$= e^{iax}\int (\cos ax \operatorname{cosec} ax - i \cancel{\sin ax}\, \cancel{\operatorname{cosec} ax})\, dx.$$

$$= e^{iax}\int \cot ax - i \; dx.$$

$$= e^{iax}\left[\log \frac{\sin ax}{a} + ix\right]$$

consider PI_2:

$$y_{P_2} = \frac{1}{D+ia} \operatorname{cosec} ax$$

$$= e^{-iax} \int e^{iax} \operatorname{cosec} ax \, dx$$

Replace 'i' as '-i' in PI_1

$$y_{P_2} = e^{-iax}\left[\log \frac{\sin ax}{a} + ix\right]$$

sub PI_1, PI_2 in ③

$$y_p = \frac{1}{2ia}\left[e^{iax}\left(\log \frac{\sin ax}{a} - ix\right) - e^{-iax}\left(\log \frac{\sin ax}{a} + ix\right)\right]$$

$$= \frac{1}{2ia^2}\left[e^{iax}\left(\log \sin ax - iax\right) - e^{-iax}\left(\log \sin ax + iax\right)\right]$$

$$= \frac{1}{2ia^2}\left[e^{iax}\log \sin ax - iax e^{iax} - e^{-iax}\log \sin ax - ie^{-iax} ax\right]$$

$$= \frac{1}{2ia^2}\left[\log \sin ax\,(e^{iax} - e^{-iax}) - iax(e^{iax} + e^{-iax})\right]$$

$$= \frac{1}{2ia^2}\left[\log \sin ax\, 2i \sin ax - iax(2\cos ax)\right]$$

$$= \frac{\cancel{2i}}{\cancel{2i}a^2}\left[\sin ax \log \sin ax - ax \cos ax\right]$$

$$= \frac{1}{a^2}\left[\sin ax \log \sin ax - ax \cos ax\right]$$

$$\therefore y = y_c + y_p$$

$$= e^{0x}\left[c_1 \cos ax + c_2 \sin ax\right] + \frac{1}{a^2}\left[\sin ax \log \sin ax - ax \cos ax\right]$$

3. solve $(D^2+a^2)y = \tan ax$

Sol: To find C.F:

Given eqn is of form $f(D)y=0$

It's A.E i.e, $f(m)=0$

$m^2+a^2=0 \Rightarrow m=\pm ia$

$y_c = e^{0x}[c_1\cos ax + c_2 \sin ax]$

To find P.I:

$$y_p = \frac{1}{f(D)} X(x)$$

$$= \frac{1}{D^2+a^2} \tan ax$$

$$= \frac{1}{(D+ia)(D-ia)} \tan ax$$

$$= \frac{1}{2ia}\left[\frac{1}{D-ia} - \frac{1}{D+ia}\right] \tan ax$$

PI₁

$$= \frac{1}{2ia}\left[\underbrace{\frac{1}{D-ia}\tan ax}_{PI_1} - \underbrace{\frac{1}{D+ia}\tan ax}_{PI_2}\right] \quad (1)$$

consider PI_1:

$$y_{P_1} = \frac{1}{D-ia}\tan ax \left[\frac{1}{D-m} X(x) = e^{mx}\int e^{-mx}X(x)\,dx\right]$$

$$= e^{iax}\int e^{-iax}\tan ax\, dx$$

$$= e^{iax}\int (\cos ax - i\sin ax)\frac{\sin ax}{\cos ax}\, dx.$$

$$= e^{iax}\int\left(\sin ax - i\,\frac{\sin^2 ax}{\cos ax}\right) dx$$

$$= e^{iax}\int \sin ax - i\,\frac{(1-\cos^2 ax)}{\cos ax}\, dx$$

$$= e^{iax}\int \sin ax - i(\sec ax - \cos ax)\, dx$$

$$= e^{iax}\left[-\frac{\cos ax}{a} - i\left[\frac{\log(\sec ax + \tan ax)}{a} - \frac{\sin ax}{a}\right]\right] dx$$

$$= -\frac{e^{iax}}{a}\left[\cos ax + i\left[\log(\sec ax + \tan ax) - \sin ax\right]\right]$$

consider PI_2:

$$y_{P_2} = \frac{1}{D+ia}\tan ax$$

$$= e^{-iax}\int e^{iax}\tan ax\, dx.$$

Replace 'i' as '-i' in y_{P_1}.

$$y_{P_2} = -\frac{e^{-iax}}{a}\left[\cos ax - i\left[\log(\sec ax + \tan ax) - \sin ax\right]\right].$$

sub y_{P_1}, y_{P_2} in (1).

$$y_P = \frac{1}{2ia}\left[-\frac{e^{iax}}{a}\left(\cos ax + i\left[\log(\sec ax + \tan ax) - \sin ax\right]\right]\right.$$

$$\left.-\frac{e^{-iax}}{a}\left[\cos ax - i\left[\log(\sec ax + \tan ax) - \sin ax\right]\right]\right]$$

$$= \frac{1}{2ia^2}\left[-e^{iax}\cos ax - ie^{iax}\log(\sec ax + \tan ax) + ie^{iax}\sin ax\right.$$

$$+ e^{-iax}\cos ax - ie^{-iax}\log(\sec ax + \tan ax) + ie^{-iax}\sin ax)$$

$= \frac{1}{2ia^2}\left[-\cos ax(e^{iax}-e^{-iax}) - i\log(\sec ax+\tan ax)(e^{iax}+e^{-iax})\right.$

$\left. + \sin ax(e^{iax}+e^{-iax})\right]$

$= \frac{1}{2ia^2}\left[-\cos ax(2i\sin ax) - i\log(\sec ax+\tan ax)\, 2\cos ax + \right.$

$\left. i2\sin ax\cos ax\right]$

$= \frac{-2i}{2ia^2}\left[\cos ax\sin ax + \log(\sec ax+\tan ax)\cos ax - \right.$

$\left. \sin ax\cos ax\right]$

$= \frac{-1}{a^2}\left[\log(\sec ax+\tan ax)\cos ax\right]$

$\therefore y = y_c + y_p$

$= e^{0x}\left[c_1\cos ax + c_2\sin ax\right] - \frac{1}{a^2}\left[\log(\sec ax+\tan ax)\cos ax\right]$

4, Solve $(D^2+a^2)y = \cot ax$.

Sol: To find C.F:

Given eq^n is of form $f(D)y=0$

It's A.E i.e, $f(m)=0$.

$m^2+a^2=0 \Rightarrow m = \pm ia$.

$y_c = e^{0x}[c_1\cos ax + c_2\sin ax]$

To find P.I:

$y_p = \frac{1}{f(D)} X(x)$

$= \frac{1}{D^2+a^2}\cot ax$.

$$= \frac{1}{(D+ia)(D-ia)} \cot ax$$

$$= \frac{1}{2ia}\left[\frac{1}{D-ia} - \frac{1}{D+ia}\right]\cot ax.$$

$$= \frac{1}{2ia}\left[\underbrace{\frac{1}{D-ia}\cot ax}_{PI_1} - \underbrace{\frac{1}{D+ia}\cot ax}_{PI_2}\right] \text{———} (1)$$

consider PI_1

$$y_{P_1} = \frac{1}{D-ia}\cot ax$$

$$= e^{iax}\int e^{-iax}\cot ax\,dx.$$

$$= e^{iax}\int (\cos ax - i\sin ax)\frac{\cos ax}{\sin ax}\,dx.$$

$$= e^{iax}\int\left(\frac{\cos^2 ax}{\sin ax} - i\cos ax\right)dx.$$

$$= e^{iax}\int\frac{(1-\sin^2 ax)}{\sin ax} - i\cos ax.\,dx.$$

$$= e^{iax}\int(\operatorname{cosec} ax - \sin ax) - i\cos ax\,dx.$$

$$= e^{iax}\left[\left[\log\frac{\operatorname{cosec} ax - \cot ax}{a} + \frac{\cos ax}{a}\right] - i\frac{\sin ax}{a}\right]$$

$$= \frac{e^{iax}}{a}\left[\left[\log(\operatorname{cosec} ax - \cot ax) + \cos ax\right] - i\sin ax\right]$$

consider PI_2

$$y_{P_2} = \frac{1}{D+ia}\cot ax.$$

$$= e^{-iax}\int e^{iax}\cot ax\,dx.$$

Replace 'i' as '-i' in y_{P_1}

$$y_{P_2} = \frac{e^{-iax}}{a}\left[[\log(\operatorname{cosec} ax - \cot ax) + \cos ax] + i\sin ax\right]$$

sub y_{P_1}, y_{P_2} in ①

$$y_P = \frac{1}{2ia}\left[\left[\frac{e^{iax}}{a}\{\log(\operatorname{cosec} ax - \cot ax) + \cos ax\} - i\sin ax\right] - \left[\frac{e^{-iax}}{a}\{\log(\operatorname{cosec} ax - \cot ax) + \cos ax\} + i\sin ax\right]\right]$$

$$= \frac{1}{2ia^2}\left[e^{iax}\log(\operatorname{cosec} ax - \cot ax) + e^{iax}\cos ax - i\sin ax\, e^{iax} - e^{-iax}\log(\operatorname{cosec} ax - \cot ax) - e^{-iax}\cos ax - ie^{-iax}\sin ax\right]$$

$$= \frac{1}{2ia^2}\left[\log(\operatorname{cosec} ax - \cot ax)(e^{iax} - e^{-iax}) + \cos ax(e^{iax} - e^{-iax}) - i\sin ax(e^{iax} + e^{-iax})\right]$$

$$= \frac{1}{2ia^2}\left[\log(\operatorname{cosec} ax - \cot ax)\, 2i\sin ax + \cos ax\, 2i\sin ax - i\sin ax\, 2\cos ax\right]$$

$$= \frac{2i}{2ia^2}\left[\sin ax \log(\operatorname{cosec} ax - \cot ax) + \cancel{\sin ax \cos ax} - \cancel{\sin ax \cos ax}\right]$$

$$= \frac{1}{a^2}\left[\sin ax \log(\operatorname{cosec} ax - \cot ax)\right]$$

$$\therefore y = y_c + y_p$$

$$= e^{0x}[c_1\cos ax + c_2\sin ax] + \frac{1}{a^2}[\sin ax \log(\operatorname{cosec} ax - \cot ax)]$$

5, Solve $(D^2+3D+2)y = e^{e^x}$

Sol:- To find C.F:

Given eq^n is of form $f(D)y=0$

. It's A.F i.e, $f(m)=0$.

$$m^2+3m+2=0$$

$$m^2+2m+m+2=0$$

$$m(m+2)+1(m+2)=0$$

$$(m+2)(m+1)=0$$

$$m=-1,-2.$$

$$y_c = c_1e^{-x}+c_2e^{-2x}$$

To find P.I:-

$$y_p = \frac{1}{f(D)} X(x)$$

$$= \frac{1}{D^2+3D+2} e^{e^x}$$

$$= \frac{1}{(D+1)(D+2)} e^{e^x} \quad (1)$$

consider $\frac{1}{(D+1)(D+2)}$

i.e, $\frac{1}{(D+1)(D+2)} = \frac{A}{D+1} + \frac{B}{D+2}$

$$\frac{1}{(D+1)(D+2)} = \frac{A(D+2)+B(D+1)}{(D+1)(D+2)}$$

$$1 = A(D+2)+B(D+1)$$

put $D=-2$ put $D=-1$

$B=-1$ $A=1.$

$\therefore \frac{1}{(D+1)(D+2)} = \frac{1}{D+1} - \frac{1}{D+2}$

sub in ①.

$$y_p = \left[\frac{1}{D+1} - \frac{1}{D+2}\right] e^{e^x}$$

$$= \underbrace{\frac{1}{D+1} e^{e^x}}_{PI_1} - \underbrace{\frac{1}{D+2} e^{e^x}}_{PI_2} \quad \text{—— ②}$$

consider PI_1:-

$$y_{p_1} = \frac{1}{D+1} e^{e^x} \quad \left[\frac{1}{D+m} X(x) = e^{-mx}\int e^{mx} X(x)\,dx\right]$$

$$= e^{-x}\int e^x e^{e^x}\,dx$$

put $e^x = t$

$e^x dx = dt$

$$= e^{-x}\int e^t\,dt$$

$$= e^{-x} e^t$$

$$= e^{-x}\left[e^{e^x}\right]$$

consider PI_2:-

$$y_{p_2} = \frac{1}{D+2} e^{e^x}$$

$$= e^{-2x}\int e^{2x} e^{e^x}\,dx$$

$$= e^{-2x}\int e^x e^x e^{e^x}\,dx$$

put $e^x = t$

$e^x dx = dt$

$$= e^{-2x}\int t e^t\,dt$$

$$= e^{-2x}\left[t e^t - (1) e^t\right]$$

$= e^{-2x}\left[e^{x} e^{e^{x}} - e^{e^{x}}\right]$

$= e^{-2x} e^{e^{x}}\left[e^{x} - 1\right]$

Sub PI_1, PI_2 in (2)

$y_p = e^{-x} e^{e^{x}} - e^{-2x} e^{e^{x}}(e^{x}-1)$

$\therefore y = y_c + y_p$

$= (c_1 e^{-x} + c_2 e^{-2x}) + \left(e^{-x} e^{e^{x}} - e^{-2x} e^{e^{x}}(e^{x}-1)\right)$

Solving of linear differential equations by using method of variation of parameters:

consider the eqn $\frac{d^2y}{dx^2} + P\frac{dy}{dx} + Qy = R$.

where P, Q, R are either constants (or) functions of x.

working procedure:

→ Reduce the given D.E into standard form if necessary

→ Find the soln of $\frac{d^2y}{dx^2} + P\frac{dy}{dx} + Qy = 0$ and let the soln be $CF = c_1 u(x) + c_2 v(x)$

→ Take the P.I as $P.I = A u(x) + B v(x)$

where A and B are functions of x.

→ Now find A and B by using

$$A = -\int \frac{VR}{u\frac{dv}{dx} - v\frac{du}{dx}}\,dx \qquad B = \int \frac{UR}{u\frac{dv}{dx} - v\frac{du}{dx}}\,dx$$

→ The complete soln of given eqn

$y_s = C.S = C.F + P.I$

y $y_c = c_1 u(x) + c_2 v(x) + A u(x) + B v(x)$

$y_p =$

1. By using the method of variation of parameter

solve $\frac{d^2y}{dx^2} + y = \operatorname{cosec} x$.

Sol: Given eqn $\frac{d^2y}{dx^2} + y = \operatorname{cosec} x$ — ①

compare eqn ① with standard form

$$\frac{d^2y}{dx^2} + P\frac{dy}{dx} + Qy = R$$

$P = 0,\ Q = 1,\ \operatorname{cosec} x$.

Given eqn is of form $f(D)y = X(x)$.

To find C.F:

It's A.E i.e, $f(m) = 0$.

$m^2 + 1 = 0$

$m^2 = -1$

$m = \pm i$

$y_c = e^{0x}[c_1 \cos x + c_2 \sin x]$

$y_c = c_1 \cos x + c_2 \sin x$

$= c_1 u(x) + c_2 v(x)$

where $u(x) = \cos x$, $v(x) = \sin x$.

To find P.I:

$y_p = Au + Bv$.

$$A = -\int \frac{vR}{u\frac{dv}{dx} - v\frac{du}{dx}}\,dx.$$

$u = \cos x$, $v = \sin x$.

$\frac{du}{dx} = -\sin x$ $\quad \frac{dv}{dx} = \cos x$.

$$u\frac{dv}{dx} - v\frac{du}{dx} = \cos x(\cos x) - \sin x(-\sin x)$$
$$= \cos^2 x + \sin^2 x = 1.$$

$$A = -\int \frac{\sin x \operatorname{cosec} x}{1}\,dx.$$

$$= -\int 1\,dx = -x.$$

$$B = \int \frac{uR}{u\frac{dv}{dx} - v\frac{du}{dx}}\,dx = \int \frac{\cos x \operatorname{cosec} x}{1}\,dx$$

$$= \int \cot x\,dx = \log(\sin x)$$

$\therefore y_p = Au + Bv$

$= -x\cos x + (\log \sin x)\sin x.$

$\therefore y = y_c + y_p$

$= c_1 \cos x + c_2 \sin x - x\cos x + \sin x(\log \sin x).$

2. solve by using the method of variation of parameters $(D^2+4)y = \tan 2x$.

sol: Given eqn $(D^2+4)y = \tan 2x$ — ①

$$D^2y + 4y = \tan 2x.$$

compare eqn ① with standard form

$$\frac{d^2y}{dx^2} + P\frac{dy}{dx} + Qy = R$$

$$P=0,\ Q=4,\ R=\tan 2x.$$

Given eqn is of form $f(D)y = X(x)$

To find C.F:

it's A.E i.e, $f(m)=0$.

$$m^2+4=0$$

$$m^2=-4$$

$$m=\pm 2i.$$

$$y_c = e^{0x}[c_1\cos 2x + c_2\sin 2x]$$

$$= c_1\cos 2x + c_2\sin 2x.$$

$$= c_1 u(x) + c_2 v(x)$$

where $u(x) = \cos 2x$, $v(x) = \sin 2x$.

To find P.I:

$$y_p = Au + Bv$$

$$A = -\int \frac{vR}{u\frac{dv}{dx} - v\frac{du}{dx}}\,dx.$$

$u = \cos 2x \qquad v = \sin 2x$

$\frac{du}{dx} = -2\sin 2x \qquad \frac{dv}{dx} = 2\cos 2x$

$u\frac{dv}{dx} - v\frac{du}{dx} = \cos 2x(2\cos 2x) - \sin 2x(-2\sin 2x)$

$= 2\cos^2 2x + 2\sin^2 2x = 2(1) = 2$

$A = -\int \frac{v \sin 2x \tan 2x}{2} dx$

$= -\frac{1}{2}\int \sin 2x \frac{\sin 2x}{\cos 2x} dx$

$= \frac{-1}{2}\int \frac{\sin^2 2x}{\cos 2x} dx = \frac{-1}{2}\int \frac{1-\cos^2 2x}{\cos 2x} dx$

$= \frac{-1}{2}\int \sec 2x - \cos 2x \, dx$

$= \frac{-1}{2}\left[\log \frac{\sec 2x + \tan 2x}{2} - \frac{\sin 2x}{2}\right]$

$= \frac{-1}{4}\left[\log(\sec 2x + \tan 2x) - \sin 2x\right]$

$B = \int \frac{\cos 2x \tan 2x}{2} dx$

$\frac{-1}{2}\int \sin 2x \, dx$

$= \frac{1}{2}\left(\frac{-\cos 2x}{2}\right) = \frac{-\cos 2x}{4}$

$\therefore y_p = AU + BV$

$= \frac{-1}{4}\left[\log(\sec 2x + \tan 2x) - \sin 2x\right]\cos 2x - \frac{\cos 2x}{4}$

$\sin 2x$

$= \frac{-1}{4}\log(\sec 2x + \tan 2x)\cos 2x + \cancel{\frac{1}{4}\sin 2x\cos 2x} - \cancel{\frac{1}{4}\frac{\sin 2x}{\cos 2x}}$

$\therefore \frac{-1}{4}\log(\sec 2x + \tan 2x)\cos 2x.$

$\therefore y = y_c + y_p$

$= (c_1\cos 2x + c_2\sin 2x) - \frac{1}{4}\log(\sec 2x + \tan 2x)\cos 2x.$

3, solve $(D^2 - 2D)y = e^x \sin x.$

sol: Given eqn $(D^2-2D)y = e^x \sin x.$

$$\frac{d^2y}{dx^2} - 2\frac{dy}{dx} = e^x \sin x \quad \text{--- (1)}$$

compare (1) with standard form

$$\frac{d^2y}{dx^2} + P\frac{dy}{dx} + Qy = R.$$

$$P = -2, \; Q = 0, \; R = e^x \sin x.$$

Given eqn is of form $f(D)y = X(x)$

<u>To find C.F:</u>

It's A.E i.e, $f(m) = 0$

$m^2 - 2m = 0$

$m(m-2) = 0$

$m = 0, 2$

$y_c = c_1 e^{0x} + c_2 e^{2x}$

$= c_1 1 + c_2 e^{2x}$

$= c_1 u(x) + c_2 v(x).$

$u(x) = 1 \quad v(x) = e^{2x}$

To find P.I:

$$y_p = Au + Bv$$

$$A = -\int \frac{vR}{u\frac{dv}{dx} - v\frac{du}{dx}} dx$$

$$u = 1 \qquad v = e^{2x}$$

$$\frac{du}{dx} = 0 \qquad \frac{du}{dx} = 2e^{2x}$$

$$u\frac{dv}{dx} - v\frac{du}{dx} = 1\,2e^{2x} - e^{2x}(0)$$

$$= 2e^{2x}$$

$$A = -\int \frac{\cancel{e^{2x}}\, e^{x}\sin x}{2\cancel{e^{2x}}} dx$$

$$= \frac{-1}{2}\int e^{x}\sin x\, dx \quad \left[\int e^{ax}\sin bx = \frac{e^{ax}}{a^2+b^2}[a\sin b - b\cos bx]\right]$$

$$= \frac{-1}{2}\left[\frac{e^{x}}{1^2+1^2}(\sin x - \cos x)\right]$$

$$= \frac{-e^{x}}{4}[\sin x - \cos x]$$

$$B = \int \frac{uR}{u\frac{dv}{dx} - v\frac{du}{dx}} dx$$

$$= \int \frac{1\, e^{x}\sin x\, dx}{2e^{2x}} dx = \frac{1}{2}\int e^{-x}\sin x\, dx$$

$$= \frac{1}{2}\left[\frac{e^{-x}}{(-1)^2+1^2}(-\sin x - \cos x)\right]$$

$$= \frac{-1}{4}e^{-x}(\sin x + \cos x)$$

$\therefore y_p = Au + Bv$

$= -\frac{e^x}{4}(\sin x - \cos x)(1)\left[-\frac{1}{4}e^{-x}\sin x - \frac{1}{4}e^{-x}\cos x\right]e^{2x}$

$-\frac{e^x}{4}\sin x + \cancel{\frac{e^x}{4}\cos x} - \frac{1}{4}e^x\sin x - \cancel{\frac{e^x}{4}\cos 2x}$.

$\frac{-e^x}{2}\sin x$

$\therefore y = y_c + y_p$.

$= c_1 + c_2e^{2x} - \frac{e^x}{2}\sin x$.

Application of 2nd order L.D.E

LCR circuits:-

consider the discharge of conduction c through an induction L and resistance R. since, the voltage drop across LCR is

$$L\frac{d^2q}{dt^2},\ q/c \text{ and } R\frac{dq}{dt}$$

By kirchoff's law $L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{q}{c} = 0$.

i.e, the algebraic sum of all potential drops for any closed loop is zero.

$$L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{q}{c} = 0$$

Divide throught by 'L' on b/s.

$$\Rightarrow \frac{d^2q}{dt^2} + \frac{R}{L}\frac{dq}{dt} + \frac{1}{LC}q = 0.$$

put $R/L = 2\lambda$, $\frac{1}{LC} = \mu^2$

$$\frac{d^2q}{dt^2} + 2\lambda\frac{dq}{dt} + \mu^2 q = 0$$

$$D^2q + 2\lambda Dq + \mu^2 q = 0$$

$$(D^2 + 2\lambda D + \mu^2)q = 0.$$

$$f(D)\, q = 0$$

To find C.F:-

It's A.E i.e, $f(m) = 0$

$$m^2 + 2\lambda m + \mu^2 = 0$$

$$m = \frac{-2\lambda \pm \sqrt{4\lambda^2 - 4\mu^2}}{2}$$

$$= \frac{-2\lambda \pm 2\sqrt{\lambda^2 - \mu^2}}{2}$$

$$= -\lambda \pm \sqrt{\lambda^2 - \mu^2}.$$

Case I:-

when $\lambda > \mu$, the roots are real and distinct

$\therefore q_c = c_1 e^{m_1 t} + c_2 e^{m_2 t}$

Case 2:-

when $\lambda = \mu$, the roots are real and equal.

The soln is $q_c = [c_1 + c_2 t] e^{mt}$

case 3: when $\lambda < \mu$, the roots are complex conjugate roots. i-e, $q_c = e^{-\lambda t}[c_1 \cos pt + c_2 \sin pt]$,

$-\lambda \pm ip$ where $p^2 = \mu^2 - \lambda^2$.

1) A condenser of capacity 'c' discharged through an inductance 'L' and resistance 'R' in series and the charge 'q' at a time 't' satisfies the eq^n $L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{q}{c} = 0$. Given that $L = 0.25$ henry, $R = 250\Omega$, $C = 2\times10^{-6}F$ and that when $t = 0$, charge $q = 0.02$ coloumbs and the current $i = \frac{dq}{dt} = 0$. obtain the value of q in terms of 't'.

Sol: Given eq^n $L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{q}{c} = 0$

Divide throughout by 'L'.

$$\frac{d^2q}{dt} + \frac{R}{L}\frac{dq}{dt} + \frac{1}{LC}q = 0$$

$$\left(D^2q + \frac{R}{L}Dq + \frac{1}{LC}q\right) = 0$$

$$\left(D^2 + \frac{R}{L}D + \frac{1}{LC}\right)q = 0$$

$$f(D)q = 0$$

It's A.F i-e, $f(m) = 0$

$$m^2 + \frac{R}{L}m + \frac{1}{LC} = 0$$

$$m^2 + \frac{250}{0.25}m + \frac{1}{0.25\times2\times10^{-6}} = 0$$

$$m^2+1000m+2\times10^6=0$$

$$m=\frac{-1000\pm\sqrt{10^6-8\times10^6}}{2}$$

$$=\frac{-1000\pm1000\sqrt{8-8}}{2}$$

$$=-500\pm500\sqrt{7}i$$

$$=-500\pm1323i$$

$$q_c=e^{-500t}[c_1\cos1323t+c_2\sin1323t] \quad (1)$$

put $q=0.02$ & $t=0$ in (1)

$$0.02=c_1+0$$

$$c_1=0.02$$

Diff eqn (1) w.r.t 't'.

$$\frac{dq}{dt}=e^{-500t}\left[-c_1\sin(1323t)1323+c_2\cos(1323t)1323\right]$$

$$-500e^{-500t}[c_1\cos1323t+c_2\sin1323t] \quad (2)$$

put $\frac{dq}{dt}=i=0$ & $t=0$ in (2)

$$0=0+c_2 1323-500(0.02)$$

$$c_2(1323)=10$$

$$c_2=\frac{10}{1323}=0.007$$

sub c_1, c_2 in (1)

$$q=e^{-500t}[0.02\cos1323t+0.007\sin1323t].$$

Q.* For an LCR circuit the charge 'q' on a plate of the condenser is given by $L\frac{d^2q}{dt^2}+R\frac{dq}{dt}+\frac{q}{c}=E\sin\omega t$. where $i=\frac{dq}{dt}$. The circuit is turned to resonance so that $\omega^2=\frac{1}{LC}$. If $CR^2<4L$ and initially $q=0, i=0$. Show that $q=\frac{E}{R\omega}\left[e^{-Rt/2L}\left(\cos pt+\frac{R}{2Lp}\sin pt\right)-\cos\omega t\right]$. where $p^2=\frac{1}{LC}-\frac{R^2}{4L^2}$ and $i=\frac{E}{R}\left[\sin\omega t-\frac{1}{p\sqrt{LC}}e^{-Rt/2L}\sin pt\right]$.

Sol: Given $L\frac{d^2q}{dt^2}+R\frac{dq}{dt}+\frac{q}{c}=E\sin\omega t$

$$\frac{d^2q}{dt^2}+\frac{R}{L}\frac{dq}{dt}+\frac{1}{LC}q=\frac{E}{L}\sin\omega t$$

$$\left(D^2+\frac{R}{L}D+\frac{1}{LC}\right)q=\frac{E}{L}\sin\omega t$$

$$f(D)q=X(x)$$

To find C.F:

It's A.E i.e, $f(m)=0$

$$m^2+\frac{R}{L}m+\frac{1}{LC}=0$$

$$m=\frac{-R/L\pm\sqrt{\frac{R^2}{L^2}-4\frac{1}{LC}}}{2}$$

$$=\frac{-R/2L\pm\sqrt{\frac{R^2}{L^2}-\frac{4}{LC}}}{2}$$

$$=\frac{\frac{-R}{2L}\pm\sqrt{-\left(\frac{4}{LC}-\frac{R^2}{L^2}\right)}}{2}$$

$$=\frac{\frac{-R}{2L}\pm i\sqrt{4\left(\frac{1}{LC}-\frac{R^2}{4L^2}\right)}}{2}$$

$$= \cancel{\frac{1}{-\frac{1}{LC}+\frac{R}{L}D+\frac{1}{LC}}\frac{E}{L}}\sin\omega t \quad \left[\omega^2=\frac{1}{LC}\right]$$

$$= \frac{-R}{2L} \pm i\sqrt{\frac{1}{LC}-\frac{R^2}{4L^2}}$$

$$= \frac{-R}{2L} \pm i\sqrt{p^2} = \frac{-R}{2L} \pm ip$$

$$q_c = e^{-Rt/2L}\left[c_1\cos pt + c_2\sin pt\right].$$

To find P.I:

$$q_p = \frac{1}{f(D)}X(x)$$

$$= \frac{1}{D^2+\frac{R}{L}D+\frac{1}{LC}}\frac{E}{L}\sin\omega t$$

$$= \frac{1}{-\omega^2+\frac{R}{L}D+\frac{1}{LC}}\frac{E}{L}\sin\omega t \quad \left[\begin{array}{l} a=\omega \\ \text{put } D^2=-a^2=-\omega^2 \end{array}\right]$$

$$= \frac{1}{\cancel{-\frac{1}{LC}}+\frac{R}{L}D+\cancel{\frac{1}{LC}}}\frac{E}{L}\sin\omega t \quad \left[\omega^2=\frac{1}{LC}\right]$$

$$= \frac{E}{RD}\sin\omega t$$

$$= \frac{E}{R}\int \sin\omega t\, dt$$

$$= \frac{-E\cos\omega t}{R\omega}$$

$\therefore$ Thus G.S is $q = q_c + q_p$

$$q = e^{-Rt/2L}\left[c_1\cos pt + c_2\sin pt\right] - \frac{E}{R\omega}\cos\omega t \quad -(1)$$

put $q=0$, $t=0$ in (1).

$0 = c_1 + 0 - \frac{E}{R\omega}$

$c_1 = \frac{E}{R\omega}$

diff ① w.r.t 't'

$\frac{dq}{dt} = i = e^{-Rt/2L}\left[-c_1 p \sin pt + c_2 p \cos pt\right] - \frac{R}{2L} e^{-Rt/2L}\left[c_1 \cos pt\right.$

$\left. + c_2 \sin pt\right] + \frac{E}{R\omega}(\sin \omega t)\omega$ — ②

put $i=0$, $t=0$ in ②

$0 = c_2 p - \frac{R}{2L}(c_1)$

$= c_2 p - \frac{R}{2L} \times \frac{E}{R\omega} = 0$

$c_2 p = \frac{E}{2L\omega}$

$c_2 = \frac{E}{2Lp\omega}$

Substitute c_1, c_2 in ①

$q = e^{-Rt/2L}\left[\frac{E}{R\omega}\cos pt + \frac{E}{2Lp\omega}\sin pt\right] - \frac{E}{R\omega}\cos \omega t$

$= \frac{E}{R\omega}\left[e^{-Rt/2L}\left(\cos pt + \frac{R}{2Lp}\sin pt\right) - \cos \omega t\right]$

sub c_1, c_2 in ②

$i = e^{-Rt/2L}\left[\frac{-E}{R\omega} p \sin pt + \frac{E}{2LR\omega} p \cos pt\right] - \frac{R}{2L} e^{-Rt/2L}\left[\right.$

$\left.\frac{E}{R\omega}\cos pt + \frac{E}{2Lp\omega}\sin pt\right]$

$$= -e^{-Rt/2L}\frac{EP}{R\omega}\sin pt + \frac{e^{-Rt/2L}\cancel{E}}{2L\omega}\cos pt - e^{-Rt/2L}\frac{RE}{2L\omega R}\cos pt$$

$$-e^{-Rt/2L}\frac{RE}{2L(2LP\omega)}\sin pt + \frac{E}{R}\sin\omega t$$

$$= -e^{-Rt/2L}\frac{E}{R\omega}\sin pt\left[p + \frac{R^2}{4L^2 p}\right] + \frac{E}{R}\sin\omega t$$

$$= -e^{-Rt/2L}\frac{E}{R\omega}\sin pt\,\frac{1}{p}\left[p^2 + \frac{R^2}{4L^2}\right] + \frac{E}{R}\sin\omega t$$

$$= -e^{-Rt/2L}\frac{E}{R\omega}\sin pt\,\frac{1}{p}\left[\frac{1}{LC} - \cancel{\frac{R^2}{4L^2}} + \cancel{\frac{R^2}{4L^2}}\right] + \frac{E}{R}\sin\omega t$$

$$= -e^{-Rt/2L}\frac{E}{R}\sin pt\left[\frac{1}{\omega LCp}\right] + \frac{E}{R}\sin\omega t$$

$$= -e^{-Rt/2L}\frac{E}{R}\sin pt\left[\frac{\sqrt{LC}}{LCp}\right] + \frac{E}{R}\sin\omega t$$

$$= \frac{E}{R}\left[\sin\omega t - e^{-Rt/2L}\sin pt\,\frac{1}{p\sqrt{LC}}\right]$$

3. An uncharged condenser of capacity c is charge by appling an EMF is $E\sin\frac{t}{\sqrt{LC}}$ through leads self inductance 'L' and negligible resistance. P.T at any time 't' the charge on one of the plate is $\frac{EC}{2}\left[\sin\frac{t}{\sqrt{LC}} - \frac{t}{\sqrt{LC}}\cos\frac{t}{\sqrt{LC}}\right]$

Sol: Given eq^n $L\frac{d^2q}{dt^2} + \frac{q}{c} = E\sin\frac{t}{\sqrt{LC}}$

$$(D^2 q + \frac{q}{LC}) = \frac{E}{L} \sin \frac{t}{\sqrt{LC}}$$

$$(D^2 + \frac{1}{LC}) q = \frac{E}{L} \sin \frac{t}{\sqrt{LC}}$$

$$f(D) q = X(x).$$

To find C.F:

It's A.E i.e, $f(m) = 0$.

$$m^2 + \frac{1}{LC} = 0$$

$$m^2 = \frac{-1}{LC}$$

$$m = \pm i \frac{1}{\sqrt{LC}}$$

$$q_c = e^{0t} [c_1 \cos \frac{1}{\sqrt{LC}} t + c_2 \sin \frac{1}{\sqrt{LC}} t].$$

$$q_c = c_1 \cos \frac{1}{\sqrt{LC}} t + c_2 \sin \frac{1}{\sqrt{LC}} t$$

To find P.I:

$$q_p = \frac{1}{f(D)} X(x)$$

$$= \frac{1}{D^2 + \frac{1}{LC}} \frac{E}{L} \sin \frac{t}{\sqrt{LC}} \quad \left[\begin{array}{l} a = 1/\sqrt{LC} \\ \text{put } D^2 = -a^2 = -\frac{1}{LC} \end{array}\right]$$

$$= \frac{t}{2D} \frac{E}{L} \sin \frac{t}{\sqrt{LC}}.$$

$$= \frac{tE}{2L} \int \sin \frac{1}{\sqrt{LC}} t \, dt$$

$$= \frac{Et}{2L} \left\{ \frac{-\cos t/\sqrt{LC}}{1/\sqrt{LC}} \right\}$$

$$= \frac{-Et\sqrt{LC}}{2L} \cos t/\sqrt{LC}$$

$$= \frac{-Et\sqrt{C}}{2\sqrt{L}} \cos t/\sqrt{LC}$$

$\therefore q = q_c + q_p$

$$q = c_1 \cos\frac{1}{\sqrt{LC}}t + c_2 \sin\frac{1}{\sqrt{LC}}t - \frac{E\sqrt{C}}{2\sqrt{L}}\, t \cos\frac{t}{\sqrt{LC}} \quad \text{--- (1)}$$

put $q=0$ & $t=0$ in (1)

$$0 = c_1 + 0 - 0$$

$$c_1 = 0$$

diff eq^n (1) w.r.t 't'

$$\frac{dq}{dt} = i = -c_1\left(\sin\frac{t}{\sqrt{LC}}\right) \times \frac{1}{\sqrt{LC}} + c_2 \cos\frac{t}{\sqrt{LC}} \times \frac{1}{\sqrt{LC}} - \frac{E\sqrt{C}}{2\sqrt{L}}\Big[$$

$$t\left(-\sin\frac{t}{\sqrt{LC}} \times \frac{1}{\sqrt{LC}}\right) + (1)\cos\frac{t}{\sqrt{LC}}\Big] \quad \text{--- (2)}$$

~~sub c.~~ put $i=0$, $t=0$ in (2)

$$0 = 0 + \frac{c_2}{\sqrt{LC}} - \frac{E\sqrt{C}}{2\sqrt{L}}$$

$$\frac{c_2}{\sqrt{LC}} = \frac{E\sqrt{C}}{2\sqrt{L}}$$

$$c_2 = \frac{E\sqrt{C}\,\sqrt{\cancel{L}C}}{2\sqrt{\cancel{L}}} = \frac{EC}{2}$$

sub c_1, c_2 in (1)

$$q = 0 + \frac{EC}{2}\sin\frac{t}{\sqrt{LC}} - \frac{E\sqrt{C}}{2\sqrt{L}}\, t \cos\frac{t}{\sqrt{LC}}$$

$$q = \frac{EC}{2}\left[\sin\frac{t}{\sqrt{LC}} - \frac{t}{\sqrt{LC}}\cos\frac{t}{\sqrt{LC}}\right].$$

Unit-4

Partial Derivatives

A symbol z which has finite value for every pair of values x and y is called a function of two independent variables x and y and we write $z = f(x,y)$ (or) $\phi(x,y)$

Ex:- $z = f(x,y) = x^3+y^3+3x^2y+3xy^2+6$

* Partial Derivative :-

Let $z = f(x,y)$ be a function of two variables x and y. If we keep y as constant, x varies alone then z is a function of x only. The derivative of z w.r.t 'x' treating y as constant is called the Partial Derivative of z and is denoted by one of the following symbols

$$\frac{\partial z}{\partial x}, \frac{\partial f}{\partial x}, f_x ; D_x$$

Ily w.r.t 'y'

$$\frac{\partial z}{\partial y}, \frac{\partial f}{\partial y}, f_y , D_y$$

In General, f_x, f_y are also functions of x and y. These can be differentiated partially with respect to x and y

$$\frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial x^2} = f_{xx}$$

$$\frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial^2 f}{\partial y \partial x} = f_{yx}$$

$$\frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial x \partial y} = f_{xy}$$

$$\frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial^2 f}{\partial y^2} = f_{yy}$$

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NOTE:
Sometimes use the following notations

$\frac{dz}{dx} = p$; $\frac{dz}{dy} = q$; $\frac{d^2z}{dx^2} = r$; $\frac{d^2z}{dxdy} = s$; $\frac{\delta^2 z}{dy^2} = t$

* Find the 1st and 2nd Derivatives of $z = x^3 + y^3 - 3axy$

Sol: Given, $z = x^3 + y^3 - 3axy$

$$\frac{dz}{dx} = 3x^2 + 0 - 3ay$$

$$\frac{dz}{dy} = 0 + 3y^2 - 3ax$$

$$\frac{d^2z}{dx^2} = \frac{d}{dx}\left(\frac{dz}{dx}\right) = \frac{d}{dx}(3x^2 - 3ay) = 6x$$

$$\frac{d^2z}{dxdy} = \frac{d}{dx}\left(\frac{dz}{dy}\right) = \frac{d}{dx}(3y^2 - 3ax) = -3a$$

$$\frac{d^2z}{dy^2} = \frac{d}{dy}\left(\frac{dz}{dy}\right) = \frac{d}{dy}(3y^2 - 3ax) = 6y$$

$$\frac{d^2z}{dxdy} = \frac{d}{dy}\left(\frac{dz}{dx}\right) = \frac{d}{dy}(3x^2 - 3ay) = -3a$$

* Evaluate $\frac{dz}{dx}$ & $\frac{dz}{dy}$ if $z = \log[x^2 + y^2]$

Sol: Given $z = \log(x^2 + y^2)$

$$\frac{dz}{dx} = \frac{d}{dx}(\log(x^2+y^2)) = \frac{1}{x^2+y^2} \times 2x = \frac{2x}{x^2+y^2}$$

$$\frac{dz}{dy} = \frac{1}{x^2+y^2} \times 2y = \frac{2y}{x^2+y^2}$$

* Evaluate $\frac{dz}{dx}$ & $\frac{dz}{dy}$ if $z = x^2y - x\sin xy$

Sol: Given, $z = x^2y - x\sin xy$

$$\frac{dz}{dx} = 2xy - (x\cos xy(y) + \sin xy)$$

$$= 2xy - \sin xy - xy\cos xy$$

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$$\frac{\partial}{\partial y} = x^2 - x(\cos xy)(x)$$

$$= \boxed{x^2 - x^2\cos xy}$$

* If $v = (x^2+y^2+z^2)^{-1/2}$ Prove that $\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} = 0$.

Sol: Given $v = (x^2+y^2+z^2)^{-1/2}$

$$\frac{\partial v}{\partial x} = \frac{-1}{\cancel{2}}(x^2+y^2+z^2)^{-3/2} \times \cancel{2}x$$

$$= \Rightarrow -x(x^2+y^2+z^2)^{-3/2}$$

$$\frac{\partial^2 v}{\partial x^2} = -\left(x\left(\frac{-3}{\cancel{2}}\right)(x^2+y^2+z^2)^{-5/2} \times \cancel{2}x + (x^2+y^2+z^2)^{-3/2}\right)$$

$$= 3x^2(x^2+y^2+z^2)^{-5/2} - (x^2+y^2+z^2)^{-3/2}$$

$$\frac{\partial v}{\partial y} = \frac{-1}{\cancel{2}}(x^2+y^2+z^2)^{-3/2} \times \cancel{2}y = -y(x^2+y^2+z^2)^{-3/2}$$

$$\frac{\partial v^2}{\partial y^2} = -\left(y\left(\frac{-3}{\cancel{2}}\right)(x^2+y^2+z^2)^{-5/2} \times \cancel{2}y + (x^2+y^2+z^2)^{-3/2}\right)$$

$$= 3y^2(x^2+y^2+z^2)^{-5/2} - (x^2+y^2+z^2)^{-3/2}$$

$$\frac{\partial v}{\partial z} = \frac{-1}{\cancel{2}}(x^2+y^2+z^2)^{-3/2} \times \cancel{2}z = -z(x^2+y^2+z^2)^{-3/2}$$

$$\frac{\partial^2 v}{\partial z^2} = -\left(z\left(\frac{-3}{\cancel{2}}\right)(x^2+y^2+z^2)^{-5/2} \times \cancel{2}z + (x^2+y^2+z^2)^{-3/2}\right)$$

$$= 3z^2(x^2+y^2+z^2)^{-5/2} - (x^2+y^2+z^2)^{-3/2}$$

$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} = 3x^2(x^2+y^2+z^2)^{-5/2} - (x^2+y^2+z^2)^{-3/2}$$
$$+ 3y^2(x^2+y^2+z^2)^{-5/2} - (x^2+y^2+z^2)^{-3/2}$$
$$+ 3z^2(x^2+y^2+z^2)^{-5/2} - (x^2+y^2+z^2)^{-3/2}$$

$$= (x^2+y^2+z^2)^{-5/2}(3x^2+3y^2+3z^2) - 3(x^2+y^2+z^2)^{-3/2}$$

$$= 3\cancel{(x^2+y^2+z^2)}^{-3/2} - 3\cancel{(x^2+y^2+z^2)}^{-3/2} = 0 \quad \therefore \boxed{\text{Proved}}$$

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NOTE :-

1) if $\frac{d^2v}{dx^2} = \frac{d^2v}{dx^2} + \frac{d^2v}{dy^2} + \frac{d^2v}{dz^2} = 0$, then the equation is said to be Laplacian Equation

2) A function be satisfying the laplace equation then the function is said to be harmonic function

Homogeneous Function :-

If $f(kx, ky) = k^n f(x, y)$ then $f(x, y)$ is called f. homogeneous function of degree 'n'

Euler's theorem :-

If 'u' is a homogeneous function of degree 'n' in x and y Then $x\frac{dv}{dx} + y\frac{dv}{dy} = nu$ where n is the degree of the function.

1. Show that $x\frac{dv}{dx} + y\frac{dv}{dy} = 2u\log u$ .. where $\log u = \frac{x^3+y^3}{3x+4y}$

Sol Given $x\frac{dv}{dx} + y\frac{\partial v}{dy} = 2u\log u$.

Let $f(x,y) = \frac{x^3+y^3}{3x+4y} = f$

$$f(kx, ky) = \frac{k^3x^3 + k^3y^3}{3kx+4ky}$$

$$= \frac{k^{2}(x^3+y^3)}{k(3x+4y)}$$

$$= k^2 f(x, y)$$

It is homogeneous function of degree '2'

By Eulers Theorem :-

$$x\frac{du}{dx} + y\frac{dv}{dy} = nf.$$

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$$x\frac{d}{dx}(\log u) + y\frac{d}{dy}(\log u) = 2\log u$$

$$x\frac{1}{u}\frac{du}{dx} + y\frac{1}{u}\frac{du}{dy} = 2\log u$$

$$\therefore \boxed{x\frac{du}{dx} + y\frac{du}{dy} = 2u\log u}$$

Show that $x\frac{du}{dx} + y\frac{du}{dy} = \tan u$ where $u = \sin^{-1}\left(\frac{x^2+y^2}{x+y}\right)$

Given $x\frac{du}{dx} + y\frac{du}{dy} = \tan u$ $\qquad \sin u = \frac{x^2+y^2}{x+y}$

$$f(x,y) = \sin^{-1}\left(\frac{x^2+y^2}{x+y}\right)$$

$$f(kx,ky) = \sin^{-1}\left(\frac{k^2x^2+k^2y^2}{kx+ky}\right)$$

$$= \sin^{-1}\left(\frac{k^2(x^2+y^2)}{k(x+y)}\right)$$

$$= \sin^{-1} k\, f(x,y)$$

It is homogeneous function of degree '1'

By Euler's theorem:-

$$x\frac{du}{dx} + y\frac{du}{dy} = nf$$

$$x\frac{d}{dx}(\sin u) + y\frac{d}{dy}(\sin u) = f$$

$$x\cos u\frac{du}{dx} + y\cos u\frac{du}{dy} = \sin u$$

$$\boxed{x\frac{du}{dx} + y\frac{du}{dy} = \tan u}$$

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3. If $\sin u = \frac{x^2y^2}{x+y}$ P.T $x\frac{du}{dx} + y\frac{du}{dy} = 3\tan u$

Sol $\sin u = \frac{x^2y^2}{x+y}$

$f(kx, ky) = \frac{k^2x^2y^2(k^2)}{kx+ky}$

$= \frac{k^4(x^2y^2)}{k(x+y)}$

$= \frac{k^3(x^2y^2)}{x+y}$

$= k^3 f(x,y)$

It is Homogeneous function of Degree '3'

By Euler's theorem,

$x\frac{du}{dx} + y\frac{du}{dy} = nf$

$x\frac{d}{dx}(\sin u) + y\frac{d}{dy}(\sin u) = 3\sin u$

$x\cos u\frac{du}{dx} + y\cos u\frac{du}{dy} = 3\sin u$

$\therefore \boxed{x\frac{du}{dx} + y\frac{du}{dy} = 3\tan u}$

4. If $u = \sin^{-1}\left(\frac{x+2y+3z}{x^8+y^8+z^8}\right)$ Find the Value of $x\frac{du}{dx} + y\frac{du}{dy} + z\frac{du}{dz}$?

Sol: Given.

$u = \sin^{-1}\left(\frac{x+2y+3z}{x^8+y^8+z^8}\right)$

$\sin u = \frac{x+2y+3z}{x^8+y^8+z^8}$

$f(x,y) = \frac{x+2y+3z}{x^8+y^8+z^8}$

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$$f(kx, ky) = \frac{kx+2ky+3kz}{k^8x^8+k^8y^8+k^8z^8}$$

$$= \frac{k}{k^8}\left[\frac{x+2y+3z}{x^8+y^8+z^8}\right]$$

$$= k^{-7}\left(\frac{x+2y+3z}{x^8+y^8+z^8}\right)$$

It is homogeneous function of Degree (-7)

By Euler's theorem;

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = nf.$$

$$x\frac{\partial}{\partial x}(\sin u) + y\frac{\partial}{\partial y}(\sin u) = -7\sin u$$

$$x\cos u\frac{\partial u}{\partial x} + y\cos u\frac{\partial u}{\partial y} = -7\sin u$$

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = -7\tan u.$$

5: Show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 2u\log u$ where $u = e^{x^2+y^2}$

Sol: Given,

$$u = e^{x^2+y^2}$$

$$\log u = \log e^{x^2+y^2} = x^2+y^2$$

$$f(kx, ky) = k^2x^2+k^2y^2$$

$$= k^2(x^2+y^2)$$

$$= k^2 f(x, y)$$

It is homogeneous function of Degree '2'

By Euler's theorem.

$$x\frac{\partial u}{\partial x} + y\cdot\frac{\partial u}{\partial y} = nf.$$

$$x\frac{\partial}{\partial x}(\log u) + y\frac{\partial}{\partial x}(\log u) = 2\log u$$

$$x\frac{1}{u}\frac{\partial u}{\partial x} + y\frac{1}{u}\frac{\partial u}{\partial y} = 2\log u.$$

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 2u\log u.$$

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MODEL-PAPER

solutions

1) a) Find the rank of the Matrix $\begin{bmatrix} 1 & 2 & 1 & 0 \\ -2 & 4 & 3 & 0 \\ 1 & 0 & 2 & -8 \end{bmatrix}$ by reducing it to canonical form.

Given $A = \begin{bmatrix} 1 & 2 & 1 & 0 \\ -2 & 4 & 3 & 0 \\ 1 & 0 & 2 & -8 \end{bmatrix}$

$R_2 \to R_2 + 2R_1 ; R_3 \to R_3 - R_1$

$\sim \begin{bmatrix} 1 & 2 & 1 & 0 \\ 0 & 8 & 5 & 0 \\ 0 & -2 & 1 & -8 \end{bmatrix}$

$C_2 \to C_2 - 2C_1$

$C_3 \to C_3 - C_1$

$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 8 & 5 & 0 \\ 0 & -2 & 1 & -8 \end{bmatrix}$

$R_3 \to 4R_3 + R_2$

$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 8 & 5 & 0 \\ 0 & 0 & 9 & -32 \end{bmatrix}$

~~$C_2 \to 8C$~~

$C_3 \to 8C_3 - 5C_2$

$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 8 & 0 & 0 \\ 0 & 0 & 72 & -32 \end{bmatrix}$

$C_3 \to C_3/72 \quad C_2 \to C_2/8$

$C_4 \to -C_4/72$

$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$

$C_4 \to C_4 - C_3$

$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$

$\therefore [I_3, 0]$

$\rho(A) = 3$

b) Solve the system of equations $x+3y-2z=0$, $2x-y+4z=0$, $x-11y+14z=0$

The given system of ~~giv~~ equations in matrix notation be

$$\begin{bmatrix} 1 & 3 & -2 \\ 2 & -1 & 4 \\ 1 & -11 & 14 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$A \cancel{X} = 0.$

consider $[A] = \begin{bmatrix} 1 & 3 & -2 \\ 2 & -1 & 4 \\ 1 & -11 & 14 \end{bmatrix}$

Reduce $[A]$ into Echelon form by using Row operations

$R_2 \to R_2 - 2R_1$; $R_3 \to R_3 - R_1$

$$\sim \begin{bmatrix} 1 & 3 & -2 \\ 0 & -7 & 8 \\ 0 & -14 & 16 \end{bmatrix}$$

$R_3 \to R_3/2$

$$\sim \begin{bmatrix} 1 & 3 & -2 \\ 0 & -7 & 8 \\ 0 & 7 & -8 \end{bmatrix}$$

$R_3 \to R_3 + R_2$

$$\sim \begin{bmatrix} 1 & 3 & 2 \\ 0 & -7 & 8 \\ 0 & 0 & 0 \end{bmatrix}$$

$\rho(A) = 2 \quad n = 3$

$r < n$ The system posses non-trivial sol^n & $\exists\, n-r = 3-2 = 1$ Linearly independent sol^n

$$\begin{bmatrix} 1 & 3 & -2 \\ 0 & -7 & 8 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$x+3y-2z=0$

$-7y+8z=0$

$0=0$

Let $z=k$

$-7y=-8z$

$-7y=-8k$

$y=\frac{8}{7}k.$

$x+3y-2z=0$

$x=-3y+2z$

$=-3\left(\frac{8}{7}\right)k+2k$

$=-\frac{24}{7}k+\frac{2k}{1}$

$=\cancel{\frac{-34k}{7}}$ $=\frac{-24k+14k}{7}$

$=-10k/7$

$$\therefore X=\begin{bmatrix}x\\y\\z\end{bmatrix}=\begin{bmatrix}-10k/7\\ \frac{8k}{7}\\ k\end{bmatrix}=k\begin{bmatrix}-10/7\\8/7\\1\end{bmatrix}$$

By giving different values to 'k' we get infinite no. of solutions.

2) a) Test for consistency and solve the equations $5x+3y+7z=4$, $3x+26y+2z=9$, $7x+2y+10z=5$

The given system of equations in matrix form be

$$\begin{bmatrix}5&3&7\\3&26&2\\7&2&10\end{bmatrix}\begin{bmatrix}x\\y\\z\end{bmatrix}=\begin{bmatrix}4\\9\\5\end{bmatrix}$$

$A\ X=B$

consider $[A/B]=\begin{bmatrix}5&3&7&:&4\\3&26&2&:&9\\7&2&10&:&5\end{bmatrix}$

Reduce [A] in to Echelon form using Row operations

$R_2\to 5R_2-3R_1$

$R_3\to 5R_3-7R_1$

$26 \times 5 = 130$

$10-21$

45

$10-21$

$25-63=38$

56

$$\sim \begin{bmatrix} 5 & 3 & 7 & : & 4 \\ 0 & 121 & -11 & : & 33 \\ 0 & -11 & \cancel{38} & : & 38 \end{bmatrix}$$

$R_3 \to \cancel{R_3} - 11R_3 + R_2$

$$\sim \begin{bmatrix} 5 & 3 & 7 & : & 4 \\ 0 & 121 & -11 & : & 33 \\ 0 & 0 & 385 & : & \end{bmatrix}$$

$$= \sim \begin{bmatrix} 5 & 3 & 7 & : & 4 \\ 0 & 121 & -11 & : & 33 \\ 0 & -11 & 1 & : & -3 \end{bmatrix}$$

$R_3 \to 11R_3 + R_2$

$$\sim \begin{bmatrix} 5 & 3 & 7 & : & 4 \\ 0 & 121 & -11 & : & 33 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$$

$R_2 \to R_2/11$

$$\sim \begin{bmatrix} 5 & 3 & 7 & : & 4 \\ 0 & 11 & -1 & : & 3 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$$

$\rho(A) = r = 2$

$\rho(AB) = 2$

$\therefore \rho(A) = \rho(AB)$

It is consistent

The system posses sol'.

$r = 2 \quad n = 3$

$r < n$.

$\exists\, n - r = 3 - 2 = 1$ orbitary constant

The system reduces to

$$\begin{bmatrix} 5 & 3 & 7 & : & 4 \\ 0 & 11 & -1 & : & 3 \\ 0 & 0 & 0 & : & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$\cancel{5x+3y}$

$$\begin{bmatrix} 5 & 3 & 7 \\ 0 & 11 & -1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \\ 0 \end{bmatrix}$$

$5x + 3y + 7z = 4$

$11y - z = 3$

$0 = 0$

Let $z = k$

$11y = 3 + z$

$11y = 3 + k$

$y = \frac{3+k}{11}$

$5x + 3y + 7z = 4$

$5x = 4 - 3y - 7z$

$= 4 - 3\left(\frac{3+k}{11}\right) - 7(k)$

$= \frac{4}{1} - \frac{(9+3k)}{11} - \frac{7k}{1}$

$= \frac{44 - 9 - 3k - 77k}{11}$

$5x = \frac{35 - 80k}{11}$

$x = \frac{35 - 80k}{11 \times 5} = \frac{\cancel{5}(7 - 16k)}{11 \times \cancel{5}}$

$= \frac{7 - 16k}{11}$

$x+3y+2z=0$

b) Solve the system of equations by using Gauss-Jordan method

$$9x-2y+z-t=50$$
$$x-7y+3z+t=20$$
$$2x+2y+7z+2t=22$$
$$x+y-2z+6t=18$$

The given system of equations in matrix notation be

$$\begin{bmatrix} 9 & -2 & 1 & -1 \\ 1 & -7 & 3 & 1 \\ 2 & 2 & 7 & 2 \\ 1 & 1 & -2 & 6 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ t \end{bmatrix} = \begin{bmatrix} 50 \\ 20 \\ 22 \\ 18 \end{bmatrix}$$

$$A \quad X = B$$

The augmented matrix

$$[A|B] = \left[\begin{array}{cccc|c} 9 & -2 & 1 & -1 & 50 \\ 1 & -7 & 3 & 1 & 20 \\ 2 & 2 & 7 & 2 & 22 \\ 1 & 1 & -2 & 6 & 18 \end{array}\right]$$

$R_1 \leftrightarrow R_2$

$$\sim \left[\begin{array}{cccc|c} 1 & -7 & 3 & 1 & 20 \\ 9 & -2 & 1 & -1 & 50 \\ 2 & 2 & 7 & 2 & 22 \\ 1 & 1 & -2 & 6 & 18 \end{array}\right]$$

$R_2 \to R_2 - 9R_1$; $R_3 \to R_3 - 2R_1$ $R_4 \to R_4 - R_1$

$$\sim \left[\begin{array}{cccc:c} 1 & -7 & 3 & 1 & 20 \\ 0 & 61 & -26 & -10 & -130 \\ 0 & 16 & -1 & 0 & -18 \\ 0 & 8 & -5 & 5 & -2 \end{array}\right]$$

7-6 18 : 22-40

$R_3 \to R_3 - 2R_4$

$$\sim \left[\begin{array}{cccc:c} 1 & -7 & 3 & 1 & 20 \\ 0 & 61 & -26 & -10 & -130 \\ 0 & 0 & 11 & -10 & -14 \\ 0 & 8 & -5 & 5 & -2 \end{array}\right]$$

~~$R_4 \to R_1$~~ $R_1 \to 8R_1 + 7R_4$

$$\sim \left[\begin{array}{cccc:c} 8 & 0 & -11 & 43 & 146 \\ 0 & 61 & -26 & -10 & -130 \\ 0 & 0 & 11 & -10 & -14 \\ 0 & 8 & -5 & 5 & -2 \end{array}\right]$$

$R_4 \to 61R_4 - 8R_2$

$$\sim \left[\begin{array}{cccc:c} 8 & 0 & -11 & 43 & 146 \\ 0 & 61 & -26 & -10 & -130 \\ 0 & 0 & 11 & -10 & -14 \\ 0 & 0 & -97 & 385 & 918 \end{array}\right]$$

$R_1 \to R_1 + R_3$

$$\sim \left[\begin{array}{ccccc} 8 & 0 & 0 & 33 & 132 \\ 0 & 61 & -26 & -10 & -130 \\ 0 & 0 & 11 & -10 & -14 \\ 0 & 0 & -97 & -385 & 918 \end{array}\right]$$

-26-11 = -37

-130 + 14 = -116

$R_2 \to R_2 - R_3$

$$\sim \left[\begin{array}{cccc:c} 8 & 0 & 0 & 33 & 132 \\ 0 & 61 & -37 & 0 & -116 \\ 0 & 0 & 11 & -10 & -14 \\ 0 & 0 & -97 & -385 & 918 \end{array}\right]$$

~~$R_2 \to 11R_2 + 37R_3$~~

~~$\sim \left[\begin{array}{ccccc} 8 & 0 & 0 & 33 & 132 \\ 0 & 671 & 0 & 370 & \end{array}\right]$~~

$R_4 \to 11R_4 + 97R_3$

$$\sim \left[\begin{array}{cccc:c} 8 & 0 & 0 & 33 & 132 \\ 0 & 61 & -37 & 0 & -116 \\ 0 & 0 & 11 & -10 & -14 \\ 0 & 0 & 0 & 37455 & 88,892 \end{array}\right]$$

146 - 14 = 132

9 -2 1 -1 50

0 -2 + 63 1-27 -1-9 50-180

9

67 -110 37345

61 × 11

-154 + 89046

problem wrong

3) a) Determine the Eigen values and Eigen vectors of the matrix $A = \begin{bmatrix} 3 & 1 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 5 \end{bmatrix}$

Given matrix $[A] = \begin{bmatrix} 3 & 1 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 5 \end{bmatrix}$

The characterstic matrix $(A-\lambda I)$

$$= \begin{bmatrix} 3-\lambda & 1 & 4 \\ 0 & 2-\lambda & 6 \\ 0 & 0 & 5-\lambda \end{bmatrix} \quad - (1)$$

The characterstic equ $|A-\lambda I| = 0$

$$\begin{vmatrix} 3-\lambda & 1 & 4 \\ 0 & 2-\lambda & 6 \\ 0 & 0 & 5-\lambda \end{vmatrix} = 0$$

$$(3-\lambda)\left[(2-\lambda)(5-\lambda)-0\right] - 1\left[0\right] + 4\left[0\right] = 0$$

$$(3-\lambda)(2-\lambda)(5-\lambda) = 0$$

$$\lambda = 2, 3, 5$$

<u>case 1</u>: The Eigen value corresponding to Eigen vector $\lambda = 2$ is in (1).

$$(A-\lambda I)X = 0$$

$$(A-2I)X = 0$$

$$\begin{bmatrix} 1 & 1 & 4 \\ 0 & 0 & 6 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$A \ X = 0$$

$[A] = \begin{bmatrix} 1 & 1 & 4 \\ 0 & 0 & 6 \\ 0 & 0 & 3 \end{bmatrix}$ Reduce $[A]$ into Echelon form by using Row-operations

$R_3 \to 6R_3 - 3R_2$

$$\sim \begin{bmatrix} 1 & 1 & 4 \\ 0 & 0 & 6 \\ 0 & 0 & 0 \end{bmatrix}$$

$r = 2 \quad n = 3$

$r < n \ \exists \ n - r = 3 - 2 = 1$ L.I sol.

$$\begin{bmatrix} 1 & 1 & 4 \\ 0 & 0 & 6 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$x + y + 4z = 0$

$6z = 0$

$0 = 0$

Let $y = k$

$z = 0$

$x = -y - 4z$

$= -k$

$$\therefore X_1 = \begin{bmatrix} -k \\ k \\ 0 \end{bmatrix} = k \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

Case 2: put $\lambda = 3$ in ①

$(A - \lambda I)X = 0$

$(A - 3I)X = 0$

$$\begin{bmatrix} 0 & 1 & 4 \\ 0 & -1 & 6 \\ 0 & 0 & -2 \end{bmatrix}$$

$R_2 \to R_2 + R_1$

$$R \sim \begin{bmatrix} 0 & 1 & 4 \\ 0 & 0 & 2 \\ 0 & 0 & 2 \end{bmatrix}$$

$R_3 \to R_3 - R_2$

$$\sim \begin{bmatrix} 0 & 1 & 4 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

$r = 2 \quad n = 3$

$r < n \ \exists \ n - r = 3 - 2 = 1$ L.I sol.

$$\begin{bmatrix} 0 & 1 & 4 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$y + 4z = 0$

$2z = 0$

$0 = 0$

Let $x = k$.

$2z = 0$

$z = 0$

$y = -4z$

$y = 0$

$$\therefore X_2 = \begin{bmatrix} k \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

Case 3: put $\lambda = 5$ in ①

$(A - \lambda I)X = 0$

$(A - 5I)X = 0$

$$\begin{bmatrix} -2 & 1 & 4 \\ 0 & -3 & 6 \\ 0 & 0 & 0 \end{bmatrix}$$

$r = 2 \quad n = 3 \quad r < n$

$\exists \ n - r = 3 - 2 = 1$ L.I sol.

$$\begin{bmatrix} -2 & 1 & 4 \\ 0 & -3 & 6 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$-2x + y + 4z = 0$

$-3y + 6z = 0$

$0 = 0$

Let $z = k$

$-3y + 6z = 0$

$3y = +6z$

$3y = +6k$

$y = +2k$

$\cancel{2x = +y - 4z}$

$\cancel{= 2k - 4k}$

$\cancel{2x = -2k}$

$x = -k$

$-2x = -y - 4z$

$2x = y + 4z$

$2x = 2k + 6k$

$x = 3k$

$$\therefore X_3 = \begin{bmatrix} 3k \\ 2k \\ k \end{bmatrix} = k \begin{bmatrix} \cancel{3} \\ \cancel{2} \\ 1 \end{bmatrix} = k \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$$

b) verify cayley Hamilton theorem for $A = \begin{bmatrix} 1 & 0 & 3 \\ 2 & -1 & -1 \\ 1 & -1 & 1 \end{bmatrix}$

given $[A] = \begin{bmatrix} 1 & 0 & 3 \\ 2 & -1 & -1 \\ 1 & -1 & 1 \end{bmatrix}$

The characteristic Matrix $(A-\lambda I) = \begin{bmatrix} 1-\lambda & 0 & 3 \\ 2 & -1-\lambda & -1 \\ 1 & -1 & 1-\lambda \end{bmatrix}$

The characteristic equation $|A-\lambda I| = 0$

$$\begin{vmatrix} 1-\lambda & 0 & 3 \\ 2 & -1-\lambda & -1 \\ 1 & -1 & 1-\lambda \end{vmatrix} = 0$$

$\Rightarrow (1-\lambda)\left[(-1-\lambda)(1-\lambda)+1\right] - 0 + 3\left[-2-(-1-\lambda)\right] = 0$

$\Rightarrow (1-\lambda)\left[-1+\lambda-\lambda+\lambda^2+1\right] + 3\left[-2+1+\lambda\right] = 0$

$\Rightarrow (1-\lambda)\left[\lambda^2-2\right] + 3\left[\lambda-1\right] = 0$

$\Rightarrow \lambda^2-2-\lambda^3+2\lambda+3\lambda-3 = 0$

$\Rightarrow -\lambda^3+\lambda^2+5\lambda-5 = 0$

$\Rightarrow \lambda^3-\lambda^2-5\lambda+5 = 0$

$\cancel{(\lambda-1)(\lambda^2-5)=0}$

$$\begin{array}{c|cccc} 1 & 1 & -1 & -5 & 5 \\ & 0 & 1 & 0 & -5 \\ \hline & 1 & 0 & -5 & 0 \end{array}$$

C.H.T states that every square Matrix satisfies its characteristic equ^n.

To S.T $A^3 - A^2 - 5A + 5I = 0$

$A^3 = \begin{bmatrix} 4 & -3 & 21 \\ 9 & -8 & 1 \\ 5 & -5 & 5 \end{bmatrix}$ $A^2 = \begin{bmatrix} 4 & -3 & 6 \\ -1 & 2 & 6 \\ 0 & 0 & 5 \end{bmatrix}$

$$\begin{bmatrix} 4 & 3 & 21 \\ 9 & -8 & 1 \\ 5 & -5 & 5 \end{bmatrix} - \begin{bmatrix} 4 & -3 & 6 \\ -1 & 2 & 6 \\ 0 & 0 & 5 \end{bmatrix} - 5\begin{bmatrix} 1 & 0 & 3 \\ 2 & -1 & -1 \\ 1 & -1 & 1 \end{bmatrix} + 5\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Thus C.H.T is verified

To find A^{-1}: $A^3 - A^2 - 5A + 5I = 0$

Mul.- b/s by A^{-1}

$$A^2 - A - 5I + 5A^{-1} = 0$$

$$5A^{-1} = -A^2 + A + 5I$$

$$A^{-1} = \frac{1}{5}\left[-A^2 + A + 5I\right]$$

$$= \frac{1}{5}\left[\begin{bmatrix} -4 & 3 & -6 \\ 1 & -2 & -6 \\ 0 & 0 & -5 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 3 \\ 2 & -1 & -1 \\ 1 & -1 & 1 \end{bmatrix} + \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix}\right]$$

$$= \frac{1}{5}\begin{bmatrix} 2 & 3 & -3 \\ 3 & 2 & -7 \\ 1 & -1 & 1 \end{bmatrix}$$

4) Reduce the quadratic form $3x^2 + 5y^2 + 3z^2 - 2xy - 2yz + 2zx$ to the canonical form by orthogonal reduction. And hence state that, nature, rank, index & signature.

The matrix of the given quadratic form is

$$A = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$$

The characterstic matrix $(A - \lambda I) = \begin{bmatrix} 3-\lambda & -1 & 1 \\ -1 & 5-\lambda & -1 \\ 1 & -1 & 3-\lambda \end{bmatrix}$

The characterstic equation $(A - \lambda I) = 0$ $\begin{vmatrix} 3-\lambda & -1 & 1 \\ -1 & 5-\lambda & -1 \\ 1 & -1 & 3-\lambda \end{vmatrix} = 0$

The ~~characterstic~~

$$(3-\lambda)\left[(5-\lambda)(3-\lambda) - 1\right] + 1\left[-1(3-\lambda) + 1\right] + 1\left[1 - (5-\lambda)\right] = 0$$

$$\Rightarrow (3-\lambda)\left\{\lambda^2 - 8\lambda + 14\right\} + (\lambda - 2) + (\lambda - 4) = 0$$

$\Rightarrow 3\lambda^2 - 24\lambda + 42 - \lambda^3 + 8\lambda^2 - 14\lambda + \lambda - 2 + \lambda - 4 = 0$

$\Rightarrow -\lambda^3 + 11\lambda^2 - 36\lambda + 36 = 0$

$\Rightarrow \lambda^3 - 11\lambda^2 + 36\lambda - 36 = 0$

$\Rightarrow (\lambda-2)(\lambda^2 - 9\lambda + 18) = 0$

$\Rightarrow (\lambda-2)(\lambda-3)(\lambda-6) = 0$

$\lambda = 2, 3, 6$

<u>case I</u>: The Eigen ~~value~~ vector corresponding to Eigen value $\lambda = 2$ is $(A - \lambda I) = 0$

$$\begin{bmatrix} 1 & -1 & 1 \\ -1 & 3 & -1 \\ 1 & -1 & 1 \end{bmatrix}$$

Reduce (A) in to Echelon form using Row-operation

$R_2 \to R_2 + R_1$, $R_3 \to R_3 - R_1$

$$\sim \begin{bmatrix} 1 & -1 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$r = 2$ $n = 3$ $r < n$

The system posses non-trivial sol^n.

$\exists$ $n - r = 3 - 2 = 1$ L.I sol^n.

$$\begin{bmatrix} 1 & -1 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \therefore X_1 = \begin{bmatrix} -k \\ 0 \\ k \end{bmatrix} = k \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

$x - y + z = 0$

$2y = 0$

$0 = 0$

Let $z = k$

$y = 0$

$x = y - z$

$= 0 - k$

$= -k$

case2: Put $\lambda = 3$ in characteristic matrix

$$\begin{bmatrix} 0 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$A \quad X = 0$

Consider $[A] = \begin{bmatrix} 0 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 0 \end{bmatrix}$

Reduce $[A]$ into echelon form using Row-operations

$R_1 \leftrightarrow R_2$

$$\sim \begin{bmatrix} -1 & 2 & -1 \\ 0 & -1 & 1 \\ 1 & -1 & 0 \end{bmatrix} \qquad -1-2$$

$R_3 \to R_3 + R_1$ $\quad -1-2$ $\quad -1+2$

$$\sim \begin{bmatrix} -1 & 2 & -1 \\ 0 & -1 & 1 \\ 0 & \text{[illegible]} & 0 \end{bmatrix}$$

$R_3 \to R_3 + R_2$

$$\sim \begin{bmatrix} -1 & 2 & -1 \\ 0 & -1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

~~$R_3 \to R_3 - 2R_2$~~

$$\cancel{\sim \begin{bmatrix} 1 & 2 & -1 \\ 0 & -1 & 1 \\ 0 & 0 & \end{bmatrix}}$$

$r = 2 \quad n = 3 \quad r < n$

$\exists \; n - r = 3 - 2 = 1$ L.I sol^{n}.

$$\begin{bmatrix} -1 & 2 & -1 \\ 0 & -1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$-x + 2y - z = 0$

$-y + z = 0$

$0 = 0$

Let $z = k$

$-y = -z$

$y = k$

$-x + 2y - z = 0$

$-x = -2y + z$

$x = 2y - z$

$= 2k - k$

$= k$

$$X_2 = \begin{bmatrix} k \\ k \\ k \end{bmatrix} = k \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

case3: put $\lambda = 6$ in characteristic matrix

$$\begin{bmatrix} -3 & -1 & 1 \\ -1 & -1 & -1 \\ 1 & -1 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$[A] = \begin{bmatrix} -3 & -1 & 1 \\ -1 & -1 & -1 \\ 1 & -1 & -3 \end{bmatrix}$$

$R_1 \leftrightarrow R_2$

$$\sim \begin{bmatrix} -1 & -1 & -1 \\ -3 & -1 & 1 \\ 1 & -1 & -3 \end{bmatrix}$$

$R_2 \to R_2 - 3R_1$; $R_3 \to R_3 + R_1$

$$\sim \begin{bmatrix} -1 & -1 & -1 \\ 0 & 2 & 4 \\ 0 & -2 & -4 \end{bmatrix}$$

$R_3 \to R_2 + R_3$

$$\sim \begin{bmatrix} -1 & -1 & -1 \\ 0 & 2 & 4 \\ 0 & 0 & 0 \end{bmatrix}$$

$r = 2 \quad n = 3 \quad r < n$

$\exists \; n - r = 3 - 2 = 1$ L.I sol^{n}

$$\begin{bmatrix} -1 & -1 & -1 \\ 0 & 2 & 4 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$-1+$ $\quad 11+3$

$-x - y - z = 0$

$2y + 4z = 0$

$0 = 0$

Let $z = k$

$2y = -4z$

$y = -2k$

$-x - y - z = 0$

$-x = y + z$

$x = -y - z$

$x = +2k - k = +k$

$$X_3 = \begin{bmatrix} k \\ -2k \\ k \end{bmatrix} = k \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

Here $x_1 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$ $x_2 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ $x_3 = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$

$x_1 x_2^T = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} = -1+0+1 = 0$

$x_2 x_3^T = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & -2 & 1 \end{bmatrix} = 1-2+1 = 0$

$x_1 x_3^T = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & -2 & 1 \end{bmatrix} = -1+0+1 = 0$

Here x_1, x_2, x_3 are pairwise orthogonal

The Modal Matrix $P = \begin{bmatrix} -1 & 1 & 1 \\ 0 & 1 & -2 \\ 1 & 1 & 1 \end{bmatrix}$

The normalized modal matrix is $P = [x_1, x_2 x_3] = \begin{bmatrix} -1/\sqrt{2} & 1/\sqrt{3} & 1/\sqrt{6} \\ 0 & 1/\sqrt{3} & -2/\sqrt{6} \\ 1/\sqrt{2} & 1/\sqrt{3} & 1/\sqrt{6} \end{bmatrix}$

Diagonalization $D = P^{-1} A P$

To S.T $P^{-1} A P = D$

$P^T A P = D$ $\left\{ \begin{array}{l} \text{since } P \text{ being orthogonal} \\ P^{-1} = P^T \end{array} \right\}$

$$\begin{bmatrix} -1/\sqrt{2} & 1/\sqrt{3} & 1/\sqrt{6} \\ 0 & 1/\sqrt{3} & -2/\sqrt{6} \\ 1/\sqrt{2} & 1/\sqrt{3} & 1/\sqrt{6} \end{bmatrix} \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix} \begin{bmatrix} -1/\sqrt{2} & 1/\sqrt{3} & 1/\sqrt{6} \\ 0 & 1/\sqrt{3} & -2/\sqrt{6} \\ 1/\sqrt{2} & 1/\sqrt{3} & 1/\sqrt{6} \end{bmatrix}$$

$$\begin{bmatrix} -1/\sqrt{2} & 0 & 1/\sqrt{2} \\ 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \\ 1/\sqrt{6} & -2/\sqrt{6} & 1/\sqrt{6} \end{bmatrix} P$$

$$= \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 6 \end{bmatrix} = D$$

Canonical form $= Y^T D Y = [y_1\ y_2\ y_3] \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 6 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = 2y_1^2 + 3y_2^2 + 6y_3^2$

rank = 3

index $p = 3$

signature = 3

5)

a) solve $(3y^2+4xy-x)dx + x(x+2y)dy = 0$

Given equ is of form $Mdx+Ndy=0$ —①

$M = 3y^2+4xy-x \quad N = x^2+2xy$

$\frac{\partial M}{\partial y} = 6y+4x \quad \frac{\partial N}{\partial x} = 2x+2y$

$\therefore \frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$

Given equ is non-homogenous function in x,y

$\frac{\frac{\partial M}{\partial y}-\frac{\partial N}{\partial x}}{N} = \frac{6y+4x-2x-2y}{x^2+2xy} = \frac{4y+2x}{x^2+2xy}$

$= \frac{2(\cancel{2y+x})}{x(\cancel{2y+x})} = \frac{2}{x}$

$\therefore I\cdot F = e^{\int f(x)dx} = e^{\int 2/x\, dx} = e^{2\log x} = e^{\log x^2} = x^2$

Mult. b/s of equ ① by x^2

$x^2(3y^2+4xy-x)dx + x^2(x^2+2xy)dy = 0$

$(3x^2y^2+4x^3y-x^3)dx + (x^4+2x^3y)dy = 0$ —②

equ ② is of form M_1dx+N_1dy

$M_1 = 3x^2y^2+4x^3y-x^3 \quad N_1 = x^4+2x^3y$

$\frac{\partial M_1}{\partial y} = 6x^2y+4x^3 \quad \frac{\partial N_1}{\partial x} = 4x^3+6x^2y$

$\frac{\partial M_1}{\partial y} = \frac{\partial N_1}{\partial x}$

equ ② is an Exact D.E

$\therefore$ G.S $\int M_1 dx + \int N_1 dy = C$

y constant eliminate x term

$\int(3x^2y^2+4x^3y-x^3)dx+\int(4x^3+6x^2y)dy=c$

$\frac{\cancel{3}x^3}{\cancel{3}}y^2+\cancel{4}\frac{x^4}{\cancel{4}}y-\frac{x^4}{4}=c$

$x^3y^2+x^4y-\frac{x^4}{4}=c$ is required G.S.

b) A body is originally at 80^0c cools down to 60^0C in 20 min the temperature of the air being 40^0c. what will be the temperature of the body after 40 min from the original

$\theta_0=40^0C$

$\theta=80^0 \quad t=0 \quad -(1)$

$\theta=60^0C \quad t=20 \quad -(2)$

$\theta=? \quad t=40 \quad -(3)$

According to newton's law of cooling

$\log(\theta-\theta_0)=-kt+c \quad -(4)$

Apply equ (1) in (4)

$\log(80-40)=-k(0)+C$

$c=\log 40$

Apply equ (2) in (4)

$\log(60-40)=-k(20)+\log 40$

$\log 20-\log 40=-20k$

$\log\left(\frac{20}{40}\right)=-20k$

$k=-\frac{1}{20}\log\left(\frac{2}{4}\right)$

Apply equ ③ in ④

$\log(\theta-40) = -\cancel{k}(\frac{7}{20}\log(1/2)).40 + \log 40$

$\log(\theta-40) = 2\log(\frac{1}{2}) + \log 40$

$= \log\frac{1}{4} + \log 40$

$\log(\theta-40) = \log(\frac{1}{4}\times \cancel{40}^{10})$

$\theta - 40 = 10$

$\theta = 50^{\circ}C$

6) a) solve $x\log x\frac{dy}{dx} + y = 2\log x$.

Given equ: $x\log x\frac{dy}{dx} + y = 2\log x$

Divide on b/s by $x\log x$

$\frac{\cancel{x\log x}}{\cancel{x\log x}}\frac{dy}{dx} + \frac{y}{x\log x} = \frac{2\cancel{\log x}}{x\cancel{\log x}}$

$\frac{dy}{dx} + \left(\frac{1}{x\log x}\right)y = \frac{2}{x}$ - ①

equ ① is of form L.D.E in terms of y

i.e $\frac{dy}{dx} + P(x)y = Q(x)$

$P(x) = \frac{1}{x\log x}$ $\quad Q(x) = \frac{2}{x}$

$I.F = e^{\int P(x)dx} = e^{\int\frac{1}{x\log x}dx} = e^{\log(\log x)}$ $\quad \left\{\int\frac{f'(x)}{f(x)}dx = \log|f(x)|\right.$

$= \log|x|$ $\quad \Rightarrow y\cancel{\log x =}$

$G.S: \; y(I.F) = \int I.F\, Q(x)\,dx + C$ $\quad$ Put $\log x = t$, $\frac{1}{x}dx = dt$

$y\log|x| = \int \log|x|\frac{2}{x}dx + C$

$y \log x = 2\int t\,dt + C$

$= \cancel{2}\frac{t^2}{\cancel{2}} + C$

$y \log x = (\log x)^2 + C$

~~b) solve $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 3y$~~

b) find the O.T's family of curves $x^2+y^2=a^2$

Given family of curves $x^2+y^2=a^2$ — ①

Diff ① w.r.t 'x' we get

$$2x + 2y\frac{dy}{dx} = 0$$

$$x + y\frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -x/y \text{ — ② is D.E of family of curves}$$

now by replacing $\frac{dy}{dx}$ as $-\frac{dx}{dy}$ we get

$\cancel{-}\frac{dx}{dy} = \cancel{-}\frac{x}{y}$

$\frac{dx}{dy} = \frac{x}{y}$ is D.E of O.T's

on solving we get

$\frac{dx}{x} = \frac{dy}{y}$

on I.B.S

$\int \frac{1}{x}dx = \int \frac{1}{y}dy + C$

$\log x = \log y + \log C$

$\log x - \log y = \log C$

$\log(x/y) = \log C$

$x/y = C$ is required family of O.T's

7) b) solve $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 3y = e^x \sin 2x$

Given equation in operator form be

$(D^2y - 4Dy + 3y) = e^x \sin 2x$

$(D^2 - 4D + 3)y = e^x \sin 2x$

<u>To find C.F:</u>

$f(D)y = 0$

$(D^2 - 4D + 3)y = 0$

It's A.E is i.e $f(m) = 0$

$m^2 - 4m + 3 = 0$

$m^2 - 3m - m + 3 = 0$

$m(m-3) - 1(m-3) = 0$

$m = 1, 3$

$y_c = c_1 e^x + c_2 e^{3x}$

<u>To find P.I:</u>

$y_p = \frac{1}{D^2 - 4D + 3} e^x \sin 2x$

$= e^x \frac{1}{(D+1)^2 - 4(D+1) + 3} \sin 2x$

$= e^x \frac{1}{D^2 + 1 + 2D - 4D - 4 + 3} \sin 2x$

$= e^x \frac{1}{D^2 - 2D} \sin 2x$

$= e^x \frac{1}{D^2 - 2D} \sin 2x \quad \left[\begin{array}{l} a = 2 \\ \text{put } D^2 = -a^2 = -4 \end{array}\right]$

$= e^x \frac{1}{-4 - 2D} \sin 2x$

$= e^x \frac{1}{-(4+2D)} \sin 2x$

$= e^x \frac{1}{-(4+2D)} \times \frac{4-2D}{4-2D} \sin 2x$

$= \frac{e^x (4-2D)}{-[4^2 - (2D)^2]} \sin 2x$

$= \frac{e^x (4-2D)}{-16 + 4D^2} \sin 2x \quad \left\{ \text{put } D^2 = -a^2 = -4 \right\}$

$= \frac{e^x (4-2D) \sin 2x}{-16 + 4(-4)}$

$= \frac{e^x [4 \sin 2x - 2D(\sin 2x)]}{-32}$

$= \frac{e^x}{-32} [4 \sin 2x - 4 \cos 2x]$

8) a) Solve the D.E $(D^2+4)y = \sec 2x$ by the method of variation of parameters

Given equ is $(D^2+4)y = \sec 2x$

Here $P = 0$ $Q = 4$ $R = \sec 2x$

A.E is $m^2+4=0 \Rightarrow m = \pm 2i$

The roots are complex conjage $\left[\frac{d^2y}{dx^2}+P\frac{dy}{dx}+Qy\right]=R$

$y_c = C.F = e^{0x}[C_1\cos 2x + C_2\sin 2x]$

$C.F = C_1 u(x) + C_2 v(x)$

$\cancel{y_p = P.I = \frac{1}{f(D)}x(x)}$

$\cancel{= \frac{1}{D^2+4}\sec 2x}$

let $y_p = P.I = A\cos 2x + B\sin 2x$

Here $u = \cos 2x$ $v = \sin 2x$ $R = \sec 2x$

$\frac{du}{dx} = -2\sin 2x$ $\frac{dv}{dx} = 2\cos 2x$

$$u\frac{dv}{dx} - v\frac{du}{dx} = (\cos 2x)(2\cos 2x) - \sin 2x(-2\sin 2x)$$

$$= 2\cos^2 2x + 2\sin^2 2x$$

$$= 2(\cos^2 2x + \sin^2 2x)$$

$$= 2$$

A & B

$$A = -\int \frac{vR}{u\frac{dv}{dx} - v\frac{du}{dx}} = -\int \frac{(\sin 2x)(\sec 2x)}{2} = -\frac{1}{2}\int \tan 2x\, dx$$

$$= \frac{1}{2}\frac{\log(\cos 2x)}{2}$$

$$A = \frac{\log|\cos 2x|}{4}$$

$$B = \int \frac{uR}{u\frac{dv}{dx} - v\frac{du}{dx}} = \int \frac{\cos 2x \sec 2x}{2} = \frac{1}{2}\int \cos 2x \frac{1}{\cos 2x}\, dx$$

$$= \frac{1}{2}[x]$$

$$P.I = \frac{\log|\cos 2x|}{4}\cos 2x + \frac{x}{2}\sin 2x$$

$\therefore$ The C.S $y = y_c + y_p = C_1\cos 2x + C_2\sin 2x + \frac{\cos 2x}{4}\log|\cos 2x| + \frac{x}{2}\sin 2x$

b) solve $(D^2+9)y = \cos 3x + \sin 2x$

$1+\frac{y}{x}$

$\frac{x^2}{x}\left(\frac{x+y}{x}\right)$

Given equ $(D^2+9)y = 0$

<u>To find C.F:</u>

Given equ is of form $f(D)y=0$

$(D^2+9)y=0$

It's Auxillary equ i.e $f(m)=0$

$m^2+9=0$

$m^2=-9 \Rightarrow m=\pm 3i$

$C.F = y_c = e^{0x}\{c_1\cos 3x + c_2\sin 3x\}$

<u>To find P.I:</u> $y_p = \frac{1}{f(D)}X(x)$

$= \frac{1}{D^2+9}(\cos 3x + \sin 2x)$

$= \frac{1}{(D^2+9)}\cos 3x + \frac{1}{D^2+9}\sin 2x$

$\left\{\begin{array}{l} a=3 \\ \text{put } D^2=-a^2=-9 \\ f(D)=0 \end{array}\right.$ $\left.\begin{array}{l} a=2 \\ \text{put } D^2=-a^2 \\ =-4 \end{array}\right\}$

$= \cancel{\frac{1}{D^2+9}} = \frac{1}{-9+9}\cos 3x$

$= \frac{x}{2D}\cos 3x + \frac{1}{-4+9}\sin 2x$

$= \frac{x}{2}\int \cos 3x\,dx + \frac{1}{5}\sin 2x$

$= \frac{x}{2}\frac{\sin 3x}{3} + \frac{1}{5}\sin 2x$

$= \frac{x\sin 3x}{6} + \frac{\sin 2x}{5}$

$G.S = C.F + P.I = c_1\cos 3x + c_2\sin 3x + \frac{x\sin 3x}{6} + \frac{\sin 2x}{5}$

9.a) If $u=\tan^{-1}\left(\frac{x^3+y^3}{x-y}\right)$, P.T $x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}=\sin 2u$

$\therefore u=\tan^{-1}\left(\frac{x^3+y^3}{x-y}\right)$, Here u is not a homogenous function

so, we write $\tan u=\frac{x^3+y^3}{x-y}$ (or) $\frac{(kx)^3+(ky)^3}{kx-ky}$

$=x^2\left[\frac{1+(y/x)^3}{1+y/x}\right]$ $\quad = \frac{k^3(x^3+y^3)}{k(x-y)}$

$=x^2\phi(y/x)$ $\quad = k^2\left\{\frac{x^3+y^3}{x-y}\right) = k^2 f(x,y)$.

$=f(x,y)$

$\therefore f(x,y)$ is a function of x & y of degree 2

By Euler's theorem $x\frac{\partial f}{\partial x}+y\frac{\partial f}{\partial y}=2f$

$\Rightarrow x\frac{\partial(\tan u)}{\partial x}+y\frac{\partial}{\partial y}(\tan u)=2\tan u.$

$\Rightarrow x\sec^2u\frac{\partial u}{\partial x}+y\sec^2u\frac{\partial u}{\partial y}=2\tan u$

$\Rightarrow \sec^2u\left\{x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}\right]=2\tan u$

$\Rightarrow x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}=\frac{2\tan u}{\sec^2 u}$

$=\cancel{2\tan u \times}$

$=2\frac{\sin u}{\cos u}\times\cos^2 u$

$x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y} \quad =\sin 2u$

b) In spherical polar co-ordinates $x=r\sin\theta\cos\phi$ $y=r\sin\theta\sin\phi$ $z=r\cos\theta$ S.T $\frac{\partial(x,y,z)}{\partial(r,\theta,\phi)}=r^2\sin\theta$

from spherical polar co-ordinates, we have

$x=r\sin\theta\cos\phi \qquad y=r\sin\theta\sin\phi \qquad z=r\cos\theta$

$\frac{\partial x}{\partial r}=\sin\theta\cos\phi \qquad \frac{\partial y}{\partial r}=\sin\theta\sin\phi \qquad \frac{\partial z}{\partial r}=\cos\theta$

$\frac{\partial x}{\partial\theta}=r\cos\theta\cos\phi \qquad \frac{\partial y}{\partial\theta}=r\cos\theta\sin\phi \qquad \frac{\partial z}{\partial\theta}=-r\sin\theta$

$\frac{\partial x}{\partial\phi}=-r\sin\theta\sin\phi \qquad \frac{\partial y}{\partial\phi}=r\sin\theta\cos\phi \qquad \frac{\partial z}{\partial\phi}=\phi$

$$\frac{\partial(x,y,z)}{\partial(r,\theta,\phi)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} & \frac{\partial x}{\partial \phi} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} & \frac{\partial y}{\partial \phi} \\ \frac{\partial z}{\partial r} & \frac{\partial z}{\partial \theta} & \frac{\partial z}{\partial \phi} \end{vmatrix} = \begin{vmatrix} \sin\theta\cos\phi & r\cos\theta\cos\phi & -r\sin\theta\sin\phi \\ \sin\theta\sin\phi & r\cos\theta\sin\phi & r\sin\theta\cos\phi \\ \cos\theta & -r\sin\theta & 0 \end{vmatrix}$$

$= \cos\theta\{[(r\cos\theta\cos\phi)(r\sin\theta\cos\phi) + (r\cos\theta\sin\phi)(r\sin\theta\sin\phi)]$

$+ r\sin\theta\{[(\sin\theta\cos\phi)(r\sin\theta\cos\phi) + (\sin\theta\sin\phi)(r\sin\theta\sin\phi)]$

$= \cos\theta[r^2\sin\theta\cos\theta(\cos^2\phi+\sin^2\phi)] + r\sin\theta\{r\sin^2\theta(\cos^2\phi+\sin^2\phi)$

$= r^2\sin\theta\cos^2\theta + r^2\sin^3\theta$

$= r^2\sin\theta(\cos^2\theta+\sin^2\theta)$

$= r^2\sin\theta$

$\therefore \frac{\partial(x,y,z)}{\partial(r,\theta,\phi)} = r^2\sin\theta.$

10) a) Find the minimum value of $x^2+y^2+z^2$, given that $xyz=a^3$

~~Let $f(x,y,z) = x^2+y^2+z^2 = \cancel{0} = \cancel{a}$ - (1)~~

~~Given $xyz=a^3$ - (2)~~

from 2 =

Let $f(x,y,z) = x^2+y^2+z^2$ - (1)

$\phi = xyz - a^3$ - (2)

Consider Legranges function

$F(x,y,z) = f(x,y,z) + \lambda\phi(x,y,z)$

$= (x^2+y^2+z^2) + \lambda(xyz-a^3)$ - (3)

Now $\frac{\partial F}{\partial x}=0 \Rightarrow \frac{\partial f}{\partial x} + \lambda\frac{\partial\phi}{\partial x} = 2x+\lambda yz = 0 \Rightarrow \lambda = \frac{-yz}{2x}$ - (4)

$\frac{\partial F}{\partial y}=0 \Rightarrow \frac{\partial f}{\partial y}+\lambda\frac{\partial \phi}{\partial y} \Rightarrow 2y+\lambda xz=0 \Rightarrow \lambda=-\frac{xz}{2y}$ (5)

$\frac{\partial F}{\partial z}=0 \Rightarrow \frac{\partial f}{\partial z}+\lambda\frac{\partial \phi}{\partial z} \Rightarrow 2z+\lambda xy=0 \Rightarrow \lambda=-\frac{xy}{2z}$ (6)

from (4), (5) & (6)

$$\frac{-yz}{2x}=\frac{-xz}{2y}=\frac{-xy}{2z}$$

$\therefore \frac{\cancel{+}y\cancel{z}}{\cancel{2}x}=\frac{\cancel{+}x\cancel{z}}{\cancel{2}y}$ | $\frac{\cancel{+}\cancel{x}z}{\cancel{2}y}=\frac{\cancel{+}\cancel{x}y}{\cancel{2}z}$ | $\frac{\cancel{+}\cancel{y}z}{\cancel{2}x}=\frac{\cancel{+}x\cancel{y}}{\cancel{2}z}$

$y^2=x^2$ | $y^2=z^2$ | $z^2=x^2$

$\therefore x^2=y^2=z^2$

$x=y=z$

$\therefore$ sub in (2)

$xyz=a^3$

$(x)(x)(x)=a^3$

$x^3=a^3$

$x=a$

$\therefore x=y=z=a$

$\therefore$ Minimum value $x^2+y^2+z^2=f$

$a^2+a^2+a^2=f$

$\therefore f=3a^2$

10) b) Expand the function $f(x,y) = e^x \log(1+y)$ in terms of x & y up to terms of 3rd degree using Taylor's theorem

Taylor's series Expansion

$$f(x,y) = f(a,b) + \frac{1}{1!}\left[x\frac{\partial f}{\partial x}(a,b) + y\frac{\partial f}{\partial y}(a,b)\right] + \frac{1}{2!}\left[x^2\frac{\partial^2 f}{\partial x^2}(a,b) + 2xy\frac{\partial^2 f}{\partial x \partial y}(a,b) + y^2\frac{\partial^2 f}{\partial y^2}(a,b)\right] + \cdots$$

~~this is known~~ Given that $x=0;\ y=0$ i.e $a=0;\ b=0$

Given that $f(x,y) = e^x \log(1+y)$; $f(0,0) = 0$

$$\frac{\partial f}{\partial x} = e^x \log(1+y) = \frac{\partial f}{\partial x}(a,b) = \frac{\partial f}{\partial x}(0,0) = 0$$

$$\frac{\partial f}{\partial y} = e^x \frac{1}{1+y} \ ; \ \frac{\partial f}{\partial y}(a,b) = \frac{\partial f}{\partial y}(0,0) = 1$$

$$\frac{\partial^2 f}{\partial x \partial y} = e^x \frac{1}{1+y} \ ; \ \frac{\partial^2 f}{\partial x \partial y}(a,b) = \frac{\partial^2 f}{\partial x \partial y}(0,0) = 1$$

$$\frac{\partial^2 f}{\partial x^2} = e^x \frac{1}{1+y} \ ; \ \frac{\partial^2 f}{\partial x^2}(0,0) = 0$$

$$\frac{\partial^2 f}{\partial y^2} = -e^x\left(\frac{1}{(1+y)^2}\right) = \frac{\partial^2 f}{\partial y^2}(0,0) = -1$$

$$\frac{\partial^3 f}{\partial x^2 \partial y} = \frac{\partial}{\partial y}\left(\frac{\partial^2 f}{\partial x^2}\right) = -e^x \frac{1}{(1+y)^2} = \frac{\partial^3 f}{\partial x^2 \partial y}(0,0) = 0$$

$$\frac{\partial^3 f}{\partial x^3} = \frac{\partial}{\partial x}\left(\frac{\partial^2 f}{\partial x^2}\right) = e^x \log(1+y) = \frac{\partial^3 f}{\partial x^3}(0,0) = 0$$

$$\frac{\partial^3 f}{\partial y^3} = \frac{\partial}{\partial y}\left(\frac{\partial^2 f}{\partial y^2}\right) = 2e^x \frac{1}{(1+y)^3} = \frac{\partial^3 f}{\partial y^3} = 2$$

$\therefore$ Taylor's Expansion

$$f(x,y) = f(0,0) + \frac{1}{1!}\left\{x\frac{\partial f}{\partial x}(0,0) + y\frac{\partial f}{\partial y}(0,0)\right\} + \frac{1}{2!}\left\{x^2\frac{\partial^2 f}{\partial x^2}(0,0) + \right.$$

$$\left. 2xy\frac{\partial^2 f}{\partial x\partial y}(0,0) + y^2\frac{\partial^2 f}{\partial y^2}(0,0)\right] + \frac{1}{3!}\left[x^3\frac{\partial^3 f}{\partial x^3}(0,0) + 3x^2y\frac{\partial^3 f}{\partial x^2\partial y}(0,0)\right.$$

$$\left. + 3xy^2\frac{\partial^3 f}{\partial x\partial y^2}(0,0) + y^3\frac{\partial^3 f}{\partial y^3}(0,0)\right]$$

$$e^x\log(1+y) = 0+0+0+y+\frac{1}{2!}\left[0+2xy-y^2\right] + \frac{1}{3!}\left[0+3x^2y-3xy^2 + 2y^3\right]$$

$$\cancel{+2y^3\}+}$$

$$e^x\log(1+y) = y + xy - \frac{y^2}{2} + \frac{x^2y}{2} - \frac{xy^2}{2} + \frac{y^3}{3}$$

9) a) Another form of 9(a) question.

If $U = \tan^{-1}\left[\frac{x^3+y^3}{x+y}\right]$ then S.T $x\frac{\partial U}{\partial x} + y\frac{\partial U}{\partial y} = \sin 2U$

let $\tan U = z$

$$\tan\{\tan^{-1}\left(\frac{x^3+y^3}{x+y}\right)\} = z$$

$$\frac{x^3+y^3}{x+y} = z$$

$$\frac{k^3x^3+k^3y^3}{kx+ky} = z$$

It is a homogenous fun/- of degree '2'

$$\frac{k^3(x^3+y^3)}{k(x+y)} = z$$

$$\frac{k^2(x^3+y^3)}{x+y}$$

By Euler's theorem

$$x\frac{\partial z}{\partial x}+y\frac{\partial z}{\partial y}=nz \quad (1)$$

$z = \tan u$

$$\frac{\partial z}{\partial x}=\frac{\partial z}{\partial u}\frac{\partial u}{\partial x}=\sec^2 u\frac{\partial u}{\partial x} \quad (2)$$

$$\frac{\partial z}{\partial y}=\frac{\partial z}{\partial u}\frac{\partial u}{\partial y}=\sec^2 u\frac{\partial u}{\partial y} - (3)$$

sub (2) & (3) in (1)

$$x\sec^2 u\frac{\partial u}{\partial x}+y\sec^2 u\frac{\partial u}{\partial y}=2\tan u$$

$$x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}=2\frac{\tan u}{\sec^2 u}$$

$$=2\frac{\sin u}{\cancel{\cos u}}\times\cos^{\cancel{2}} u$$

$$x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}=\sin 2u$$

7) a) Solve $(D^2+2D+3)y = 5x^2+3e^{3x}$

Given equation $(D^2+2D+3)y = 5x^2+3e^{3x}$

It is of form $f(D)y = X(x)$

<u>To find C.F :-</u>

$f(D)y = 0$

It is A.E i.e $f(m) = 0$

$m^2+2m+3=0$

$m = \frac{-2 \pm \sqrt{4-12}}{2} = \frac{-2 \pm \sqrt{-8}}{2} = \frac{-2 \pm 2\sqrt{2}i}{2} = -1 \pm \sqrt{2}i$

$y_c = e^{-x}\{c_1 \cos\sqrt{2}x + c_2 \sin\sqrt{2}x\}$

<u>To find P.I :-</u>

$y_p = \frac{1}{f(D)} X(x)$

$= \frac{1}{D^2+2D+3} 5x^2+3e^{3x}$

$= \underbrace{\frac{1}{D^2+2D+3} 5x^2}_{PI_1} + \underbrace{\frac{1}{D^2+2D+3} 3e^{3x}}_{PI_2}$ — (1)

<u>consider PI_1 :-</u>

$y_{p_1} = \frac{1}{D^2+2D+3} 5x^2$

$= \frac{1}{3\left[1+\left(\frac{D^2+2D}{3}\right)\right]} 5x^2$

$= \frac{1}{3}\left[\left\{1+\left(\frac{D^2+2D}{3}\right)\right\}\right]^{-1} 5x^2$

$$= \frac{5}{3}\left[1+\left(\frac{D^2+2D}{3}\right)\right]^{-1} x^2 \qquad \left[\because (1+x)^{-1} = 1-x+x^2-x^3+\cdots\right]$$

$$= \frac{5}{3}\left[1-\left(\frac{D^2+2D}{3}\right)+\left(\frac{D^2+2D}{3}\right)^2-\left(\frac{D^2+2D}{3}\right)^3+\cdots\right]x^2$$

$$= \frac{5}{3}\left[1-\frac{D^2}{3}-\frac{2D}{3}+\frac{4D^2}{9}\right]x^2 \quad \text{neglect powers} > 2$$

$$= \frac{5}{3}\left[\cancel{1-D}\, x^2 - \frac{D^2}{3}(x^2) - \frac{2}{3}D(x^2) + \frac{4}{9}D^2(x^2)\right]$$

$$= \frac{5}{3}\left[x^2 - \frac{2}{3} - \frac{2}{3}(2x) + \frac{4}{9}(2)\right] \qquad \left\{\begin{array}{l} D(x^2) = 2x \\ D^2(x^2) = 2 \end{array}\right\}$$

$$-\frac{2}{3}+\frac{8}{9} \qquad \frac{-6+8}{9}$$

$$= \frac{5}{3}\left[x^2 - \frac{4x}{3} + \frac{2}{9}\right]$$

$$= \frac{5x^2}{3} - \frac{20x}{9} + \frac{10}{27} \quad -(2)$$

consider $P.I_2$

$$y_{P_2} = \frac{1}{D^2+2D+3}\, 3e^{3x} \qquad \left[\begin{array}{l} a=3 \\ \text{put } D=a=3 \\ f(D) = 18 \neq 0 \end{array}\right]$$

$$= \frac{1}{3^2+2(3)+3}\, 3e^{3x}$$

$$= \frac{1}{9+6+3}\, 3e^{3x}$$

$$= \frac{1}{18}\, 3e^{3x} = \frac{e^{3x}}{6} \quad -(3)$$

sub (2) & (3) in (1)

$$y_P = \frac{5x^2}{3} - \frac{20x}{9} + \frac{10}{27} + \frac{e^{3x}}{6}$$

$\therefore$ $C.S = C.F + P.I$

$$y_G = y_C + y_P$$

$$= e^{-x}\left[C_1\cos\sqrt{2}x + C_2\sin\sqrt{2}x\right] + \frac{5x^2}{3} - \frac{20x}{9} + \frac{10}{27}$$

UNIT-I-IMP QUESTIONS

[redacted]

1) Reduce the matrix A= $\begin{bmatrix} 1 & 2 & 3 & 0 \\ 2 & 4 & 3 & 2 \\ 3 & 2 & 1 & 3 \\ 6 & 8 & 7 & 5 \end{bmatrix}$ into Echelon form and hence find its rank.

2) Define the rank of the matrix and find the rank of the following matrix

$$\begin{bmatrix} 2 & 1 & 3 & 5 \\ 4 & 2 & 1 & 3 \\ 8 & 4 & 7 & 13 \\ 8 & 4 & -3 & -1 \end{bmatrix}.$$

3) Reduce the matrix $\begin{bmatrix} 0 & 1 & 2 & -2 \\ 4 & 0 & 2 & 6 \\ 2 & 1 & 3 & 1 \end{bmatrix}$ to normal form and hence find the rank.

4) Find the rank of matrix A= $\begin{bmatrix} 2 & -2 & 0 & 6 \\ 4 & 2 & 0 & 2 \\ 1 & -1 & 0 & 3 \\ 1 & -2 & 1 & 2 \end{bmatrix}$ by reducing it to canonical form.

5) Find the non singular matrices P and Q such that PAQ in normal form where

A= $\begin{bmatrix} 3 & 2 & -1 & 5 \\ 5 & 1 & 4 & -2 \\ 1 & -4 & 11 & -19 \end{bmatrix}$ and hence find its rank.

6) Find the non singular matrices P and Q such that PAQ in normal form where

A= $\begin{bmatrix} 1 & 3 & 6 & -1 \\ 1 & 4 & 5 & 1 \\ 1 & 5 & 4 & 3 \end{bmatrix}$ and hence find its rank.

[redacted]

7) Find whether the following equations are consistent, if so solve them

$x + y + 2z = 4,\ \ 2x - y + 3z = 9,\ \ 3x - y - z = 2.$

8) Find the value of p and q so that the equations $2x + 3y + 5z = 9$,

$7x + 3y + 2z = 8,\ \ 2x + 3y + pz = q.$

(i) No Solution

(ii) Unique solution

(iii) An infinite number of solutions

9) Solve the system of equations

$x+3y-2z=0$, $2x-y+4z=0$, $x-11y+14z=0$.

10) Determine b such that the system of homogeneous equations $2x+y+2z=0$, $x+y+3z=0$, $4x+3y+bz=0$ has trivial and non trivial solutions. Find Non trivial solution.

11) Solve the system of equations $3x+y+2z=3$, $2x-3y-z=-3$, $x+2y+z=4$.

By Gauss elimination method.

12) Solve the system of equations $10x-y-z=13$, $x+10y+z=36$, $-x-y+10z=35$.

By Gauss -Jordan method.

13) Use Gauss-Seidal iteration method to solve the system.

$10x+y+z=12$, $2x+10y+z=13$, $2x+2y+10z=14$.

IMP QUESTIONS OF UNIT 2

EIGEN VALUES AND EIGEN VECTORS

1. Find the Eigen values and Eigen vector of the matrix $A = \begin{bmatrix} 3 & 10 & 5 \\ -2 & -3 & -4 \\ 2 & 5 & 7 \end{bmatrix}$

2. Find the Eigen roots and Eigen vectors of the matrix $A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$

3. Determine the characteristic roots and corresponding characteristic vectors of the matrix

$$A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$$

4. Verify Cayley – Hamilton theorem for the matrix $A = \begin{bmatrix} 1 & 0 & 3 \\ 2 & -1 & -1 \\ 1 & -1 & 1 \end{bmatrix}$ and hence find A^{-1}?

5. State Cayley – Hamilton theorem and verify Cayley – Hamilton theorem for the matrix

$A = \begin{bmatrix} 2 & -1 & 2 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$ and hence find A^{-1} and A^4

6. Varify Cayley – Hamilton theorem for the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$ and hence find A^{-1}.

7. Diagonalize the matrix $A = \begin{bmatrix} 8 & -8 & -2 \\ 4 & -3 & -2 \\ 3 & -4 & 1 \end{bmatrix}$

8. Diagonalize the matrix $A = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$ and find A^4 using modal matrix P?

9. Show that the matrix $A = \begin{bmatrix} -9 & 4 & 4 \\ -8 & 3 & 4 \\ -16 & 8 & 7 \end{bmatrix}$ is diagonalizable. Also find the diagonal form and a Diagonalizing matrix P.

10. Reduce the quadratic form $3{x_1}^2 + 3{x_2}^2 + 3{x_3}^2 + 2x_1x_2 + 2x_1x_3 - 2x_2x_3$ into canonical form by an Orthogonal Transformation.

11. Reduce the quadratic form $3x^2 + 2y^2 + 3z^2 - 2xy - 2yz$ to the normal form by Orthogonal Transformation.

12. Reduce the quadratic form $2x^2 + 2y^2 + 2z^2 - 2xy - 2yz - 2zx$ to the sum of squares form by Orthogonal Transformation and hence find rank, index, signature and nature of the quadratic form.

13. Prove that two eigen vectors corresponding to the two different eigen values are linearly independent.

14. If A and B are n rowed square matrices and if A is invertible show that $A^{-1}B$ and BA^{-1} have same eigen values.

III unit – model questions

1. solve $\frac{dy}{dx} + x\sin 2y = x^3 cos^2 y$
2. solve $(1-x^2)\frac{dy}{dx} + xy = y^3 \sin^{-1} x$
3. solve $x\frac{dy}{dx} + y = x^3y^6$
4. solve $(x+y+1)\frac{dy}{dx} = 1$
5. solve $2xy\,dy - (x^2+y^2+1)dx = 0$
6. solve $(y^4+2y)dx + (xy^3+2y^4-4x)dy = 0$
7. solve $y(1+xy)dx + x(1-xy)dy = 0$
8. solve $x^2y\,dx - (x^3+y^3)dy = 0$
9. solve $(2x-y+1)\,dx + (2y-x-1)\,dy = 0$
10. solve $\frac{dy}{dx} + \frac{y}{x} = y^2 x \sin x$

11. Find the orthogonal trajectories of the family of curves $x^{2/3} + y^{2/3} = a^{2/3}$ where 'a' is parameter

12. Find the orthogonal trajectories of the family of cardioids $r = a(1-\cos\theta)$ where 'a' is parameter

13. A body is originally at 80°C and cools down to 60°C in 20 minutes. If the temperature of the air is 40°C, find the temperature of the body after 40 minutes.

14. A bacterial culture growing exponentially increases from 100 to 400 grams in 10 Hrs. How much was present after 3 Hrs from the initial instant.

15. Radium decomposes at a rate proportional to the amount present. If 5% of the original amount disappears in 50 years, how much will remain after 100 years

16. Prove that the system of parabolas $y^2 = 4a(x+a)$ is self orthogonal

17. Find the orthogonal trajectories of the family of curves $r^n \sin n\theta = a^n$ where 'a' is parameter

18. If the temperature of a body is changing from 100°C to 70°C in 15 minutes. Find when the temperature will be 40°C, if the temperature of the air is at 30°C

19. Uranium disintegrates at a rate proportional to the amount present at any instant. If M_1 and M_2 are grams of uranium that are present at times T_1 and T_2 respectively, find the half-life of uranium.

20. A bacterial culture, growing exponentially, increases from 200 to 500 grams in the period from 6 AM to 9 AM. How many grams will be present at noon.

Unit – 4

IMP QUESTIONS

[redacted]

1. Solve $(D^2+4D+13)y = 2e^{-x}$
2. Solve $(4D^2-4D+1)y = 100$
3. Solve $(D^2-3D+2)y = \cosh x$
4. Solve $(D^2-4)y = 2\cos^2 x$
5. Solve $(D^2-4D+3)y = \sin 3x \cos 2x$
6. Solve $(D^2-4D+4)y = 8x^2+e^{2x}$
7. Solve $(D^3+2D^2+D)y = e^{2x}+x^2+x+\sin 2x$
8. Solve $(D^2-5D+6)y = e^x \sin x$
9. Solve $(D^2-2D+4)y = e^{2x}\cos x$
10. Solve $(D^3+1)y = \cos(2x-1)+x^2e^{-x}$

[redacted]

1. Solve $(D^2+4)y = x\sin x$
2. Solve $(D^2+3D+2)y = xe^x \sin x$
3. Solve $(D^2-4d+4)y = 8x^2e^{2x}\sin 2x$
4. Solve $(D^4+2D^2+1)y = x^2\cos x$
5. Solve $(D^2+a^2)y = \tan ax$
6. Solve $(D^2+4D+3)y = e^{e^{x}}$
7. Solve $(D^2+a^2)\ y = \sec ax$

8. Apply the method of variation of parameters to solve $(D^2+1)y = \text{cosec}x$

9. Solve by the method of variation of parameters $(D^2-2D)y = e^x \sin x$

10. Solve the differential equation $(D^2+4)Y = \text{Sec}2x$ by the method of variation parameters